REPUBLIC OF NAMIBIA

MINISTRY OF ENVIRONMENT, FORESTRY AND TOURISM

Namibia's Third Nationally Determined Contribution (NDC 3.0)

2026



A CLEANER | GREENER | MORE RESILIENT NAMIBIA

Acknowledgement

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The NDC 3.0 was developed through a collaborative process involving a multidisciplinary team of national and international consultants working closely with MEFT, UNDP, government ministries, agencies, development partners, the private sector, academia, civil society organisations, and other stakeholders. Their technical expertise, stakeholder engagement, and valuable contributions have strengthened the ambition, inclusiveness, and implementability of Namibia's updated climate commitments.

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FOREWORD



Climate change remains one of Namibia's most pressing development challenges. As one of the driest countries in sub-Saharan Africa, Namibia is increasingly exposed to drought, rising temperatures, variable rainfall, floods and wildfires. These impacts affect water and food security, biodiversity, infrastructure, health and livelihoods, particularly among communities and groups with limited capacity to respond.

It is therefore my honour, on behalf of the Government of the Republic of Namibia, to present Namibia's Third Nationally Determined Contribution (NDC 3.0) under the Paris Agreement. This contribution builds on the progress and lessons of our previous NDCs and reflects our continued commitment to climate action that supports sustainable and inclusive national development.

Namibia contributes less than 0.03% of global greenhouse gas emissions and therefore remains a net carbon sink. Our climate response must therefore both reduce emissions from a growing economy and protect and strengthen the natural systems that continue to remove carbon from the atmosphere, while building the resilience of our people and productive sectors.

NDC 3.0 strengthens mitigation ambition across energy, agriculture, forestry and land use, industrial processes and waste, while placing adaptation at the centre of the national response. Priority areas include water and food security, climate-resilient agriculture, biodiversity and ecosystems, fisheries, health, infrastructure, coastal areas and urban settlements. It also places greater emphasis on gender equality and social inclusion.

Our ambition must be matched by implementation. Namibia has committed domestic resources to climate action, but the scale of investment required exceeds what can be mobilised nationally. Approximately 88% of the estimated financing required for NDC 3.0 is conditional on international climate finance, technology transfer and capacity-building support. Strong partnerships with development partners, climate funds, financial institutions and the private sector will therefore be essential.

The preparation of NDC 3.0 has been a collaborative national process involving Government institutions, development partners, the private sector, academia, civil society and other stakeholders. I extend my sincere appreciation to all who contributed their knowledge, time and resources to this process.

As we move from commitment to implementation, our focus must remain on delivery, accountability and partnerships. Climate action must strengthen Namibia's resilience, protect our natural heritage, create opportunities for our people and support a sustainable economy for present and future generations.

MINISTRY OF ENVIRONMENT, FORESTRY AND TOURISM
25 SEP 2026
OFFICE OF THE MINISTER
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Hon. Indileni Daniel, MP
Minister of Environment, Forestry and Tourism
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List of Abbreviations and Acronyms

Acronym / Abbreviation	Meaning
%	Percent
AC	Air Conditioning
AEB	Atomic Energy Board
AFD	Agence Française de Développement (French Development Agency)
AFOLU	Agriculture, Forestry and Other Land Use
AR5	Fifth Assessment Report (IPCC)
BAU	Business-As-Usual
BCBU	Bush Control and Biomass Utilisation (GIZ programme; BCBU III denotes Phase III)
BESS	Battery Energy Storage System
BEV	Battery Electric Vehicle
BTR / BTR1 / BTR2	Biennial Transparency Report (first/second)
CA	Conservation Agriculture
CBNRM	Community-Based Natural Resource Management
CC	Climate Change
CCAP	Comprehensive Conservation Agriculture Programme
CDM	Clean Development Mechanism
CEM I / CEM II	Cement type classifications under standard EN 197-1 (CEM I = Portland cement; CEM II = Portland-composite cement); "LL" denotes limestone content
CH_4	Methane
CIF	Climate Investment Funds
CMA	Conference of the Parties serving as the meeting of the Parties to the Paris Agreement
CO_2 / CO_2e	Carbon Dioxide / Carbon Dioxide equivalent
COP / CP	Conference of the Parties
COP28	28th Conference of the Parties (Dubai, 2023)
CRAVE	Climate Resilient Agriculture in three of the Vulnerable Extreme northern crop-growing regions (GCF/EIF project)
CSA	Climate-Smart Agriculture
DBN	Development Bank of Namibia
DDRM	Directorate of Disaster Risk Management
DOM	Dead Organic Matter
DRI	Direct Reduced Iron (as in H_2-DRI)
DSM	Demand-Side Management
ECB	Electricity Control Board
ECC	Environmental Clearance Certificate
EDA	Enhanced Direct Access (GCF financing modality)
EE	Energy Efficiency
EIA	Environmental Impact Assessment
EIF	Environmental Investment Fund (of Namibia)
EPR	Extended Producer Responsibility
EOC	Emergency Operations Centre

ETF	Enhanced Transparency Framework
EU	European Union
EV	Electric Vehicle
EWS	Early Warning System
FAO	Food and Agriculture Organisation of the United Nations
FEED	Front-End Engineering Design
FFEM	Fonds Français pour l'Environnement Mondial (French Facility for Global Environment)
FSC	Forest Stewardship Council
GA	General Aviation
GACMO	Greenhouse Gas Abatement Cost Model
GBV	Gender-Based Violence
GCF	Green Climate Fund
GDP	Gross Domestic Product
GEF	Global Environment Facility
GESI	Gender Equality and Social Inclusion
GHG	Greenhouse Gas
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit (German Agency for International Cooperation)
GJ	Gigajoule
GSE	Ground Support Equipment
GST	Global Stocktake
GWh	Gigawatt-hour
GWP	Global Warming Potential
H_2	Hydrogen
H_2-DRI	Hydrogen-based Direct Reduced Iron
ha	Hectare
HCFC	Hydrochlorofluorocarbon
HDI	Human Development Index
HFC	Hydrofluorocarbon
HFO	Heavy Fuel Oil
HIV	Human Immunodeficiency Virus
HNAP	Health National Adaptation Plan
HPMP	HCFC Phase-out Management Plan
HVO	Hydrotreated Vegetable Oil
HWC	Human-Wildlife Conflict
IAEA	International Atomic Energy Agency
IAS	Invasive Alien Species
ICCBRLP	Integrated Climate Resilient Community Based Rangeland and Livestock Value Chain Project
ICE	Internal Combustion Engine
ICT	Information and Communication Technology
INCTU	Information (Necessary) to Facilitate Clarity, Transparency and Understanding
ICZM	Integrated Coastal Zone Management
IDP	Industry Decarbonisation Programme (Climate Investment Funds)
IPCC	Intergovernmental Panel on Climate Change
IPP	Independent Power Producer

IPPU	Industrial Processes and Product Use
ISAP	Implementation Strategy and Action Plan (see NDC-ISAP)
IWRM	Integrated Water Resources Management
kWh	Kilowatt hours
KfW	Kreditanstalt für Wiederaufbau (German Development Bank)
kt CO2e	Kilotonnes of Carbon Dioxide equivalent
LED	Light-Emitting Diode
LLPBN	Livestock and Livestock Products Board of Namibia
LT-LEDS	Long-Term Low Emission Development Strategy
LULUCF	Land Use, Land-Use Change and Forestry
MAFWLR	Ministry of Agriculture, Fisheries, Water and Land Reform
MEFT	Ministry of Environment, Forestry and Tourism
MEIYSAC	Ministry of Education, Innovation, Youth, Sport, Arts and Culture
MEL	Monitoring, Evaluation and Learning
MEPS	Minimum Energy Performance Standards
MGECW	Ministry of Gender Equality, Poverty Eradication and Social Welfare
MHAISS	Ministry Of Home Affairs, Immigration, Safety and Security
MHSS	Ministry of Health and Social Services
MIME	Ministry of Industries, Mines and Energy
MISP	Minimum Initial Service Package
MJ	Megajoule
MRV	Measurement, Reporting and Verification
MSB	Modified Single Buyer (electricity market model)
$MtCO_2e$	Million metric tonnes of Carbon Dioxide equivalent
MURD	Ministry of Urban and Rural Development
MW / MWh	Megawatt / Megawatt-hour
MWT	Ministry of Works and Transport
N_2O	Nitrous Oxide
N/A	Not applicable
NAB	Namibian Agronomic Board
NAC	Namibia Airports Company
NAMRA	Namibia Revenue Agency
NAMS	Namibian Standard(s)
NATIS	Namibia Traffic Information System
NAU	Namibia Agricultural Union
NBSAP	National Biodiversity Strategy and Action Plan
NC5	Fifth National Communication
NCAA	Namibia Civil Aviation Authority
NCAP	National Cooling Action Plan
NCCC	National Climate Change Committee
NCCGAP	National Climate Change Gender Action Plan
NDC	Nationally Determined Contribution
NDC-ISAP	NDC Implementation Strategy and Action Plan
NDP / NDP6	National Development Plan / Sixth National Development Plan
NDRMS	National Disaster Risk Management System
NEPIO	Nuclear Energy Programme Implementation Organisation

NERA	Namibia Energy Regulatory Authority (proposed, under the NERA Bill)
NGH2P	Namibia Green Hydrogen Programme
NGO	Non-Governmental Organisation
NH_3	Ammonia
NID	National Inventory Document
NILALEG	Namibia Integrated Landscape Approach for Enhancing Livelihoods and Environmental Governance to Eradicate Poverty
NIR	National Inventory Report
NIRP	National Integrated Resource Plan
NMS	Namibia Meteorological Service
NNFU	Namibia National Farmers Union
NOU	National Ozone Unit
NPC	National Planning Commission
NQA	Namibia Qualifications Authority
NRPA	National Radiation Protection Authority
NSA	Namibia Statistics Agency
NSI	Namibian Standards Institution
NSWMS	National Solid Waste Management Strategy
NWR	Namibia Wildlife Resorts
NTA	Namibia Training Authority
O&M	Operations and Maintenance
ODS	Ozone-Depleting Substance(s)
OPM	Office of the Prime Minister
PA	Paris Agreement
PHEOC	Public Health Emergency Operations Centre
PPA	Power Purchase Agreement
PPP	Public-Private Partnership
PV	Photovoltaic
R290	Propane (a low Global Warming Potential hydrocarbon refrigerant)
RACHP	Refrigeration, Air Conditioning and Heat Pump (sector)
RDF	Refuse-Derived Fuel
RE	Renewable Energy
RA	Roads Authority
RC	Regional Council
RFA	Road Fund Administration
RMNCAH	Reproductive, Maternal, Neonatal, Child and Adolescent Health
SADDD	Sex, Age, and Disability Disaggregated Data
SAF	Sustainable Aviation Fuel
SAPP	Southern African Power Pool
SASSCAL	Southern African Science Service Centre for Climate Change and Adaptive Land Management
SCM	Supplementary Cementitious Material
SDG	Sustainable Development Goal
SHS	Solar Home System(s)
SOC	Soil Organic Carbon

SOLTRAIN	Southern African Solar Thermal Training and Demonstration Initiative
SRF	Solar Revolving Fund
SRH / SRHR	Sexual and Reproductive Health / Sexual and Reproductive Health and Rights
SSF	Small-scale fisheries
SWH	Solar Water Heater / Heating
t	Tonnes (1000kg)
TACCC	Transparency, Accuracy, Completeness, Comparability, Consistency
TEES	Transmission Expansion and Energy Storage (World Bank/NamPower project)
TJ	Terajoule
ToU	Time-of-Use
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
USD	United States Dollar
VAT	Value Added Tax
VTC	Vocational Training Centre
WASH	Water, Sanitation and Hygiene
WBBC	Windhoek Waste Buy-Back Centre
WtE	Waste-to-Energy
WWTP	Wastewater Treatment Plant
Y/N	Yes/No
ZAMCO	Zambezi Meat Corporation

EXECUTIVE SUMMARY

BACKGROUND

Namibia's Third Nationally Determined Contribution (NDC 3.0) sets out the country's climate commitments for the period 2026–2035, with 2030 serving as an interim milestone and 2035 as the principal target year. It builds on Namibia's previous NDCs, lessons from implementation of NDC 2.0, the latest national greenhouse gas inventory and the outcomes of the first Global Stocktake under the Paris Agreement.

Namibia remains one of the world's most arid and climate-vulnerable countries. Increasing temperatures, prolonged droughts, erratic rainfall, floods and wildfires threaten water security, agriculture, biodiversity, public health, infrastructure and livelihoods. These impacts are particularly significant for communities whose livelihoods depend directly on land, water and other natural resources, and can deepen existing inequalities affecting women, youth, children, older persons, persons with disabilities and poorer households.

At the same time, Namibia makes a very small contribution to global greenhouse gas emissions and remains a net carbon sink, largely because of removals from its land and forest resources. NDC 3.0 therefore follows a dual approach: reducing greenhouse gas emissions while protecting and enhancing the country's carbon sink, alongside strengthening Namibia's resilience to the impacts of climate change.

MITIGATION CONTRIBUTION

Namibia's mitigation programme covers the Energy; Agriculture, Forestry and Other Land Use (AFOLU); Industrial Processes and Product Use (IPPU); and Waste sectors. Key priorities include expanding renewable energy and energy efficiency; cleaner transport and fuels; sustainable livestock and agricultural production; reducing wildfires; restoring forests and degraded land; increasing soil carbon; reducing clinker use in cement production; improving waste management, recycling and resource recovery; and reducing methane emissions.

By 2035, the measures identified in NDC 3.0 are estimated to deliver approximately 7.158 $MtCO_2e$ in mitigation under the unconditional scenario and 15.608 $MtCO_2e$ under the conditional scenario. The larger conditional contribution depends on access to international climate finance, technology and capacity support.
The Energy Sector represents a major component of this transition, including renewable electricity generation, grid modernisation, energy efficiency and cleaner transport. AFOLU remains equally important because Namibia must not only reduce emissions but also maintain the ecosystems and landscapes responsible for its substantial carbon removals.

ADAPTATION AND RESILIENCE

Adaptation is central to NDC 3.0. Namibia has identified nine adaptation priority areas comprising 46 measures and 110 actions: Agriculture and Food Security; Water Re-

sources; Terrestrial Biodiversity and Ecosystems; Marine Biodiversity, Fisheries and Aquaculture; Health and Sanitation; Cross-cutting Issues; Infrastructure; Coastal Zones; and Urban Settlements.

These priorities seek to strengthen food and water security, climate-resilient agriculture, water infrastructure and alternative water sources, healthy ecosystems and community-based natural resource management, sustainable fisheries and aquaculture, climate-resilient health systems and settlements, disaster preparedness and resilient infrastructure. Water security receives particular attention through improved water-resource planning and management, riparian restoration, desalination, water infrastructure, rainwater harvesting and water recycling.

NDC 3.0 also recognises that adaptation cannot eliminate all climate impacts. Improved assessment of loss and damage will therefore support disaster risk management, social protection, public finance and future access to climate finance.

AN INCLUSIVE CLIMATE RESPONSE

Gender equality and social inclusion are integrated across mitigation and adaptation. NDC 3.0 strengthens the use of disaggregated indicators and seeks greater participation and access to climate-resilient livelihoods, productive resources and decision-making for groups that may be disproportionately affected by climate change. These commitments will be further operationalised through the NDC Implementation Strategy and Action Plan and the adaptation Monitoring, Evaluation and Learning system.

FINANCING AND IMPLEMENTATION

Delivering NDC 3.0 will require substantial investment. The current estimated implementation requirement is approximately USD 14.07 billion, comprising about **USD 6.42 billion for mitigation and USD 7.65 billion for adaptation.** Approximately USD 1.66 billion (12%) is classified as unconditional, while approximately USD 12.42 billion (88%) is conditional on international climate finance, technology transfer and capacity-building support.

Implementation will be coordinated by the Ministry of Environment, Forestry and Tourism in collaboration with sector ministries, the National Planning Commission, the Environmental Investment Fund, regional and local authorities, the private sector, civil society, communities and development partners. An updated NDC Implementation Strategy and Action Plan (NDC-ISAP) will translate the commitments into implementation arrangements, while strengthened measurement, reporting, verification and adaptation monitoring systems will track progress and support transparency.

NDC 3.0 ultimately seeks to ensure that Namibia's response to climate change supports, rather than constrains, its broader development aspirations. It provides a pathway towards a climate-resilient, low-carbon and inclusive economy, while safeguarding the natural resources and ecosystems on which Namibia's people and economy depend.

1. INTRODUCTION

Namibia, as a Party to the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement, hereby communicates its third Nationally Determined Contribution (NDC 3.0). This contribution represents a progression from Namibia's First NDC (2015) and its NDC 2.0 (submitted in 2021 and updated in 2023), reflecting the country's commitment to the highest possible ambition in light of its common but differentiated responsibilities and respective capabilities, and its particular national circumstances as one of the most arid and climate-vulnerable countries in sub-Saharan Africa.

Namibia is, and is projected to remain, a net carbon sink, owing to extensive removals in the Agriculture, Forestry and Other Land Use (AFOLU) sector. The country's NDC 3.0 therefore pursues a dual objective: reducing gross greenhouse gas (GHG) emissions across all sectors while protecting and enhancing the national sink. NDC 3.0 covers all IPCC-defined sectors, Energy, Industrial Processes and Product Use (IPPU), AFOLU, and Waste, and all greenhouse gases reported in the national inventory (CO_2, CH_4, N_2O and HFCs), with a target end-year of 2035.

This NDC 3.0 has been informed by the outcomes of the first Global Stocktake (GST) concluded at COP28 (December 2023), and is grounded in a comprehensive national evidence base, including the NDC 2.0 Stocktake Report and Revision Roadmap (2026), Namibia's first Biennial Transparency Report (BTR1) and Fifth National Communication (NC5) submitted in 2024, the National Inventory Document (NID), and the Long-Term Low Emission Development Strategy (LT-LEDS) Vision document. Its preparation has been guided by the NDC Partnership's NDC 3.0 Navigator and the UNDP Quality Assurance Checklist for NDCs 3.0 across the five dimensions of country ownership and inclusivity, ambition, just transition and sustainable development, clarity and transparency, and feasibility.

The NDC is aligned with Namibia's Vision 2030, the National Development Plan (NDP6), and the National Climate Change Policy (2011)[1] and its Strategy and Action Plan, and relevant sectoral policies and legal frameworks.

1.1. Salient Features of Namibia

- Namibia covers approximately 825,000 km² and has a population of just over 3.02 million, growing at about 3.0% annually. More than 71% of the population is under 35 years of age, with an almost even urban-rural distribution (49.5/50.5%) (NSA, 2024). Rapid urbanisation is increasing demand for climate-resilient infrastructure, services and disaster risk management.

- Namibia's Gross Domestic Product (GDP) stood at N$269.8 billion (approximately USD 16.4 billion) in 2025 (NSA, 2026), with real GDP growth of 2.6% in 2026 (BoN, 2026). Sustaining economic growth will increasingly depend on cli-

[1] The National Climate Change Policy was under review at the time of preparation of this document

mate-resilient infrastructure, sustainable natural resource management and investment in low-carbon industries.

- Namibia is one of the world's driest countries, receiving less than 25 mm to over 600 mm of rainfall annually. Water scarcity remains a major constraint to socio-economic development and increases vulnerability to climate change.

- Namibia's economy remains highly dependent on climate-sensitive natural resources, making it vulnerable to droughts, floods and commodity price fluctuations. NDP6 prioritises economic diversification, value addition and climate-resilient growth.

- Manufacturing, mining and mineral beneficiation remain central to Namibia's economic transformation agenda. The expansion of value addition, green industrialisation and low-carbon manufacturing presents significant opportunities to decouple industrial growth from greenhouse gas emissions while strengthening economic resilience.

- Climate change disproportionately affects women and vulnerable groups due to existing socio-economic inequalities. Gender-responsive planning therefore remains central to Namibia's climate change response. Namibia remains one of the world's most unequal countries, with high levels of inequality increasing vulnerability to climate-related risks and reducing adaptive capacity.

- Bush encroachment affects an estimated 45 million hectares (about 60% of Namibia), reducing rangeland productivity and biodiversity while creating opportunities for sustainable biomass utilisation, ecosystem restoration and climate change mitigation.

- Namibia remains a net sink of greenhouse gases and is expected to remain so through 2035. Despite contributing less than 0.001% of global greenhouse gas emissions, Namibia is highly vulnerable to the impacts of climate change.
- Agriculture supports the livelihoods of more than 70% of the population and directly employs about 21% of the workforce, making it one of the sectors most vulnerable to climate change.

- Livestock production remains the largest contributor to Namibia's agricultural sector, accounting for approximately 59% of total agricultural output. The sector supports rural livelihoods, food security and export earnings but is highly vulnerable to climate variability, reinforcing the need for climate-resilient rangeland management and sustainable livestock production systems.

- By the end of 2024, Namibia had 86 registered communal conservancies and 47 community forests, collectively managing more than 185,800 km² (approximately 22% of the country's land area). These community-managed landscapes

play a critical role in biodiversity conservation, sustainable natural resource management, ecosystem-based adaptation, and carbon sequestration.

- Namibia possesses globally competitive renewable energy resources, including high solar irradiation, favourable wind regimes, sustainable biomass resources and significant green hydrogen potential, providing opportunities for green industrialisation and a low-carbon and climate-resilient economy.

- Namibia has an installed electricity generation capacity of approximately 730–750 MW, with peak electricity demand exceeding 700 MW and a significant proportion of electricity still imported from the Southern African Power Pool (SAPP). Rapid expansion of renewable energy generation, battery storage and transmission infrastructure is therefore central to achieving energy security and the country's long-term decarbonisation objectives.

- Namibia hosts globally significant biodiversity and approximately 40% of its terrestrial area is under some form of conservation, providing important ecosystem services that support climate adaptation and carbon sequestration.

2. NDC REVISION PROCESS

2.1. Overview

Namibia's NDC 3.0 has been developed through a structured, inclusive and nationally owned process led by the Ministry of Environment, Forestry and Tourism (MEFT) as the national climate change focal point, in coordination with the United Nations Development Programme (UNDP), and sectoral line ministries.

The revision was underpinned by dedicated technical workstreams:

i) the NDC 2.0 stocktake assessing implementation progress, gaps and barriers across mitigation and adaptation;
(ii) an update of the GHG inventory and Business-as-Usual (BAU) scenario;
(iii) revision of measures, actions and Measurement, Reporting and Verification (MRV)-ready indicators; and
(iv) costing and conditionality analysis.

Gender Equality and Social Inclusion (GESI) considerations were integrated and would be further elaborated through the forthcoming National Climate Change Gender Action Plan (NCCGAP). The revised measures will be operationalised through an NDC Implementation Strategy and Action Plan (NDC-ISAP) designating lead ministries, reporting responsibilities and coordinating bodies.

2.2. Lessons Learnt from NDC 2.0

The NDC 2.0 stocktake highlighted several lessons that have directly informed the development of NDC 3.0:

- **MRV and indicator design:** NDC 2.0 indicators were largely input-based (e.g. installed capacity in MW, units installed) rather than output or outcome-based. NDC 3.0 reformulates indicators towards measurable outcomes (e.g. GWh generated, tCO_2e avoided, litres of fuel displaced), consistent with the UNFCCC Enhanced Transparency Framework (ETF).

- **Inventory data consistency and transparency:** The NID is consistent and compliant with the MPGs (paragraphs 17 to 58) and a tabular summary of GHG emissions and removals by sector are included in the CRTs, labelling of GWPs and inclusion of standardised sectoral emissions and removals. The NID aligns with the mandatory requirement under the Modalities, Procedures and Guidelines for the Enhanced Transparency Framework (Article 13 of the Paris Agreement), Section E, paragraph 91.

- **Measure-design coherence:** Several NDC 2.0 measures aggregated fundamentally different technologies (for example, solar, wind, biomass, hydropower and the Anixas II heavy fuel oil peaking plant under a single renewable-substitution measure). NDC 3.0 separates mitigation actions from enabling or reliability infrastructure to avoid overstating mitigation progress.

- **Realistic, phased ambition:** Targets for hydrogen mobility (road, rail and marine) were found to exceed near-term policy, infrastructure and market readiness. NDC 3.0 recalibrates these towards phased, pilot-to-scale pathways while retaining long-term aspirational targets on a conditional basis.

- **Conditionality and finance tracking:** NDC 2.0 lacked systematic tracking of conditional versus unconditional actions and of international support received. NDC 3.0 designates the conditionality of each action and provides for tracking of finance, technology and capacity support.

- **GESI and institutional clarity:** Gender-disaggregated data and clearly assigned institutional ownership were largely absent. NDC 3.0 embeds GESI-disaggregated indicators (where applicable) and assigns lead and coordinating institutions for each measure.

3. MITIGATION

3.1. Overview

Namibia's mitigation contribution is framed by its status as a net carbon sink. The mitigation strategy therefore both reduces gross emissions across the Energy, IPPU, AFOLU and Waste sectors and enhances national removals. NDC 3.0 mitigation actions respond directly to the calls of the first Global Stocktake, including tripling renewable energy capacity, doubling the rate of energy-efficiency improvement, accelerating the transition away from fossil fuels in the energy system, and reducing methane and transport emissions, and are aligned with Namibia's LT-LEDS vision.

Mitigation actions are classified as unconditional (financed from domestic resources) or conditional (dependent on international climate finance, technology transfer and capacity building). The mitigation timeframe runs from 2027 to 2035, with 2035 as the target year. This builds on the NDC 2.0 baseline under which Namibia targeted a reduction of projected national emissions of 7.669 Mt CO_2e together with an increase in removals of 4.233 Mt CO_2e, for a total mitigation potential of 11.902 Mt CO_2e and an enhancement of the net sink from the BAU level of −90.713 Mt CO_2e to −102.615 Mt CO_2e by 2030.

3.2. National Emission Profile

Namibia is a net sink of greenhouse gases. Total emissions do not show a clear increasing or decreasing trend over the time series but stayed rather stable at slightly above 20,000 kt CO_2e. However, a slight increase of 11% is observed when considering the national emissions of 2022 compared to those of 1990. Removals increased from 89,977 kt CO_2e in 1990 to 122,411 kt CO_2e in 2022, resulting in an increase of 32,434 kt CO_2e (45%) in net removals also for the same period[2].

The most recent official emissions data are reported in BTR1 and NC5 (2024) under the Enhanced Transparency Framework, with the National Inventory Document (NID) providing category-level detail. The inventory covers CO_2, CH_4, N_2O and HFCs.

For NDC 3.0, the Business-as-Usual scenario and sectoral emission profile are being updated using 2022 as the most recent inventory year derived from the NID, extending projections to 2035.

3.2.1. Business as Usual Emissions

Figure 1 outlines the projected sectoral BAU emissions between 2027 and 2035. Projections were estimated using 2022 emissions derived from the NID. Annual sectoral emission projections were estimated as an average of emissions obtained

[2] Source: Namibia's BTR1, and NC5 to UNFCCC, and the Paris Agreement page 23 and First National Inventory Document (1990 – 2022); December 2024

by applying a GDP growth rate of 4.3% and a population growth rate of 3%. The energy sector is projected to increase from 4,740.8 kt CO_2e in 2027 to 6,307.3 kt CO_2e in 2035. Agriculture, IPPU and Waste are projected to increase to 12,558.5 kt CO_2e, 831.0 kt CO_2e, and 323.0 kt CO_2e, respectively, by 2035. Net removals are projected to increase from -90,595.1 kt CO_2e in 2027 to -67,378.2 kt CO_2e in 2035.

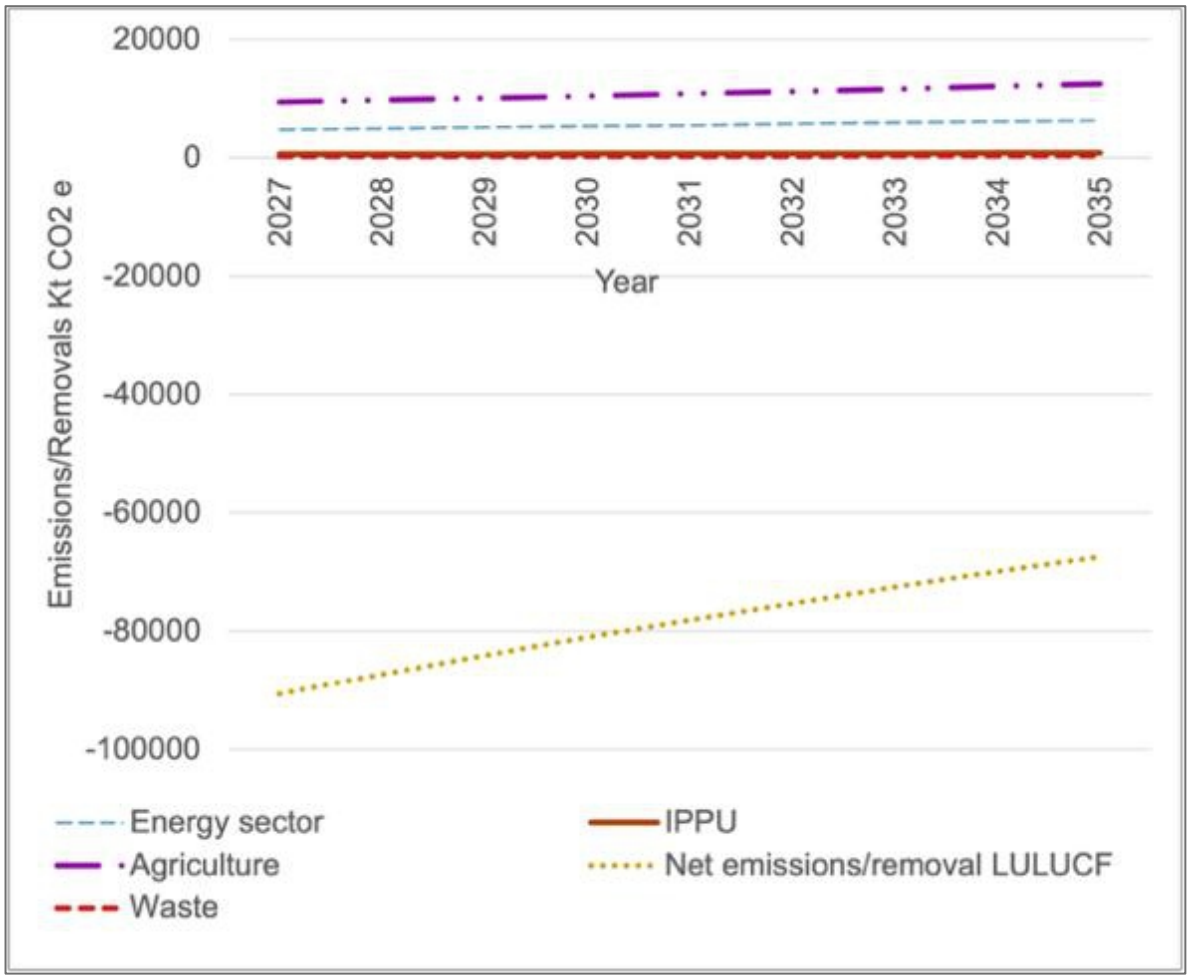


Figure 1. Projected baseline emissions (kt CO_2e)

3.3. National Mitigation Potential

As a Party to the Paris Agreement, Namibia has invested, and will continue to invest, in reducing its emissions and increasing its removals to contribute to the global effort to hold the temperature increase well below 2°C.

Namibia targets to reduce emissions from all sectors under unconditional mitigation measures amounting to 7.158 $MtCO_2e$ by 2035. Under conditional measures, Namibia targets reducing emissions by 15.608 $MtCO_2e$ by 2035. The conditional target increased the ambition by 7.939 $MtCO_2e$ compared to a target of 7.669 $MtCO_2$ in the NDC 2.0.

Table 1. NDC 3.0 Mitigation Measures and Actions

Sector	No. of Measures	No. Actions	Total emission Reduction ($MtCO_2e$)-Conditional	Total emission Reduction ($MtCO_2e$)-Unconditional
Energy	15	57	5.196	0.927
AFOLU	4	20	10.354	6.203
IPPU	4	15	0.019	0.01
Waste	3	14	0.039	0.019
Total	**26**	**106**	**15.608**	**7.158**

An overview of mitigation measures is outlined in the sections below. The complete set of mitigation measures is provided in Annex II.

3.4. Energy Sector

Namibia's Energy Sector mitigation addresses electricity generation, energy efficiency, transport, household and institutional energy access, agriculture, and marine decarbonisation. The sector comprises 15 measures, delivered through 57 specific actions, and represents the country's principal contribution to national and global emission reduction efforts over the period to 2030 and beyond. These measures target the expansion of renewable electricity generation, grid modernisation, energy efficiency, low-carbon transport, clean household energy, renewable energy in agriculture, and maritime decarbonisation. Together, these measures promote a diversified, secure and low-carbon energy system while supporting Namibia's long-term energy transition. Refer to Annex II.

3.4.1 Renewable Electricity Generation

Renewable electricity generation, comprising utility-scale solar photovoltaic, hydropower, wind and biomass, together with battery storage and grid modernisation, forms the anchor of the sector's mitigation efforts. This capacity is financed predominantly through domestic public revenue, utility balance sheets and existing concessional credit lines, and delivers the majority of the sector's quantified emission reductions. Strategic long-term generation options, including scaled green hydrogen deployment supporting renewable electricity generation and grid stability, extend this ambition into the 2030s, subject to securing climate finance.

3.4.2 Transport Decarbonisation

The transport sub-sector is addressed through a phased and technically sequenced set of measures spanning road, rail, aviation and maritime transport and is elaborated as follows:

- Road Transportation: Namibia will target 15% of new light-vehicle imports to be electric or hybrid by 2030, supported by expanded charging infrastructure and fiscal incentives.

- Railways: Heavy-duty freight decarbonisation and rail conversion to green hydrogen proceed through structured pilot phases ahead of any fleet-scale commitment, guided by the outcomes of the Walvis Bay–Windhoek hydrogen-diesel locomotive pilot.

- Civil Aviation: A dedicated Sustainable Aviation Fuel and synthetic e-fuels measure brings domestic civil aviation into the national GHG inventory, targeting a 50% sustainable fuel blending rate for domestic aviation by 2030.

- Water-borne Navigation: Marine decarbonisation proceeds through the Phased Marine Decarbonisation Programme, comprising institutional frameworks for green bunkering, demonstration of commercially viable low-carbon marine fuels (including biofuels, hydrogen and ammonia) and fuel-efficiency technologies, and reduction of NamPort's own operational emissions through on-site solar PV and shore-side power (cold ironing) at Walvis Bay and Lüderitz, ahead of any large-scale green hydrogen bunkering and export investment.

3.4.3 Nuclear Energy

Nuclear energy is included as a long-term strategic option, following Cabinet's approval of Namibia's Nuclear Industry Strategy and the establishment of the Nuclear Energy Programme Implementation Organisation (NEPIO). This pathway extends beyond the current NDC's mitigation horizon and is at the institutional and regulatory planning stage, reflecting Namibia's ambition to diversify its long-term low-carbon baseload generation.

3.4.4 Oil and Gas

Namibia's upstream oil and gas sector continues to expand on the strength of major offshore discoveries in the Orange Basin. While upstream production falls outside the scope of this NDC's mitigation accounting, its growth underscores the strategic importance of the Energy Sector's parallel decarbonisation measures, particularly transport fuel diversification, electric and hydrogen vehicle uptake, and sustainable aviation fuel development, in moderating the sector's associated emissions and economic trajectory.

3.4.5 Emerging Green Hydrogen and Green Industrial Developments

Namibia's green hydrogen sector is progressing from early demonstration towards commercial-scale development. Privately led projects are testing applications across green ammonia, fertiliser, green iron, clean transport and renewable baseload power. While not all are presented as specific NDC actions, these developments support Namibia's broader mitigation and green-industrialisation ambitions:

- **Hyphen Hydrogen Energy:** Flagship large-scale green hydrogen and

ammonia project near Lüderitz, targeting about 2 million tonnes of green ammonia annually. Project preparation is advancing, supported by a USD 10 million AfDB/SEFA facility for FEED and related studies.

- **Zhero Molecules:** Proposed large-scale green ammonia development near Walvis Bay. SDG Namibia One committed up to USD 5.15 million in development funding in 2025, supporting feasibility and preparation towards a targeted investment decision.

- **Dâures Green Hydrogen Village:** Pilot project progressing towards commercial green ammonia and fertiliser production. Phase 2 proposes 60 MW solar, 10 MW wind and 40 MW electrolysis, targeting commercial operations around 2030.

- **Hylron Oshivela:** Operational green-iron demonstration facility using renewable energy and green hydrogen for iron production. The initial phase targets about 15,000 tonnes of direct reduced iron annually, demonstrating opportunities for low-carbon industrialisation and mineral beneficiation.

- **Cleanergy Solutions Namibia:** Operational green hydrogen production and refuelling facility near Walvis Bay, demonstrating hydrogen applications in heavy transport, industrial equipment and off-grid power, alongside local skills development.

- **Renewstable Swakopmund:** Proposed renewable power facility combining solar generation, hydrogen storage and fuel cells to provide dispatchable renewable electricity. Project implementation is subject to securing a Power Purchase Agreement (PPA) with NamPower.

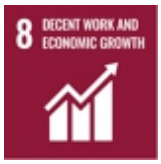
These developments demonstrate how Namibia's renewable resources could support industrial decarbonisation, clean fuels, mineral beneficiation, transport and energy security.

The country's development trajectory is reflected in the Green Industrialisation Blueprint and the Sixth National Development Plan (NDP6), which emphasizes economic transformation, employment creation, youth empowerment, environmental sustainability, and value addition. It is therefore critical that Namibia develops a skilled workforce required to support green industrialisation. Failure to do so could result in increased dependence on imported skills, missed employment opportunities for Namibians, reduced local content participation, and diminished economic benefits from green investments.

3.4.6 Co-benefits and Sustainable Development

In addition to climate action-related benefits, the implementation of the Energy Sector measures will generate substantial co-benefits aligned with the Sustainable Development Goals (SDGs):

Table 2. Energy sector co-benefits

SDG	Co-benefits
	**Air quality and health improvements** Reduces particulate matter and indoor smoke by phasing out diesel, paraffin, and traditional wood.
	**Energy Security** Cuts reliance on imported electricity from >60% to under 30% by 2030, building energy sovereignty.
	**Foreign Exchange savings:** Replaces imported liquid fossil fuels with domestic power, keeping capital within the national economy **Job creations:** Generates technical jobs across construction, operations, bush-harvesting, and hydrogen logistics.
	**Industrial Development** Positions Walvis Bay and Lüderitz as competitive global hubs for green steel, green ammonia, and e-bunkering.

3.5. Overview: AFOLU Sector

Namibia's AFOLU Sector mitigation addresses livestock methane, sustainable land and forest management, rangeland restoration, and integrated carbon sequestration. The sector comprises four measures, delivered through 20 actions, drawing on the country's rangeland, forestry and agricultural landscapes to deliver measurable emission reductions alongside land restoration and rural livelihood co-benefits.

3.5.1. Livestock and Rangeland Management

Livestock methane mitigation will focus on developing and scaling lower-emission, climate-resilient livestock production systems, informed by a Namibia-specific assessment of the net GHG balance of feedlot, free-range and regenerative systems. Priority interventions include improved cattle finishing and regenerative rangeland management, supported by sustainable domestic fodder production and strengthened monitoring of livestock emissions intensity.

Implementation will build linkages with existing climate-resilient rangeland and livestock programmes and strengthen GHG reporting and performance standards for livestock operations. Gender-responsive delivery will progressively increase the participation of women, youth and other vulnerable groups in supported livestock programmes.

3.5.2. Sustainable Land and Forest Management

Sustainable land and forest management reduces forest fire disturbances and unsustainable wood removal by accelerating rural and peri-urban electrification and clean cooking access, promoting alternative construction materials, and strengthening national bush biomass information systems.

Namibia will further strengthen its terrestrial carbon sink through implementation of the National Strategy for Tree Planting, scaling up agroforestry, reforestation, assisted natural regeneration and restoration of degraded land to increase woody biomass and associated carbon removals. Remote-sensing-based monitoring of forestland cover change and woody biomass stock underpins the sector's evidentiary base, alongside the promotion of agroforestry, regenerative agriculture and ecosystem restoration on degraded croplands and grasslands.

3.5.3. Carbon Sequestration and Carbon Markets

Integrated carbon sequestration will be advanced through a national biochar, biofertiliser and compost deployment roadmap. Regenerative agriculture will be scaled up, including through the Regenerative Agriculture Project led by the Namibia National Farmers Union (NNFU), alongside improved pasture and rangeland management. These efforts will be complemented by the reinstatement of urban greening, consistent with the National Strategy for Tree Planting (2026-2035).

A national carbon market framework will establish the architecture for registering, authorising and monitoring high-integrity AFOLU carbon activities, anchored to the national GHG inventory and Biennial Transparency Report to ensure transparent accounting, application of corresponding adjustments where required, and avoidance of double counting. A dedicated AFOLU sector monitoring, reporting and verification (MRV) system will progressively strengthen national data on land cover, soil organic carbon (SOC) stocks and stock changes, management practices and associated removals.

SOC sequestration will be quantified using nationally approved MRV approaches consistent with IPCC guidance and the national GHG inventory, supported by appropriate combinations of field measurements, soil sampling, remote sensing and country-specific data. Carbon-market activities may apply recognised international crediting methodologies and standards, subject to national approval and applicable carbon-market and Article 6 requirements.

Conditional AFOLU mitigation outcomes are not automatically designated for international transfer. Eligibility and authorisation for Article 6 transfer will be determined through Namibia's national carbon-market framework and applicable Article 6 procedures, taking into account NDC achievement, environmental integrity and corresponding-adjustment requirements.

3.5.4. Land Use, Land-Use Change and Forestry (LULUCF)

The NDC Implementation Action Plan for Biodiversity & Ecosystems of 2022 seeks to implement the following mitigation actions related to forestry:

a) Afforestation and Reforestation

b) Production and use of biochar

c) Improve SOC in grasslands and croplands- use of compost and biochar

d) Enhance soil carbon sequestration using biochar and compost in cropland

e) Sustainable land management (including integrated rangeland management)

f) Fire management plans

Table 3. Emission factors and assumptions (Afforestation/ Reforestation)

Monitored area (ha)	71000
Start year	2026
End year	2036
Biomass C at start (t C/ha)	0.5
Biomass C at end (t C/ha)	8.4
Dead Organic Matter (DOM) C at start (t C/ha)	0.2
DOM C at end (t C/ha)	0.7
Soil Organic Carbon (SOC) at start (t C/ha)	18
Soil Organic Carbon (SOC) at end (t C/ha)	19
Total C stock change (t C)	667400
Annual net mitigation (tCO_2e)	**244,644.2**

Emission factors used to estimate emissions for forest fire disturbances are provided in Table 4.

Table 4. Emission factors (Forest Fires)

Area affected by disturbances(ha/yr)	Forest land
Average above-ground biomass of area affected (tonnes/ha) (Bw)	22.63
Ratio of below-ground biomass to above-ground biomass (R)	0.28
Carbon Fraction of Dry matter (tonne C/ tonne) (CF)	0.47
Fraction of biomass lost in disturbance (fd)	0.15

The Directorate of Forestry (DoF) seeks to reduce areas affected by fire disturbances by 50%. The emissions estimates assume a 5% annual escalation factor from 2027 to reach 50% by 2035 for the conditional target, while a 3% annual increment reaching 30% by 2035 is assumed for the unconditional target. Projected areas affected by forest fire disturbances were then deduced from the 2024 measured values while applying the escalation factors.

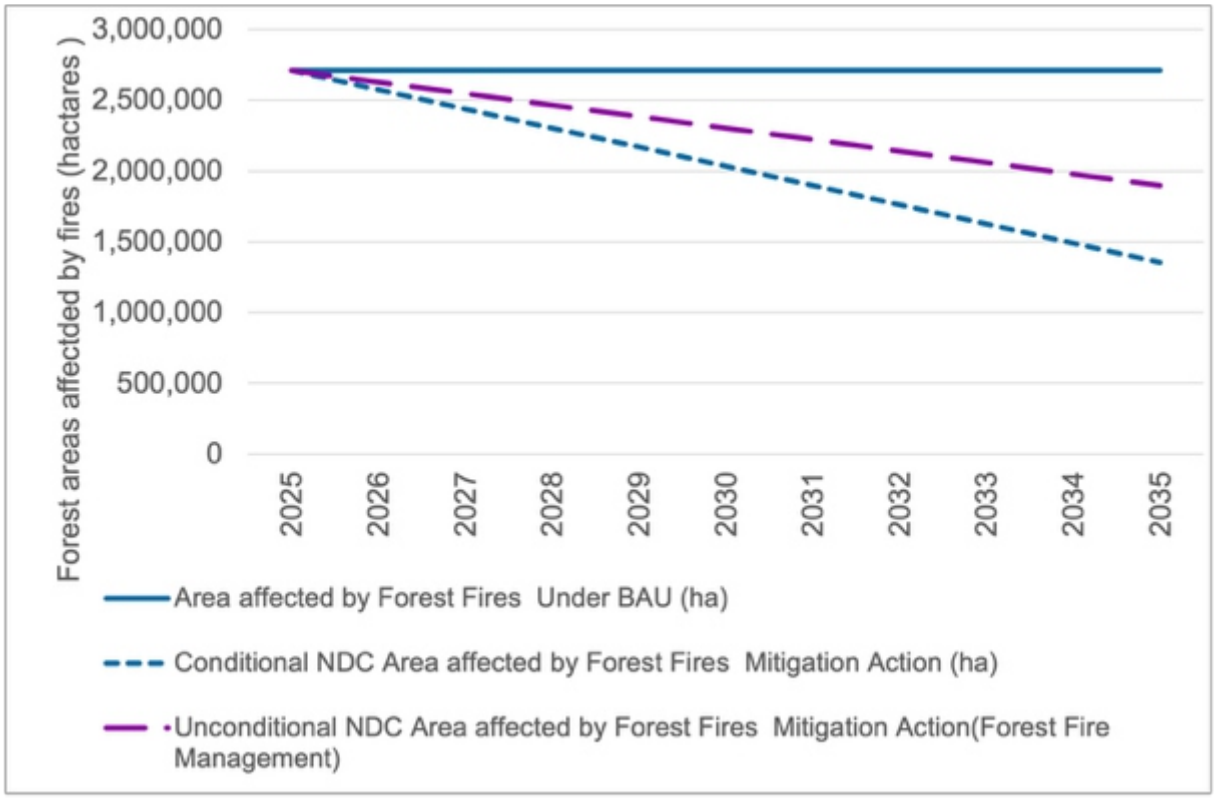


Figure 2. Projected area affected by forest fire disturbances. Source: DoF and analysis

3.5.5. LULUCF Emissions reduction

Emissions reduction from forest fire management and afforestation is presented in Figure 3. Emissions under the conditional NDC are expected to reduce to 10,294,926.00 tCO_2e compared with 6,186,675.00 tCO_2e for unconditional mitigation actions in 2035.

Namibia will enhance LULUCF removals through implementation of the National Strategy for Tree Planting (Afforestation and Reforestation) 2026–2035. The Strategy targets planting one (1) million trees by 2030 and two (2) million trees by 2035 (200,000 trees per annum), through agroforestry, reforestation, bush control and replanting, assisted natural regeneration and urban greening, with emphasis on survival, sustainable land management and long-term carbon sequestration. These interventions complement broader NDC actions on restoration of degraded lands and strengthening of the national carbon sink.

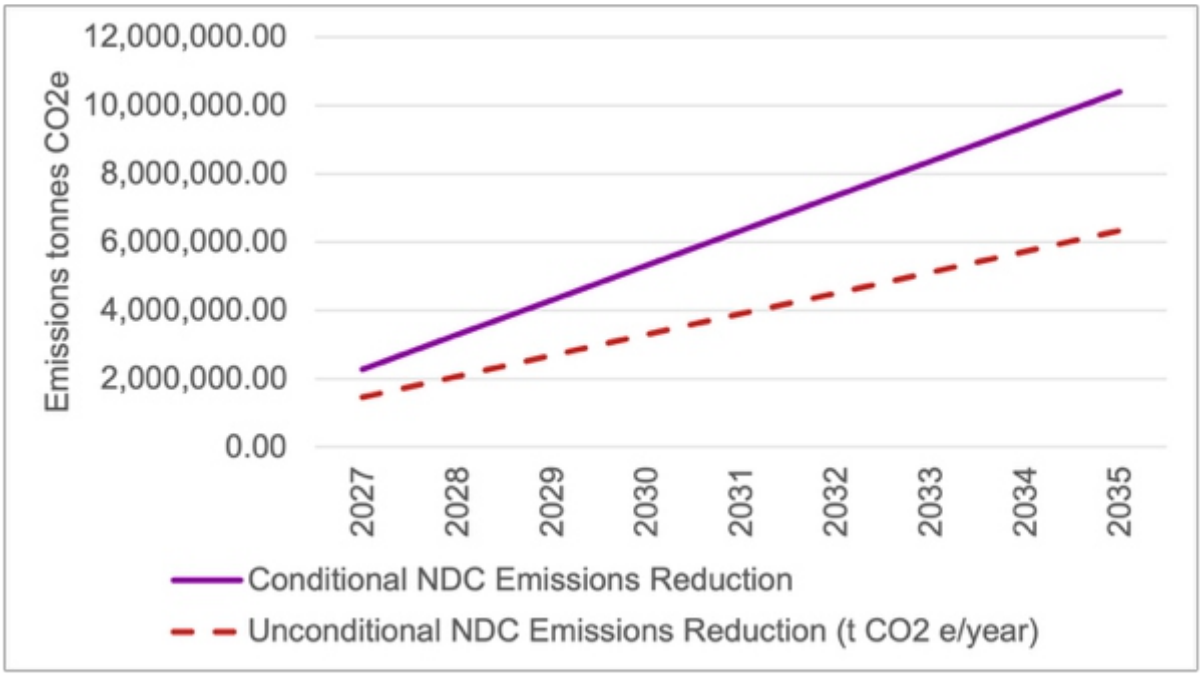


Figure 3. GHG emissions reduction from fire management and afforestation/reforestation (tCO_2e)

3.5.6. Agriculture

Namibia has been promoting Conservation agriculture (CA) as part of climate-smart agriculture (CSA) aimed at improving yield and income, food security and livelihood through optimising land use based on timely land preparation and improved crop management. The Ministry of Agriculture, Fisheries, Water and Land Reform (MAFWLR) and stakeholders implemented a five-year Comprehensive Conservation Agriculture Programme (CCAP) between 2015/2016–2019/2020. Over the implementation period, the CA adoption rate among farmers practising at least one of the CA principles increased from 40% to 88.6%.

Building on the achievements of CCAP I, the Ministry has formulated a Comprehensive Conservation Agriculture Programme II (CCAP II) for the period 2025/2026 to 2029/2030, which aims to further promote the adoption of CA as a foundation for sustainable crop production and productivity, and enhanced food and nutrition security at the national and household levels.

a) Assumptions and parameters

Table 5. Percentage of farmers with access to CA-oriented mechanised services for tillage.

Year	2023	2025	2026	2027	2028	2029
Percentage of farmers (%)	1	5	10	15	20	25

Source: Namibia's Comprehensive Conservation Agriculture Programme II (CCAP II) 2025/2026–2029/2030

Areas under conservation farming were estimated as the product of the percentage of farmers accessing an average area allocated for CA by individual farmers. Provided in Table 6 are areas under conservation farming.

Table 6. Estimated areas under CA

Year	Percentage of farmers with access to CA (%)	Unconditional NDC		Conditional NDC	
		Estimated Number of Farmers	Estimated Area under Conservation Farming (ha)	Estimated Number of Farmers	Estimated Area under Conservation Farming (ha)
2023	1	1,200	1,200	2,000	2,000
2025	5	6,000	6,000	10,000	10,000
2026	10	12,000	12,000	20,000	20,000
2027	15	18,000	18,000	30,000	30,000
2028	20	24,000	24,000	40,000	40,000
2029	25	30,000	30,000	50,000	50,000
2030	30	36,000	36,000	60,000	60,000
2031	35	42,000	42,000	70,000	70,000
2032	40	48,000	48,000	80,000	80,000
2033	45	54,000	54,000	90,000	90,000
2034	50	60,000	60,000	100,000	100,000
2035	55	66,000	66,000	110,000	110,000

Source: Namibia's Comprehensive Conservation Agriculture Programme II (CCAP II) 2025/2026–2029/2030- Programme Logical Framework, and Country report to the FAO international technical conference on plant genetic resources (Leipzig 1996)

Further assumptions and parameters for conservation farming emissions estimates are indicated in Table 7.

Table 7. Assumptions and parameters

Parameters	Unconditional NDC	Conditional NDC	Source
Typical cultivated areas (ha)	2 to 4	2 to 4	Source: Country report to the FAO international technical conference on plant genetic resources (Leipzig)
Area allocated to CA per farmer (ha)	1	1	
Annual soil carbon increment (t C/ha/year)	0.05	0.05	Martinsena, V., et.al., 2019: Significant build-up of soil organic carbon under climate-smart conservation farming in Sub-Saharan Acrisols
Target Farmers	120,000	200,000	Percentage of farmers, Source: Namibia's Comprehensive Conservation Agriculture Programme II (CCAP II) 2025/2026–2029/2030. Programme Logical Framework

b) Emissions reduction from agriculture

Mitigation under agriculture comprises Conservation Agriculture, improved livestock finishing systems, and regenerative agriculture and sustainable rangeland management to enhance soil organic carbon sequestration.

For Conservation Agriculture, emissions reductions are projected to increase from 3,300 tCO_2e in 2027 to 12,100 tCO_2e in 2035 under the unconditional pathway, and from 5,500 tCO_2e in 2027 to 18,333 tCO_2e in 2035 under the conditional pathway. This builds on the CCAP II (2025/26–2029/30).

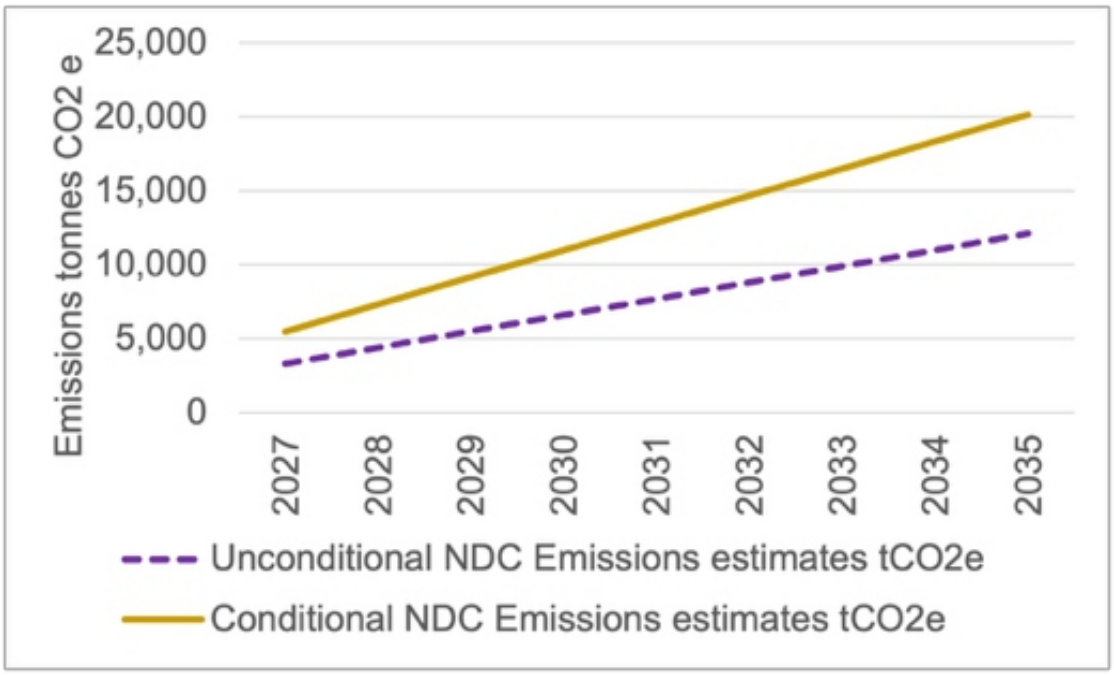


Figure 4. GHG emissions reduction for CA

Building on Conservation Agriculture under CCAP II (2025/26–2029/30), NDC 3.0 introduces livestock methane mitigation through improved livestock finishing and management systems, supported by strengthened monitoring of emissions across production systems. The measure contributes an additional estimated mitigation potential of 0.0043 $MtCO_2e$/year under the unconditional scenario and 0.038 $MtCO_2e$/year under the conditional scenario.

3.6. Overview: IPPU Sector

Namibia's IPPU mitigation programme addresses cement sector decarbonisation, industrial and mineral manufacturing, lubricant-related emissions, and the refrigeration and air-conditioning (RACHP) sub-sector. The programme comprises four measures, delivered through fifteen specific actions.

3.6.1. Cement and Industrial Decarbonisation

Cement sector decarbonisation expands clinker substitution beyond soda ash to validated supplementary cementitious materials, sets minimum blended cement content requirements for public procurement, and improves kiln energy efficiency through alternative fuel substitution. Mandatory plant-level reporting and a National Industrial Emissions Monitoring and Reporting Framework, aligned to TACCC principles, underpin the sector's transparency, alongside a bankable decarbonisation

project concept, with gender equality and social inclusion criteria embedded, prepared for submission to green financing institutions.

Decarbonisation across other manufacturing and mineral industries promotes low-carbon production technologies, energy efficiency, and on-site renewable and hybrid energy systems, including the H_2-DRI Green Iron Manufacturing pilot, which tests low-carbon production technology for Namibia's iron ore resources and links the sector to the country's green hydrogen economy.

For the IPPU sector, Namibia's NDC seeks to reduce the clinker-to-cement ratio at its two cement production facilities, namely, Ohorongo Cement and Cheetah Cement. Ohorongo Cement began production of cement in 2011, and Cheetah Cement began in 2016. Consultations revealed that only Ohorongo Cement has intentions to reduce the clinker ratio in cement. Table 8 outlines the time series data for clinker production.

Table 8: Clinker production

Plant	2017	2018	2019	2020	2021	2022
Plant 1 (Clinker production) tonnes	522,392	409,838	268,121	280,745	336,659	373,971

Source: NID

Assumptions and parameters used in the emissions estimate are provided in Table 9.

Table 9. Assumptions and parameters

Parameters	Conditional NDC	Unconditional NDC
Percentage reduction in clinker (%)	*10%*	*5%*
Emission factor	*0.52*	*0.52*

The proposed mitigation measures are estimated to achieve annual emissions reductions of 18,995 tCO_2e under the conditional measures and 9,497 tCO_2e under the unconditional measures, as outlined in Figure 5 below.

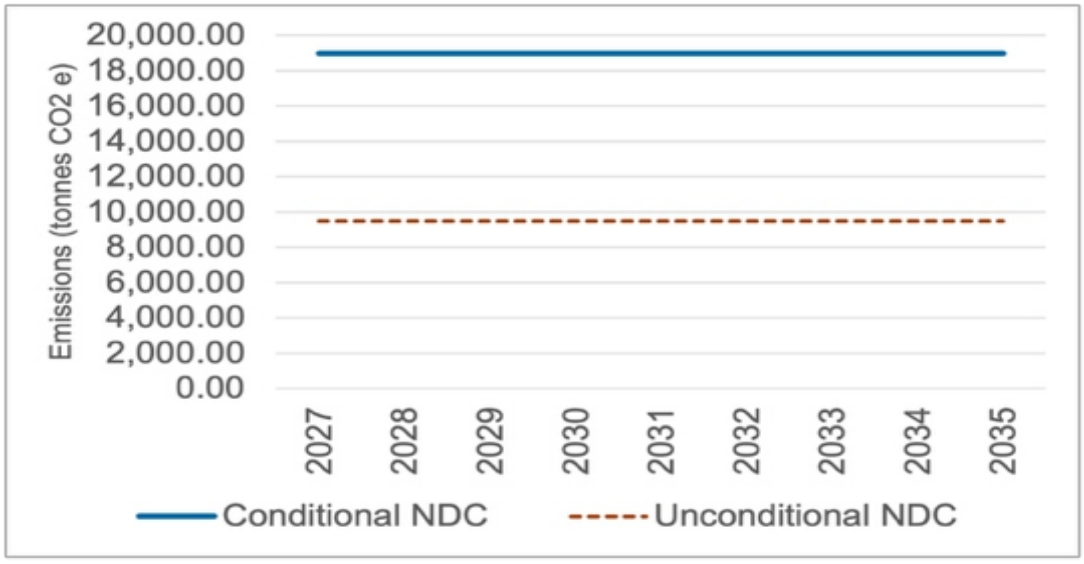


Figure 5 IPPU emissions (tCO_2e)

3.6.2. Refrigeration, Cooling and Non-Energy Product Use

A National RACHP Sector Decarbonisation and Lifecycle Refrigerant Management Programme will support the transition to low-global-warming-potential (low-GWP) refrigerants through strengthened technician certification, refrigerant recovery and destruction, and implementation of the National Cooling Action Plan and the second-stage HCFC Phase-out Management Plan.

A complementary measure will strengthen the national inventory by improving data collection and reporting on lubricant imports and consumption to support more accurate estimation and management of lubricant-related emissions.

3.7. Overview: Waste Sector

Namibia's Waste mitigation programme addresses solid waste reduction, recycling and energy recovery, open burning reduction, and wastewater management and methane recovery. The programme comprises three (3) measures, delivered through 14 specific actions.

3.7.1. Solid Waste Reduction, Recycling and Energy Recovery

The NDC Implementation Action Plan for Circular Economy (2022-2030) aspires to establish waste-to-energy turbines at the Kupferberg Landfill. NDC3 targets that the Kupferberg landfill methane capture and waste-to-energy project is advanced to investment-ready status through a bankable feasibility study, regulatory approvals and a power purchase agreement.

In comparison, sorting and recycling waste has a comparatively low mitigation potential but a very high cost attributed to it. Reducing the quantity of solid waste burned has a similarly low mitigation potential but a lower cost. Table 10 outlines the Waste Quantities at the Kupferberg landfill.

Table 10. Waste Quantities: Kupferberg landfill (Windhoek)

Year	General Waste (tonnes)	Conditional NDC		Unconditional NDC	
		Potential Waste for Waste-to-Energy (tonnes)	Potential Waste for Composting (tonnes)	Potential Waste for Waste Derived Fuel (tonnes)	Potential Waste for Composting (tonnes)
2020	79,101.87	21,358	31,641	14,950	22,149
2019	85,478.007	23,079	34,191	16,155	23,934
2018	22,371.93	6040	8,949	4,228	6,264
2017	85,589.12	23,109	34,236	16,176	23,965
2016	78,572.06	21,214	31,429	14,850	22,000
2015	76,350.6	20,615	30,540	14,430	21,378
2014	100,319.51	27,086	40,128	18,960	28,089
2013	87,040.862	23,501	34,816	16,451	24,371
2012	76,391.09	20,626	30,556	14,438	21,390
2011	71,386.27636	19,274	28,555	13,492	19,988
2010	71,859.939	19,402	28,744	13,582	20,121
2009	67,577.05	18,246	27,031	12,772	18,922
Total	**822,936.44**				
Average	73,745.84	20,296	30,068	14,207	21,048

Source: *Waste Management and Pollution Control Sub-Division, MEFT: Department of Environmental Affairs*

Assumptions for estimating emissions from composting and Refuse Derived Fuel (RDF) are provided in Table 11.

Table 11. Waste-to-Energy Parameters and assumptions

Parameters	Unconditional NDC	Conditional NDC
Calorific Value (MJ/kg)	15	15
Electrical efficiency of the plant (commonly 20% to 30% for waste-to-energy).	15%	15%
Factor to convert Megajoules (MJ) to kilowatt-hours (kWh) (1 kWh = 3.6 MJ)	3.6	3.6
Power Potential Estimates (MWh)	12,685	18,792
Percentage utilisation of waste (for waste-to-energy) %	70%	100%
Percentage utilisation of waste for compost (%)	70%	100%

Waste characterisation for solid waste in Windhoek is provided in Table 12.

Table 12: Namibia Waste Characterisation-Contents of solid waste in Windhoek

Waste Type	% Weight
Metal	4
Glass	14
Ceramics	0
Plastic containers	4
Plastic (soft)	7
Organic food products	15
Garden refuse	25
Wood/timber	1
Paper (Plain)	9
Paper (carton)	6

Source: Hasheela, 2009: Municipal Waste Management in Namibia

3.7.2. Emissions Reductions (Waste)

Emissions estimates for composting and Waste-Derived Fuel were calculated using GACMO, while deriving the quantity of compost from the percentage of organic waste based on waste characterisation.

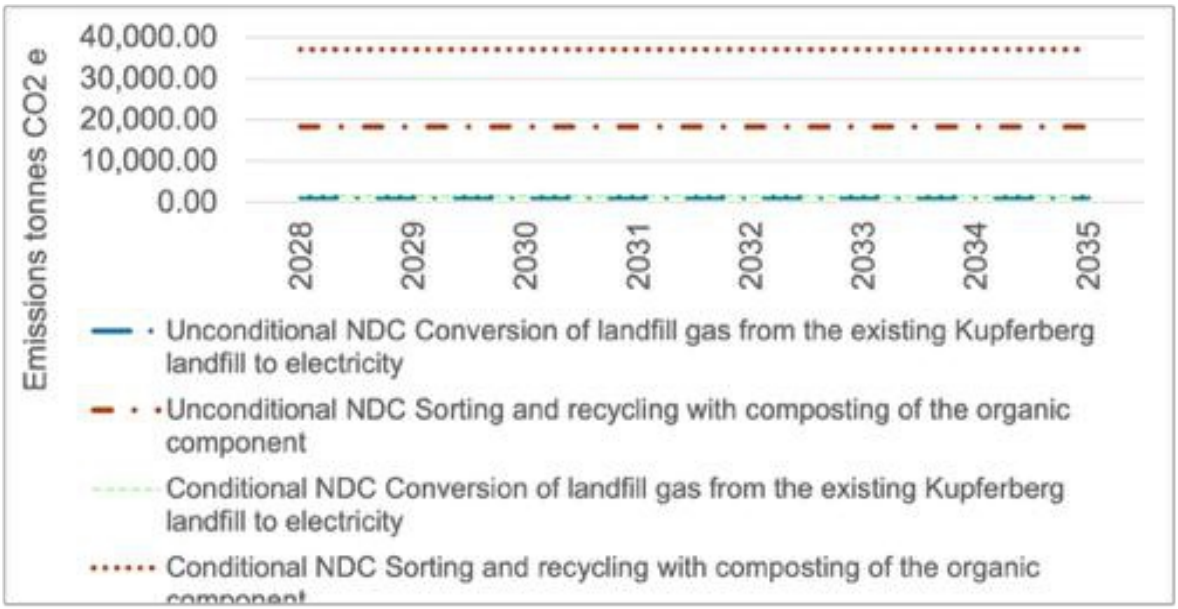


Figure 6. Emissions reduction from Waste (tCO_2e)

NDC 3.0 strengthens waste sector mitigation through an integrated National Solid Waste Reduction, Recycling and Energy Recovery Programme, complemented by measures to reduce open waste burning and improve wastewater management. Priority actions include implementation of the revised National Solid Waste Management Strategy, Extended Producer Responsibility schemes, and landfill gas recovery for energy generation.

3.7.3. Reduction of Open Burning

Open burning is addressed through a formal national tracking system, accelerated upgrading of municipal waste disposal sites to managed disposal standards, expanded formal waste collection coverage in informal settlements, and strengthened enforcement under the Environmental Management Act (Act No. 7 of 2007) and associated regulations, supported by nationally coordinated public awareness campaigns.

3.7.4. Wastewater Management and Methane Recovery

Wastewater management is strengthened through accelerated sewer network expansion in informal, peri-urban and newly urbanising areas, minimum treatment performance standards across municipal facilities, and exploration of methane recovery from wastewater treatment facilities as an additional mitigation pathway.

3.8. Summary of Emissions Reduction

The proposed conditional mitigation measures are estimated to deliver annual emissions reductions of 5.196 $MtCO_2e$ from the Energy sector, 10.295 $MtCO_2e$ from LULUCF, and 0.019 $MtCO_2e$, 0.0587 $MtCO_2e$, and 0.039 $MtCO_2e$ from the IPPU, Agriculture, and Waste sectors, respectively, by 2035.

The proposed unconditional mitigation measures are estimated to deliver annual emissions reductions of 0.927 $MtCO_2e$ from the Energy sector, 6.203 $MtCO_2e$ from LULUCF, and 0.009 $MtCO_2e$, 0.0163 $MtCO_2e$, and 0.019 $MtCO_2e$ from the IPPU, Agriculture, and Waste sectors, respectively, by 2035

Table 13 summarises the estimated annual GHG emissions reductions associated with the conditional and unconditional mitigation pathways for each sector.

Table 13: Overall conditional and unconditional Emissions Reduction ($MtCO_2e$)

	Energy	IPPU-Mineral Industry	AFOLU – Forestland/ Fire management	Agriculture	Waste	Total
Conditional	5.196	0.019	10.295	0.0587	0.039	**15.608**
Unconditional	0.927	0.009	6.187	0.0163	0.019	**7.158**

Namibia targets reducing emissions from all sectors under unconditional mitigation measures estimated at 7.158 $MtCO_2e$ by 2035. Under conditional measures, the country targets to reduce emissions estimated at 15.608 $MtCO_2e$ by 2035. The conditional target increased the ambition by 7.939 $MtCO_2e$ compared with the target in NDC 2 of 7.669 $MtCO_2e$.

4. ADAPTATION

4.1. Overview

Adaptation is a central component of Namibia's NDC 3.0 because the country's predominantly arid climate, dependence on climate-sensitive natural resources and persistent socio-economic inequalities increase its vulnerability to climate change. Climate-related shocks can deepen existing inequalities, undermine livelihoods and limit the ability of households to meet essential needs, particularly among women, children, youth, older persons, persons with disabilities and poorer communities.

Namibia's adaptation contribution therefore focuses on reducing vulnerability, strengthening the resilience of communities, ecosystems, infrastructure and productive sectors, and safeguarding national development gains. The adaptation measures and actions presented in this NDC promote inclusive, gender-responsive and locally appropriate interventions that are aligned with national development priorities, sectoral policies and Namibia's emerging National Adaptation Plan.

4.2. Climate change risks and adaptation priorities

Namibia faces multiple and interlinked climate risks, including increasingly erratic rainfall, prolonged droughts, floods, wildfires and rising temperatures. These risks have direct implications for water availability, agricultural productivity, biodiversi ty, public health, infrastructure and livelihood security.

Drought and rainfall variability are particularly significant because many rural households depend on rain-fed agriculture, livestock production and natural resources for their livelihoods. Floods and wildfires further compound these pressures by damaging infrastructure, disrupting services and degrading ecosystems. Women, youth, children, elderly people, persons with disabilities, and poorer households often have lower adaptive capacity and are therefore more exposed to climate-related shocks.

Namibia recognises that adaptation can substantially reduce climate risks but cannot eliminate them entirely. Residual climate risks may result in economic and non-economic loss and damage, including loss of livelihoods and productive assets, damage to infrastructure, livestock and crop losses, ecosystem degradation, displacement and impacts on human health and well-being.

Strengthening the documentation and assessment of such losses, including their distribution across vulnerable groups and geographic areas, will therefore complement adaptation planning and support disaster risk management, social protection, public finance and access to climate finance.

In response to these risks, Namibia has identified nine (9) adaptation priority areas to reduce vulnerability and strengthen resilience across key sectors and communi-

ties. The nine (9) adaptation priority areas comprise 46 measures and 110 actions. These measures and actions align with Namibia's long-term national and sectoral development strategies, including Vision 2030 and NDP6, as well as relevant sectoral policies, strategies and implementation frameworks. Namibia is also developing its National Adaptation Plan, which will further guide adaptation planning and implementation within the context of the NDC.

Namibia is also developing its National Adaptation Plan (NAP), which will provide the detailed framework for prioritising, costing, implementing, monitoring, evaluating and learning from adaptation interventions. The NAP and its Monitoring, Evaluation and Learning (MEL) framework will support implementation and tracking of the adaptation priorities under the NDC, while enabling periodic review of progress, emerging climate risks and adaptation needs.

Table 14: Adaptation priority areas, measures, actions

Sector	No. of measures	No. of actions
Agriculture and Food Security	11	24
Water Resources	5	11
Terrestrial Biodiversity and Ecosystems	5	13
Marine Biodiversity, Fisheries and Aquaculture	5	6
Health and Sanitation	6	19
Cross-cutting issues	6	23
Infrastructure	2	5
Coastal Zone	1	1
Urban Settlements	5	8
TOTAL	**46**	**110**

The following subsections provide a brief overview of each adaptation priority area.

4.2.1. Agriculture and Food Security

Agriculture and food security are critical adaptation priority areas for Namibia because of their central role in sustaining livelihoods, supporting rural economies and enhancing national food resilience. About 70% of Namibia's population depends on agriculture for their livelihoods, particularly through the livestock and crop value chains. However, Namibia remains a net importer of food commodities, which increases the country's exposure to food and nutrition insecurity, especially under conditions of climate variability, drought and external market shocks.

Strengthening the agriculture sector is therefore essential for reducing vulnerability and improving resilience. This requires improved access to basic agricultural inputs, including seeds, fertilisers, machinery, irrigation systems, animal feed and

veterinary medicines, to increase production, enhance competitiveness and sustain agricultural value chains.

The following are the measures under this priority area:

- Improvement of livestock disease control, prevention and treatment, including facilities
- Development of climate-resilient livestock species
- Improved fodder production and creation of a fodder bank
- Value chain development of agricultural products
- Upgrading infrastructure and monitoring systems for climate impacts
- Promote conservation agricultural practices
- Improve support to enhance adaptation to CC through better-adapted crops, crop husbandry practices, alternative crops
- Introduce new technologies to enhance/maintain crop productivity
- Diversify from traditional crops to perennials
- Promote climate-resilient urban agriculture
- Upgrading of soft and hard infrastructure for improved monitoring of climate impacts on agricultural production system

4.2.2. Water Resources

Namibia is a water-scarce country, and the limited availability of good-quality water remains a major constraint to socio-economic development. Existing water scarcity challenges are expected to intensify under climate change, as rising temperatures, prolonged droughts and variable rainfall further reduce water availability and increase pressure on already stressed water resources. For this reason, water resources management has been identified as a critical adaptation priority area. The following are the measures under this priority area:

- Develop and Update the IWRM Plan
- Develop a National Water Master Plan
- Establishment and restoration of riparian buffer
- Desalination (sea water and saline groundwater)
- Water infrastructure development, maintenance and rehabilitation (including rainwater harvesting)
- Water recycling infrastructure

4.2.3. Terrestrial Biodiversity and Ecosystems

Biodiversity and ecosystems are critical adaptation priority areas for Namibia because of their importance to community livelihoods, ecological stability and the national economy.

Namibia's biodiversity includes the variety of living organisms found across terrestrial, marine and desert ecosystems, as well as the ecological systems of which

they form part. Biodiversity management is recognised in Namibia's Constitution and reinforced by NDP6, which prioritises ecosystem protection, biodiversity conservation and the sustainable use of natural resources.

However, Namibia's ecosystems are fragile and increasingly vulnerable to climate change, particularly rising temperatures and reduced rainfall. These changes threaten ecosystem services that communities depend on and that contribute significantly to the economy. The following are the measures under this priority area:

- Sustainable land management
- Management of state protected areas
- Environmental management
- Fire management plans
- Green spaces and urban corridors
- Adaptation management of fragile natural habitats (Karoo, Wetlands and Desert)
- Disaster Risk Reduction (DRR)
- Land use planning
- Designation of flood areas/high-risk areas and development - free zones
- Promotion of landscape-based tourism
- Integrating Ecosystem-based Adaptation in Community-Based Natural Resources Management (CBNRM) programme

4.2.4. Marine Biodiversity, Fisheries and Aquaculture

Marine biodiversity, fisheries and aquaculture are important adaptation priority areas for Namibia because of their contribution to employment, export earnings, food security and economic development. The sector also supports livelihood diversification and creates opportunities for women's participation across the fisheries and aquaculture value chains.

However, climate change poses increasing risks to the sector by affecting productivity, marine ecosystems and the stability of fish stocks. In response, Namibia prioritises fisheries and aquaculture adaptation through five key measures and 15 specific actions aimed at strengthening resilience within the blue economy. In line with NDP6, the strategy also seeks to promote and upscale aquaculture production by improving productivity, enhancing food security, generating employment and supporting economic development through investment in aquaculture infrastructure. Adaptation measures include the following:

- Support sustainable use of fisheries resources by local communities
- Adaptive management and protection of aquatic ecosystems/biodiversity
- Develop mariculture within the blue economy
- Strengthen aquatic food systems
- Building Climate Change Resilience in Marine and Freshwater Fishing and Aquaculture Communities

4.2.5. Health and Sanitation

Health is a key adaptation priority for Namibia, as climate change is expected to increase public health risks. Namibia's population grew from 2.1 million in 2011 to 3.1 million in 2023, with young people making up a large share of the population. NDP6 also notes that the urban and rural population distribution now stands at about 50 percent each, reflecting increasing rural-urban migration.

These trends place pressure on health services, water, sanitation, housing and nutrition. Climate change may further intensify these challenges through heat stress, water scarcity, malnutrition, floods, and increased risks of vector-borne and water-borne diseases.

The following are the measures under this priority area:

- Climate-Resilient Health Infrastructure and Service Readiness
- Integrated Disease Surveillance, Early Warning and Health Security
- Climate-Sensitive Malaria and Vector-Borne Disease Control
- Climate-Sensitive Public Health and Disease Prevention
- Strengthening Health System Capacity for Disaster Preparedness and Continuity of SRH and GBV Services
- Health National Adaptation Plan
- Malaria eradication
- Improve disease management
- Development of an appropriate surveillance system

4.2.6. Infrastructure

Infrastructure is a critical adaptation priority area for Namibia because it supports human settlements, transport, trade, water supply, energy systems and the delivery of essential public services. Climate change is expected to increase the exposure of roads, buildings, drainage systems and other infrastructure to extreme heat, flooding, drought and severe weather events.

These risks may damage public and private assets, disrupt essential services and increase maintenance and reconstruction costs. Strengthening climate-resilient infrastructure therefore requires the integration of climate-risk information into infrastructure planning, design, construction, operation and maintenance. The following are the measures under this priority area:

- National infrastructure resilience system
- Climate-Resilient Road Infrastructure

4.2.7. Coastal Zones

Coastal zones are a key adaptation priority for Namibia due to their economic importance and sensitive ecosystems. Climate change increases risks from sea-level

rise, erosion, flooding, and storm surges, affecting communities, infrastructure, and ecosystems. Strengthening resilience requires improved planning, ecosystem and infrastructure protection, monitoring, early warning systems, and integrating climate risks into coastal development and management.
The following are the measures under this priority area:

- Integrated coastal zone management plan

4.2.8. Cross-cutting issues

Cross-cutting actions will support Namibia's social and physical infrastructure to become more resilient to the effects of the changing climate. These cross-cutting measures will facilitate synergies with initiatives from other sectors. Primarily relevant to climate adaptation, some of the actions will also play a role in building capacity to address climate mitigation concerns, and these are listed as follows:

- Disaster risk management
- Education, training and public awareness
- Research and systematic observation
- Integrated Land Use Planning
- Gender Equality, Social Inclusion and Resilient Livelihoods
- Social Protection and Shock-Responsive Support

4.3. Monitoring, Evaluation and Learning (MEL) for Adaptation

The implementation of Namibia's adaptation contribution will be supported by a functional Monitoring, Evaluation and Learning (MEL) system housed within MEFT, as the ministry responsible for coordinating climate change matters.

The system will track progress in implementing adaptation measures and actions, assess their effectiveness in reducing vulnerability and strengthening resilience, and identify implementation gaps, emerging risks and priorities. It will monitor agreed output, outcome and, where feasible, impact indicators, as well as the financial, technological, technical and capacity-building support received and required. Data should be disaggregated, where relevant, by sex, age, disability, geographic location and other vulnerability characteristics.

The adaptation MEL system will be coordinated by MEFT in collaboration with the National Planning Commission, line ministries, regional and local authorities and other implementing institutions.

Implementing entities will be responsible for collecting, verifying and submitting adaptation-related data, including information generated by communities, civil society organisations, research institutions and the private sector. MEFT will consolidate and analyse the information to assess national progress and support reporting on the implementation of the adaptation component of the NDC. The Namibia Statistics

Agency may support the standardisation, management, analysis and archiving of relevant data, where appropriate.

Learning will be integrated throughout the implementation cycle to ensure that monitoring and evaluation findings inform adaptation planning, policy review, resource allocation and implementation. Periodic reviews will assess the relevance, effectiveness, efficiency, inclusiveness and sustainability of adaptation interventions and support the revision of measures, actions, indicators and targets where necessary. This approach will enable Namibia to respond to changing climate risks, incorporate new evidence and lessons, strengthen institutional coordination and progressively improve adaptation outcomes.

5. EQUITY, FAIRNESS AND AMBITION

Namibia considers its NDC 3.0 to be equitable, fair and ambitious. It contributes directly to the ultimate objective of the UNFCCC and the Paris Agreement and reflects Namibia's common but differentiated responsibilities and respective capabilities as one of the world's most arid and climate-vulnerable, yet lowest-emitting, countries.

Fairness rests in Namibia's continued voluntary commitment to reduce emissions and enhance removals despite having historically been, and being projected to remain, a net carbon sink through 2035. NDC 3.0 extends this fairness further by broadening the scope of gases and emission categories covered relative to NDC 2.0, and by responding directly to the first Global Stocktake's call for tripled renewable energy capacity, a doubled rate of energy-efficiency improvement, accelerated transition away from fossil fuels, and reduced methane and transport emissions.

Namibia targets to reduce emissions from all sectors under unconditional mitigation measures amounting to 7.15 $MtCO_2$ e by 2035. Under conditional measures, the country targets to reduce emissions by 15.61 $MtCO_2e$ by 2035. The conditional target increased the ambition by 7.94 $MtCO_2e$ compared with the target in the NDC 2 of 7.669 $MtCO_2$

Equity is operationalised through the differentiation of unconditional actions, financed from domestic resources, and conditional actions requiring international climate finance, technology transfer and capacity building, consistent with Article 9 of the Paris Agreement. NDC 3.0 is fully aligned with Vision 2030 and NDP6, under which equity and inclusive economic growth are embedded as core national development pillars alongside environmental sustainability, and is further anchored in Namibia's ongoing National Adaptation Plan process, which guides equitable adaptation planning.

6. NDC TRACKING, MEASUREMENT, REPORTING AND VERIFICATION

The successful implementation of NDC 3.0 rests on a functional Measurement, Reporting and Verification (MRV) system enabling Namibia to track and report on mitigation and adaptation progress, including financial, technological and capacity-building support received and needed, in compliance with the Enhanced Transparency Framework under Article 13 of the Paris Agreement and its Biennial Transparency Report (BTR) cycle. The system comprises four components: tracking emissions, mitigation, support and needs, and adaptation, building on Namibia's existing Monitoring and Evaluation (M&E) system coordinated by the National Planning Commission.

The NDC 2.0 stocktake identified several MRV weaknesses that have directly shaped the design of NDC 3.0. Indicators, previously largely input-based (e.g. installed capacity, units installed), have been reformulated towards measurable outcomes (e.g. GWh generated, tCO_2e avoided, litres of fuel displaced), consistent with ETF requirements. NDC 3.0 also addresses inconsistencies identified between the BTR and the National Inventory Document, and prioritises reconciliation of sectoral figures, consistent labelling of Global Warming Potentials, and inclusion of a standardised sectoral emissions-and-removals summary table in future inventory submissions.

Systematic tracking of conditional versus unconditional actions and international support received has been introduced, alongside clearly assigned lead and coordinating institutions for each measure. The MRV system will capture gender-disaggregated data and information on environmental, social and economic co-benefits in line with the Sustainable Development Goals and will be tested and refined through the preparation of Namibia's second Biennial Transparency Report (BTR2).

6.1. Institutional arrangement for NDC Tracking

Namibia's institutional arrangements are designed to support inter-sectoral collaboration in tracking NDC progress. Key institutions, such as the Ministry of Environment, Forestry, and Tourism (MEFT), lead coordination efforts, supported by sectoral ministries for energy, agriculture, forestry, and waste management. However, limited financial and technical capacity poses challenges to fully institutionalising tracking processes. External support, including international partnerships and capacity-building initiatives, plays a crucial role in bridging these gaps.

The NDC tracking framework integrates Measurement, Reporting and Verification (MRV) across mitigation, adaptation and support received and needed, building on existing national and sectoral monitoring and evaluation systems. Relevant institutions will collect, verify and report data against agreed NDC indicators, while MEFT will coordinate the consolidation of information for national tracking and reporting under the UNFCCC Enhanced Transparency Framework.

The system will progressively strengthen the consistency, quality and use of data to support transparent assessment of progress towards Namibia's NDC commitments.

For adaptation, the broader NDC tracking framework will be complemented by the Monitoring, Evaluation and Learning (MEL) system outlined in Section 4.3, which will integrate climate risk and vulnerability information and assess progress in reducing vulnerability and strengthening resilience. In collaboration with the National Planning Commission and other relevant institutions, monitoring and evaluation findings will inform national planning, policy review, resource allocation and the periodic adjustment of adaptation measures, indicators and targets. Where relevant, GESI-disaggregated data will be captured to support inclusive and equitable adaptation action.

7. GENDER EQUITY AND SOCIAL INCLUSION

Climate change in Namibia has differentiated impacts, with women, youth, children, persons with disabilities, indigenous minorities and other vulnerable groups disproportionately affected due to existing socio-economic inequalities, unequal access to resources and assets, and limited participation in decision-making. These vulnerabilities are compounded by dependence on climate-sensitive livelihoods and natural resources, and by gendered responsibilities such as water and fuelwood collection. Namibia's gender and social inclusion response is guided by the National Gender Policy (2010–2020), the Third National Gender Equality and Equity Policy (2025–2035), and the National Social Protection Policy (2021–2030). These frameworks are complemented by the forthcoming National Climate Change Gender Action Plan (NCCGAP), which will provide sector-specific guidance for gender-responsive climate action.

Building on lessons from NDC 2.0, NDC 3.0 integrates Gender Equality and Social Inclusion (GESI) considerations across mitigation and adaptation measure design. This includes GESI-disaggregated indicators; strengthened participation and access to climate-resilient livelihoods, productive resources and decision-making; gender-responsive delivery of selected mitigation measures; and adaptation measures addressing protection, GBV risks and shock-responsive social protection for climate-vulnerable populations. GESI integration will be further operationalised through the NDC Implementation Strategy and Action Plan (NDC-ISAP) and adaptation Monitoring, Evaluation and Learning (MEL) system

8. MEANS OF IMPLEMENTATION

8.1. Implementation overview

Namibia's ability to deliver NDC 3.0 depends on a coordinated, whole-of-government approach to financing, technology transfer, institutional coordination and capacity building, consistent with Articles 9, 10 and 11 of the Paris Agreement. Mitigation measures costed to date carry a total estimated investment requirement of approximately USD 6.42 billion to 2030, comprising USD 753 million (12%) in unconditional actions financed through domestic resources, private sector investment and already-secured financing, and USD 5.67 billion (88%) in conditional actions requiring international climate finance, technology transfer and capacity building.

Adaptation costing across the nine (9) priority areas carries a total estimated investment requirement of approximately USD 7.65 billion to 2030, comprising approximately USD 0.90 billion (12%) in unconditional actions deliverable through existing government budgets, line-ministry capacity and domestic resources, and USD 6.75 billion (88%) in conditional actions requiring international climate finance, technology transfer and capacity building. Combined, Namibia's total estimated NDC 3.0 implementation cost (mitigation and adaptation) is approximately USD 14.07 billion to 2030/2035, comprising approximately USD 1.66 billion (12%) unconditional and USD 12.42 billion (88%) conditional investment requirements.

Implementation is coordinated by MEFT as the national climate change focal point, working with sectoral line ministries, the National Planning Commission and the Environmental Investment Fund of Namibia (EIF), Namibia's Green Climate Fund-accredited National Implementing Entity, together with development partners. Delivery will be operationalised through an updated NDC Implementation Strategy and Action Plan (NDC-ISAP), which designates lead and coordinating institutions, tracks conditionality and support received, and embeds GESI-responsive delivery mechanisms, consistent with the MRV arrangements described in Section 6. Priority cross-cutting needs include expanding direct-access climate finance, strengthening MRV and technical capacity, and deepening public-private collaboration to mobilise and de-risk investment at scale.

8.2. Finance requirements

The total cost for implementation of NDC 3.0 by 2030/2035 is estimated at approximately USD 14.07 billion, comprising USD 6.42 billion for mitigation measures and USD 7.65 billion for adaptation measures, of which approximately 88% is conditional overall. Table 15 summarises the mitigation priority areas, measures, actions and estimated costs.

Table 15. Finance requirements- Mitigation Measures

Sector	Measures	Unconditional costs for 2035 (USD million)	Conditional costs for 2035 (USD million)	Total costs for 2035 (USD million)
Energy	16 measures and 57 actions across supply-side generation, transport electrification, demand-side efficiency and clean cooking/heating	726.70	3,086.68	3,813.38
IPPU	4 Measures and 15 actions focusing on industry decarbonisation	3.1	793.5	796.6
AFOLU	4 measures, 20 actions across Conservation agriculture, sustainable forest utilisation and SOC	8.5	935.0	943.5
Waste	3 measures, 14 actions: Landfill gas capture, sorting/composting and wastewater management	14.5	851.0	865.5
Total estimated cost		**752.80**	**5,666.18**	**6,418.98**

The total investment required to deliver Namibia's Energy Sector mitigation ambition to 2030 is estimated at USD 3.81 billion. Consistent with the Paris Agreement's principle of common but differentiated responsibilities, Namibia's Energy Sector investment programme comprises 22 unconditional actions, to be implemented through domestic resources, private sector investment and already-secured financing. These total USD 726.7 million and are projected to deliver 0.927 $MtCO_2e$ of emissions reductions, representing Namibia's firm commitment to climate action irrespective of international support.

These include 36 conditional actions, such as utility-scale wind generation, green hydrogen scale-up, nuclear feasibility and regulatory development, and heavy-vehicle and rail electrification and hydrogen conversion programmes, that require international climate finance, technology transfer and capacity building, budgeted at USD 3.087 billion. The conditional measures account for 5.196 $MtCO_2e$, representing 86% of the sector's total emissions reduction potential and underscoring the critical importance of international support in realising Namibia's full Energy Sector mitigation ambition.

Table 16 summarises the adaptation priority areas, measures, actions and estimated costs.

Table 16. Finance requirements- Adaptation Measures

Sector	Measures	Unconditional costs for 2030 (USD million)	Conditional costs for 2030 (USD million)	Total costs for 2030 (USD million)
Agriculture and Food Security	11 measures, 24 actions across Agriculture and Food Security	292.50	1,502.50	1,795.00
Water Resources	5 measures, 11 actions sustainable water resources management	405.00	3,438.00	3,843.00
Biodiversity and Ecosystems	5 measures, 13 actions (biodiversity & ecosystems)	120.00	282.00	402.00
Fisheries and Aquaculture	5 measures, 6 actions (fisheries and aquaculture)	9.00	118.00	127.00
Health	6 measures, 19 actions across the health sector	9.15	340.50	349.65
Cross-cutting issues	6 measures, 23 actions	52.00	148.00	200.00
Infrastructure	2 measures, 5 actions	12.00	300.00	312.00
Coastal Zone	1 measure, 1 action	0.00	32.00	32.00
Urban Settlements	5 measures, 8 actions	4.50	588.00	592.50
Total estimated cost		**904.15**	**6,749.00**	**7,653.15**

9. ANNEXURES

9.1. Annexure I. Information Necessary for Clarity, Transparency and Understanding (INCTU) of Namibia's NDC

Information to facilitate clarity, transparency and understanding of nationally determined contributions, referred to in decision 1/CP.21, paragraph 28		
1.	**Quantifiable information on the reference point (including, as appropriate, a base year):**	
(a)	Reference year(s), base year(s), reference period(s) or other starting point(s);	The reference year for all GHG emissions is 2010. The 2024 National Inventory Document (NID 2024) covers the inventory period 2017–2022. BAU projections were built from 2010 base-year emissions applying a GDP growth rate of 4.3% per annum and a population growth rate of approximately 3% per annum (NSA Population and Housing Census, 2024), on a source-category basis.
(b)	Quantifiable information on the reference indicators, their values in the reference year(s), base year(s), reference period(s) or other starting point(s), and, as applicable, in the target year;	**Business-As-Usual (BAU) projections (in kt CO_2e):** - Energy: 4,740 kt CO_2e (2027) → 6,370 kt CO_2e (2035) - IPPU: ~831 kt CO_2e by 2035 - Agriculture (livestock/crops): ~12,558 kt CO_2e by 2035 - Waste: ~323 kt CO_2e by 2035 - AFOLU net removals (LULUCF): -90,595 kt CO_2e (2027) → -67,378 kt CO_2e (2035) *Namibia is projected to remain a net carbon sink through 2035, though with declining net removal capacity due to land-use pressures and population growth.*
(c)	For strategies, plans and actions referred to in Article 4, paragraph 6, of the Paris Agreement, or policies and measures as components of nationally determined contributions where paragraph 1(b) above is not applicable, Parties to provide other relevant information;	Not applicable
(d)	Target relative to the reference indicator, expressed numerically, for example in percentage or amount of reduction;	Mitigation contributions are expressed relative to the BAU scenario. Summary of quantified emission reductions by sector by 2035: **Conditional NDC (subject to international climate finance):**

		- AFOLU – Forest fire management and afforestation/reforestation: 10.295 Mt CO_2e - IPPU – Clinker reduction in cement: 0.019 Mt CO_2e - Agriculture – Conservation farming (soil carbon): 0.02 Mt CO_2e - Agriculture- Livestock finishing systems: 0.0387 Mt CO_2e - Waste – Landfill gas capture & composting: 0.039 Mt CO_2e - Energy sector mitigation: 5,195 Mt CO_2e - Subtotal: **~15.605 Mt CO_2e by 2035** **Unconditional NDC (implementable with national resources):** - AFOLU – Forest fire management and afforestation/reforestation: 6.18 Mt CO_2e - IPPU – Clinker reduction: 0.009 Mt CO_2e - Agriculture – Conservation farming (soil carbon): 0.012 Mt CO_2e - Agriculture- Livestock finishing systems: 0.0043 Mt CO_2e - Waste – Landfill gas capture & composting: 0.019 Mt CO_2e - Energy sector mitigation: 0.927 Mt CO_2e Subtotal: **~7.158 Mt CO_2e by 2035**
(e)	Information on sources of data used in quantifying the reference point(s);	- National Inventory Document (NID 2024), inventory period 2017–2022 - BTR1 and NC5 (2024) - NSA Population and Housing Census 2023 - CCAP II: Comprehensive Conservation Agriculture Programme II (2025/26–2029/30) - Directorate of Forestry fire disturbance records and satellite imagery (2024) - NIRP 2022, National Integrated Resource Plan (energy projections) - IMF GDP growth projections for Namibia (4.3% per annum) - LT-LEDS (2025): long-term emission trajectory - IPCC 2006 Guidelines and 2019 Refinements
(f)	Information on the circumstances under which the Party may update the values of the reference indicators.	Reference indicator values may be updated when: - New national GHG inventory data become available from updated NID cycles - Revised national emission factors or methodological improvements are adopted - International decisions revise IPCC guidelines or GWP values - Significant changes in national socio-economic projections or land-use trajectories occur
2.	**Time frames and/or periods for implementation:**	

(a)	Time frame and/or period for implementation, including start and end date, consistent with any further relevant decision adopted by the Conference of the Parties serving as the meeting of the Parties to the Paris Agreement (CMA);	NDC 3.0 covers the implementation period 2026–2035. The NDC is submitted in 2026; sector-level mitigation contributions are tracked annually from 2027 to 2035. The year 2030 serves as an interim milestone and 2035 is the primary target year. Mitigation measures implemented prior to 2026 (under NDC 2.0) will be reported for information purposes in NDC implementation tracking.
(b)	Whether it is a single-year or multi-year target, as applicable.	Multi-year target. Annual implementation pathways for mitigation emissions reductions are presented from 2027 through to 2035, with 2030 as an interim reporting milestone and 2035 as the primary target year. Sector-level trajectories are quantified for each year of the implementation period.
3.	**Scope and coverage:**	
(a)	General description of the target;	Namibia's NDC 3.0 aims to: - Maintain Namibia's status as a net carbon sink through the 2035 target period; - Progressively reduce gross GHG emissions across all four IPCC sectors relative to the BAU scenario; - Enhance net removals through forest fire management, conservation agriculture, and ecosystem restoration; - Extend and increase mitigation ambition beyond NDC 2.0, consistent with the first Global Stocktake (GST) outcomes. The NDC includes both unconditional measures (implementable with national resources) and conditional measures (contingent on international climate finance and technology transfer). Conditional ambition significantly exceeds unconditional, reflecting Namibia's dependence on international support for enhanced action.
(b)	Sectors, gases, categories and pools covered by the nationally determined contribution, including, as applicable, consistent with Intergovernmental Panel on Climate Change (IPCC) guidelines;	**Sectors covered:** - Energy: Fuel combustion across energy industries, road transport, rail transport, aviation, marine transport, residential, agriculture/forestry/fishing - IPPU: Cement production (clinker substitution), refrigerant gases (HFC management), non-energy product use - AFOLU: Livestock (enteric fermentation, manure), forestland (fire management, wood removal), cropland, grassland, settlements (soil carbon) - Waste: Solid waste disposal, biological treatment, wastewater (domestic and industrial) **Greenhouse gases covered:** - CO_2

		- CH_4 - N_2O - HFCs (refrigeration and air conditioning sector) **IPCC Guidelines applied:** IPCC 2006 Guidelines and 2019 Refinements; IPCC AR5 GWPs per Decision 18/CMA.1.
(c)	How the Party has taken into consideration paragraph 31(c) and (d) of decision 1/CP.21;	Coverage is consistent with Namibia's NID 2024 (BTR1/NC5). The following categories remain excluded due to data limitations: - HFCs and PFCs from fire protection equipment and solvent use - SF_6 from electrical equipment and military use - Medical waste incineration Namibia is working to progressively improve data coverage and include these categories in future inventory cycles as national capacity improves. The scope of coverage has widened compared to NDC 2.0.
(d)	Mitigation co-benefits resulting from Parties' adaptation actions and/or economic diversification plans, including description of specific projects, measures and initiatives of Parties' adaptation actions and/or economic diversification plans	Adaptation actions in NDC 3.0 that yield identified mitigation co-benefits include: - Carbon sequestration through rangeland restoration, savannah management and agroforestry (linked to Otjikoto Biomass Power Station, 40 MW, under construction) - Soil carbon improvement through conservation agriculture combined with biochar/compost application (CCAP II) - CH_4 reduction through wastewater reuse and aquifer recharging (Goreangab expansion) - Fire management and firebreak establishment in forests and rangelands - Riparian buffer restoration and urban green corridors (CO_2 sequestration) - Solar-powered water pumping replacing diesel pumps in agricultural and communal water systems - GCF/EIF Ecosystem-Based Adaptation (EbA), CRAVE, ICCBRLP, and Empower to Adapt projects supporting landscape carbon sequestration
4.	**Planning process**	
(a)	Information on the planning processes that the Party undertook to prepare its nationally determined contribution and, if available, on the Party's implementation plans, including, as appropriate:	
i.	Domestic institutional arrangements, public participation and engagement with	**Institutional arrangements:** The Ministry of Environment, Forestry and Tourism (MEFT) is Namibia's UNFCCC national focal point

	local communities and indigenous peoples, in a gender-responsive manner;	and lead coordinating agency for climate change. The National Climate Change Committee (NCCC) provides multi-stakeholder oversight, comprising representatives from key ministries (MIME, MAFWLR, MHSS, Ministry of Works and Transport, NPC), the private sector, and civil society. The Environmental Investment Fund (EIF) serves as Namibia's national implementing entity for international climate finance (GCF). DBN has been recently accredited to the GCF. **NDC 3.0 development process:** - Comprehensive NDC 2.0 Stocktake Assessment (2026) covering all mitigation and adaptation sectors, primary evidence base for NDC 3.0 - Multi-stakeholder consultations with sector ministries, private sector, civil society organisations, and development partners - Gender-responsive engagement in line with the National Climate Change Gender Action Plan (NCCGAP, under development by MEFT/UNDP) - Technical sectoral working groups by sector (Energy, AFOLU, IPPU, Waste, and eight adaptation sectors) - Guidance aligned with NDC Partnership NDC 3.0 Navigator and UNDP NDC Quality Assurance Checklist (five dimensions: country ownership, ambition, just transition/SDGs, clarity/transparency, feasibility)
ii.	Contextual matters, including, inter alia, as appropriate:	
a.	National circumstances, such as geography, climate, economy, sustainable development and poverty eradication;	- One of sub-Saharan Africa's most arid countries; rainfall ranges from <25 mm/yr (west) to ~700 mm/yr (northeast) - Population: 3,022,401 (NSA 2023 Census), growing at ~1.9% per annum; 53% urban, with rural communities highly dependent on natural resources - HDI: 0.615 (2023); Gini coefficient ~0.59, among the world's highest inequality levels - Economy dependent on climate-sensitive sectors: tourism, fisheries, agriculture, and mining; Namibia imports 48–71% of electricity - ~84% of rural households use fuelwood as primary cooking fuel; rural electrification remains low (~20% as of 2022). Access to clean cooking is estimated at ~50% (2024) - Climate-vulnerable: rising temperatures, prolonged droughts, floods, and shifting rainfall patterns disproportionately affect women, children, and smallholder farmers - Namibia is a net carbon sink due to extensive AFOLU removals from savannah, woodland, and semi-arid vegetation

		- NDP 6 (2025/26–2029/30) embeds climate change within its Environmental Sustainability pillar; total NDC investment requirement ~USD 12.7 billion to 2030
b.	Best practices and experience related to the preparation of the nationally determined contribution;	NDC 3.0 preparation was guided by: - IPCC 2006 Guidelines and 2019 Refinements for GHG inventory and emissions estimation - Decision 4/CMA.1 and 9/CMA.1 (INCTU and transparency guidance); Decision 18/CMA.1 (AR5 GWPs) - BTR1, NC5 (2024), and NID 2024 as primary data sources - NDC Partnership NDC 3.0 Navigator for global alignment - UNDP NDC Quality Assurance Checklist across five dimensions - GFDRR Vulnerability Assessment Framework for adaptation assessment - LT-LEDS (2025) as the long-term emission trajectory reference
c.	Other contextual aspirations and priorities acknowledged when joining the Paris Agreement;	Namibia's NDC 3.0 is anchored in Vision 2030 (low-carbon, prosperous nation), NDP 6, and the Harambee Prosperity Plan III. The LT-LEDS (2025) provides a long-term decarbonisation pathway consistent with Paris Agreement temperature goals. Namibia recognises its dual position as a low-emission net sink country and one of Africa's most climate-vulnerable nations. Adaptation is a co-equal priority alongside mitigation. International support, finance, technology, and capacity building, is critical to achieving conditional targets.
(b)	Specific information applicable to Parties, including regional economic integration organisations and their member States, that have reached an agreement to act jointly under Article 4, paragraph 2, of the Paris Agreement, including the Parties that agreed to act jointly and the terms of the agreement, in accordance with Article 4, paragraphs 16–18, of the Paris Agreement;	Not applicable
(c)	How the Party's preparation of its nationally determined contribution has been informed by the outcomes of the global stocktake, in accordance with Article 4, paragraph 9, of the Paris Agreement;	Namibia's NDC 3.0 has been directly informed by the outcomes of the first Global Stocktake (GST) concluded at COP28, Dubai, December 2023, which called on Parties to: - Enhance mitigation ambition in NDC 3.0 submissions relative to previous NDCs - Transition away from fossil fuels in energy systems and triple renewable energy capacity by 2030 - Strengthen finance flows to developing country Parties

		In response, Namibia: - Conducted a review and developed a comprehensive NDC 2.0 Stocktake Assessment Report (2026) as the primary evidence base for NDC 3.0 - Extended the target year from 2030 to 2035, increasing the ambition horizon - Strengthened quantification of AFOLU and agriculture mitigation targets (new in NDC 3.0) - Improved INCTU quality and specificity consistent with enhanced transparency requirements - Aligned investment frameworks with emerging climate finance mechanisms
(d)	Each Party with a nationally determined contribution under Article 4 of the Paris Agreement that consists of adaptation action and/or economic diversification plans resulting in mitigation co-benefits consistent with Article 4, paragraph 7, of the Paris Agreement to submit information on:	
i.	How the economic and social consequences of response measures have been considered in developing the nationally determined contribution;	NDC 3.0 measures are aligned with national development priorities (NDP 6, Vision 2030) and designed to generate co-benefits for growth, employment, and resilience: - Renewable energy expansion: domestic energy security, reduced import dependency, green hydrogen export opportunities, and job creation - Bush thinning and biomass energy: rural employment, encroacher bush management, improved rangeland productivity (Otjikoto Biomass Project, 40 MW) - Conservation agriculture: improved food security, soil health, and smallholder productivity (CCAP II) - Wastewater management: improved public health, water security, and sanitation access - Just transition principles incorporated to protect livelihoods in fossil fuel-dependent sectors (transport, fisheries), with targeted support for workers and small producers - Gender-responsive implementation with gender-disaggregated targets to ensure equitable benefits for women and youth
ii.	Specific projects, measures and activities to be implemented to contribute to mitigation co-benefits, including information on adaptation plans that also yield mitigation co-benefits, which may cover, but are not limited to, key sectors, such as energy, resources, water resources, coastal resources, human settlements and urban planning, agriculture and forestry; and economic diversification actions, which may cover,	**Key projects with mitigation co-benefits:** - Otjikoto Biomass Power Station (40 MW, under construction, ~2027 operation): encroacher bush harvesting (~400,000 ha over 25 years), ~0.308 Mt CO_2/yr avoided; €100M AFD loan + FFEM grant - GCF EbA programme: landscape restoration across 13 regions, supporting carbon sequestration in rangelands (>216,000 beneficiaries) - CCAP II (2025/26–2029/30): soil carbon sequestration through conservation agriculture (up to 200,000 farmers under conditional scenario) - Goreangab Water Reclamation Plant expansion (Windhoek): reduces CH_4 from wastewater, supplies ~30% of city drinking water - Kupferberg Landfill waste-to-energy: pre-feasibility complete; landfill gas capture for

	but are not limited to, sectors such as manufacturing and industry, energy and mining, transport and communication, construction, tourism, real estate, agriculture and fisheries	electricity generation planned (4 x 1 MW turbines) - Solar pump deployment: >430 solar water pumping systems in communal areas, replacing diesel pumps - Green Hydrogen Programme (NGH_2P): long-term economic diversification into hydrogen economy, including maritime decarbonisation
5	**Assumptions and methodological approaches, including those for estimating and accounting for anthropogenic greenhouse gas emissions and, as appropriate, removals:**	
(a)	Assumptions and methodological approaches used for accounting for anthropogenic greenhouse gas emissions and removals corresponding to the Party's nationally determined contribution, consistent with decision 1/CP.21, paragraph 31, and accounting guidance adopted by the CMA;	Namibia's NDC 3.0 mitigation estimates apply the following principles: - IPCC 2006 Guidelines and 2019 Refinements for all GHG estimation - AR5 GWPs for CO_2-equivalent aggregation per Decision 18/CMA.1 - Core formula: Emissions = Activity Data × Emission Factor, applied on a source-category basis - BAU scenario projected from 2022 base-year emissions (NID 2024), using GDP (4.3% per annum) and population growth (3% per annum) - Mitigation contributions estimated relative to BAU; avoided emissions approach used for renewable energy measures - TACCC principles maintained throughout (Transparency, Accuracy, Consistency, Completeness, Comparability) - Alignment with UNFCCC Enhanced Transparency Framework (ETF) and BTR1 reporting structure
(b)	Assumptions and methodological approaches used for accounting for the implementation of policies and measures or strategies in the nationally determined contribution;	**Sector-specific methodologies:** - AFOLU – **Forest fire management:** Area reduction from 2024 baseline (~2.71 million ha/yr disturbed). Conditional: 5% annual reduction to 50% by 2035; Unconditional: 3% annual reduction to 30% by 2035. IPCC emission factors applied (above-ground biomass: 22.63 t/ha; below-ground ratio: 0.28; carbon fraction: 0.47; fraction lost in disturbance: 0.15). - AFOLU – **Afforestation/Reforestation:** Based on the National Strategy for Tree Planting (2026–2035), targeting 71,000 ha by 2035. Removals estimated from 10-year carbon stock changes in biomass (0.5–8.4 t C/ha), dead organic matter (0.2–0.7 t C/ha) and SOC (18–19 t C/ha), giving ~0.245 $MtCO_2e/yr$. Unconditional: 40% of programme area (~28,400 ha); remaining scale-up conditional. - IPPU – Clinker substitution: 10% clinker reduction (conditional) or 5% (unconditional) applied to Ohorongo Cement production data. Emission factor: 0.52 t CO_2/t clinker. Annual conditional reduction: 18,995 t CO_2e; unconditional: 9,498 t CO_2e.

		- **Agriculture – Conservation farming and SOC:** Conservation farming assumes 0.05 t C/ha/year, based on CCAP II adoption targets × 1 ha/farmer: 0.012 $MtCO_2e$/year unconditional; 0.02 $MtCO_2e$/year conditional. - **Waste – Landfill gas & composting:** Mass balance from Kupferberg landfill data. Parameters: calorific value 15 MJ/kg, electrical efficiency 15%. Conditional: 100% waste utilisation; unconditional: 70%. - Energy: will apply avoided emissions = kWh displaced × SAPP or national emission factors.
(c)	If applicable, information on how the Party will take into account existing methods and guidance under the Convention to account for anthropogenic emissions and removals, in accordance with Article 4, paragraph 14, of the Paris Agreement, as appropriate;	Namibia is committed to maintaining consistency with UNFCCC reporting under the Enhanced Transparency Framework (ETF), as established under Article 4, paragraph 14. NDC 3.0 accounting aligns with BTR1 and the TACCC principles embedded in Article 4, paragraph 13 of the Paris Agreement. All accounting methodologies are consistent with those applied in the NID 2024 to ensure traceability and comparability across reporting cycles.
(d)	IPCC methodologies and metrics used for estimating anthropogenic greenhouse gas emissions and removals;	IPCC 2006 Guidelines and 2019 Refinements are used. AR5 GWPs applied per Decision 18/CMA.1. Tier 2 approaches applied in: - AFOLU: Country-specific emission factors for enteric fermentation (dairy/non-dairy cattle based on national livestock data) and forest carbon stock changes (based on forest inventories) - Road transport: Disaggregated vehicle fleet data, mileage estimates, and fuel consumption profiles (partially using adapted emission factors) - Tier 1 (IPCC defaults) applied in: energy industries, manufacturing and construction, IPPU processes (cement, refrigerants, solvents), and waste sectors.
(e)	Sector-, category- or activity-specific assumptions, methodologies and approaches consistent with IPCC guidance, as appropriate, including, as applicable:	
i.	Approach to addressing emissions and subsequent removals from natural disturbances on managed lands;	Wildfires on managed forestland are tracked using satellite imagery and reported in the GHG inventory. NDC 3.0 explicitly includes forest fire management as a quantified mitigation action in the LULUCF sector, with area reductions modelled against the 2024 baseline (~2.71 million ha/yr disturbed). Subsequent removals following natural disturbances are not yet fully captured due to limitations in biomass regeneration data. This gap will be addressed progressively as national monitoring capacity improves through the BTR cycle.

ii.	Approach used to account for emissions and removals from harvested wood products;	The IPCC recommended approach is applied, using national trade statistics. Wood removal was estimated from 1961 to date through use-rate and population data to supplement national records where historical archives are incomplete (particularly pre-independence periods). Consistent with the NIR series submitted to the UNFCCC.
iii.	Approach used to address the effects of age-class structure in forests;	Namibia's arid conditions result in very slow woody vegetation growth (trees take substantially more than 20 years to reach maturity). National emission factors and stock factors were derived from forest inventories, peer-reviewed publications, and national scientific studies, country-specific approaches that differ from IPCC defaults to reflect Namibia's unique ecological conditions. Details are documented in the NIR series submitted to the UNFCCC.
(f)	Other assumptions and methodological approaches used for understanding the nationally determined contribution and, if applicable, estimating corresponding emissions and removals, including:	
i.	How the reference indicators, baseline(s) and/or reference level(s), including, where applicable, sector-, category- or activity-specific reference levels, are constructed, including, for example, key parameters, assumptions, definitions, methodologies, data sources and models used	Reference indicators for NDC 3.0 are the total GHG emissions and net removals for base year 2022 as published in the NID 2024 (BTR1/NC5). BAU projections apply GDP (4.3% per annum) and population growth (3% per annum) to base-year emissions on a source-category basis, summed to sector and national levels, a multi-criteria approach providing greater accuracy than aggregate-level projections. Mitigation potentials are derived using the same methodology and data sources as the BAU scenario. Key parameters, assumptions, and methodologies are documented in the NID 2024 and supporting sector analyses.
ii.	For Parties with nationally determined contributions that contain non-greenhouse-gas components, information on assumptions and methodological approaches used in relation to those components, as applicable;	Not applicable
iii.	For climate forcers included in nationally determined contributions not covered by IPCC guidelines, information on how the climate forcers are estimated;	Not applicable

iv.	Further technical information, as necessary;	The energy sector mitigation contribution incorporates: - Installed renewable energy capacity (MW) by technology (solar PV, wind, biomass, hydropower) and avoided emissions (kt CO_2e) - Annual electricity generation (GWh) and fossil fuel generation displaced - Energy efficiency measures (LED streetlights: ~181,955 units; solar water heaters; MEPS standards for appliances, NAMS 109:2022, 110:2025, 111:2025) - Transport electrification: EV charging infrastructure framework (ECB, 2024); current fleet <400 EVs nationally (224 electric, and 113 hybrid by 22 May 2026 (Source: RFA)) **Treatment of imported electricity:** Namibia imports 48–71% of its electricity. Avoided emissions from domestic RE generation are calculated as electricity generated × the relevant SAPP/national grid emission factor, reflecting displacement of imported fossil-based power. This attribution approach will be explicitly documented in the final NDC 3.0.
(g)	The intention to use voluntary cooperation under Article 6 of the Paris Agreement, if applicable.	Namibia intends to use voluntary cooperation under Article 6, including Article 6.2 ITMOs and Article 6.4 mechanisms. NDC3.0 provides for a national carbon market framework governing the registration, authorisation and monitoring of high-integrity carbon projects, aligned with the national GHG inventory and BTR to ensure corresponding adjustments and avoid double counting. MEFT has initiated a consultative process to develop the framework, which will guide future engagement in carbon markets and support eligible conditional mitigation actions.
6	**How the Party considers that its nationally determined contribution is fair and ambitious in the light of its national circumstances:**	
(a)	How the Party considers that its nationally determined contribution is fair and ambitious in the light of its national circumstances;	Namibia considers NDC 3.0 fair and ambitious for the following reasons: - NDC 3.0 extends the target year from 2030 (NDC 2.0) to 2035, increasing the ambition horizon - The NDC covers all four IPCC sectors economy-wide, maintaining progression - Conditional mitigation ambition ~15.608 $MtCO_2e$ by 2035 significantly exceeds unconditional (~7.158 $MtCO_2e$) demonstrating readiness to do more with international support - Namibia is a net carbon sink with among the lowest per capita GHG emissions globally - The country's historical cumulative contribution to global atmospheric GHG concentrations is negligible relative to its high climate vulnerability - Significant adaptation costs burden the national budget, constraining available domestic resources for mitigation - NDC 3.0 is informed by the first GST and aligned with the global 1.5°C pathway as far as national circumstances allow

		- Namibia's ambitious domestic renewable energy programme (currently ~70% of installed generation capacity) demonstrates concrete action on the global energy transition
(b)	Fairness considerations, including reflecting on equity;	- Per capita GHG emissions: - Namibia's AFOLU sector provides substantial net carbon removals that benefit global climate at domestic cost - Namibia bears a disproportionate burden of climate change impacts (droughts, floods, biodiversity loss, impacts on agriculture, water, fisheries) relative to its emissions contribution - NDC 3.0 is framed within a just transition context, aligning mitigation actions with sustainable development, poverty eradication, and social inclusion (Vision 2030, NDP 6) - Gender equity is embedded across NDC 3.0 implementation, recognising that women, youth, children and persons living with disabilities face disproportionate climate impacts and should benefit equitably from climate action - Conditional and unconditional components are transparently distinguished, reflecting the equity principle that deeper cuts require international financial and technical support
(c)	How the Party has addressed Article 4, paragraph 3, of the Paris Agreement;	NDC 3.0 represents progression beyond NDC 2.0 through: - Extended target year from 2030 to 2035, representing a longer-term commitment - Explicit quantification of AFOLU forest fire management as a mitigation target (new in NDC 3.0) - Conservation farming soil carbon sequestration added as a quantified mitigation measure in the agriculture sector (new in NDC 3.0) - Strengthened linkage with BTR/MRV framework, ensuring improved accountability - Improved INCTU quality and specificity per enhanced transparency requirements - NDC 3.0 informed by the first Global Stocktake per Article 4, paragraph 9 - Increased conditional ambition compared to NDC 2.0 conditional contribution
(d)	How the Party has addressed Article 4, paragraph 4, of the Paris Agreement;	Namibia addresses Article 4, paragraph 4 by maintaining an economy-wide emission reduction approach, covering all four IPCC sectors (Energy, IPPU, AFOLU, Waste), and continuing to address both emissions reductions and enhanced removals. The energy sector remains a key focus for renewable energy expansion and fossil fuel substitution. NDC 3.0 covers the full economy, with sector-level targets, trajectories, and implementation pathways, consistent with an economy-wide target framework.
(e)	How the Party has addressed Article 4, paragraph 6, of the Paris Agreement.	Not applicable
7	**How the nationally determined contribution contributes towards achieving the objective of the Convention as set out in its Article 2:**	

(a)	How the nationally determined contribution contributes towards achieving the objective of the Convention as set out in its Article 2;	Namibia's NDC 3.0 contributes to the Convention's objective of stabilising GHG concentrations at a level preventing dangerous anthropogenic interference with the climate system by: - Maintaining Namibia's net carbon sink status through the 2035 target period, contributing net negative emissions to the global carbon balance - Progressively reducing gross GHG emissions across all IPCC sectors relative to the BAU trajectory - Enhancing net removals through explicit forest fire management, conservation agriculture, and ecosystem restoration - Building institutional, technical, and financial architecture for long-term low-emission development, anchored in the LT-LEDS (2025) - Supporting ecosystem resilience that underpins long-term carbon sequestration capacity
(b)	How the nationally determined contribution contributes towards Article 2, paragraph 1(a), and Article 4, paragraph 1, of the Paris Agreement.	Namibia's NDC 3.0 supports the Paris Agreement temperature goal (Article 2(1)(a)) and the global emissions peak and rapid reductions trajectory (Article 4(1)) through: - Net negative emission position: Namibia's national GHG balance remains negative (net sink) through 2035, contributing to the global carbon balance consistent with the 1.5°C pathway - Conditional scale-up: With international finance, Namibia commits to mitigation of **~15.608 $MtCO_2e$** mitigation by 2035, significant relative to national emissions scale - Energy transition: Domestic renewable energy capacity at ~70% of installed generation, reducing dependence on fossil fuels and imported power - LT-LEDS alignment: Namibia's LT-LEDS (2025) provides the long-term decarbonisation pathway consistent with achieving net zero or net-negative emissions in the second half of the century - Transparency: Strengthened INCTU and MRV systems support accountability under the ETF, contributing to the Paris Agreement's ambition ratchet mechanism

9.2. Annexure II Mitigation Measures

1. Energy Sector

16 measures; 57 actions; total cost ~USD 3.81 billion (USD 0.723 billion unconditional / USD 3.09 billion conditional); total estimated emissions reduction: ~5.196 $MtCO_2e$/yr (conditional) and 0.927 $MtCO_2e$/yr (unconditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
Measure 1A: Utility-Scale Renewable Electricity Generation *Total GHG emissions reduction:* ~3.13Mt tCO_2e/yr (including BESS) [Conditional]| *Total cost:* ~USD 3.45 billion								
ACTION 1: Commission and operate utility-scale solar PV plants: Omburu (20 MW), Khan (20 MW), Sores-Gaib (100 MW), Astra Solar Farm (50 MW), Rosh Pinah expansion (100 MW), National (120 MW, as 6x20 MW sites) (unconditional), Ruacana Hydropower (national legacy), and Baynes (as a long-term project) **INDICATOR:** *MW installed/operational by technology; GWh generated/yr; tCO_2e avoided/yr.*								
NamPower, MIME, IPPs	Unconditional	Solar PV operational: 408 MW; 276.3 GWh/year	2024	PV Plants operational	PV Plants operational	533	NamPower/ECB licensing & generation reports; Biennial Transparency Report (BTR) to UNFCCC	MIME; NamPower; IPP developers
ACTION 2: Commission the Otjikoto Biomass Power Station (40 MW) and establish an FSC-certified sustainable feedstock supply chain (unconditional). **INDICATOR:** *MW operational; GWh/yr from biomass; area under certified feedstock supply.*								
NamPower	Conditional	Under construction since 2024;	2024	40MW installed capacity, 300GWh/year, ~0.308 $MtCO_2$/yr offset Rehabilitate 16,200 ha/year,	145,000 ha rehabilitated	110	NamPower project commissioning reports; FSC certification records	NamPower;
ACTION 3: Commission and operate wind IPPs: Diaz (44 MW), CERIM Lüderitz (50 MW), Kerbehuk Ridge (34 MW), NamPower Lüderitz (40 MW), Ombepo Wind Farm (6 MW) (unconditional). **INDICATOR:** *MW installed/operational; GWh generated/yr; tCO_2e avoided/yr.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
NamPower, MIME, IPPs	Conditional	Operational: 174 MW; 638.0 GWh/year,	2024	638 GWh/year, 0.226 $MtCO_2e$/yr avoided	638 GWh/year, 0.226 $MtCO_2e$/yr avoided	408	NamPower/ECB generation & licensing reports; BTR to UNFCCC	NamPower; IPP developers;
Measure 1B: Battery Storage and Grid Modernisation *Total GHG emissions reduction:* ~0.0362 $MtCO_2e$/yr (0.00661 $MtCO_2e$/yr unconditional for Omburu BESS) | *Cost:* USD 340 million								
ACTION 1: Deploy utility-scale BESS paired with solar/wind to firm generation and reduce curtailment: Omburu (58 MW/72 MWh) (unconditional), OLC Solar+BESS (100 MW/170 MWh), TeraSun (40 MW/90 MWh) (conditional). **INDICATOR:** *MW/MWh storage installed and operational; SAPP import displacement (GWh/yr).*								
NamPower, IPPs, ECB	Conditional	Omburu BESS 51 MW committed; 18.6 GWh, OLC Solar (pipeline), TeraSun (pipeline)	2024	BESS operational	BESS operational	200	NamPower/ECB commissioning & generation reports	NamPower; IPP developers
ACTION 2: Expand and modernise transmission networks (transmission lines, substation upgrades, smart grid/DSM integration) through implementation of the Transmission Expansion and Energy Storage (TEES) Project **INDICATOR:** *km of transmission lines commissioned by voltage class; substations upgraded.*								
Nampower	Unconditional	ECC issued, 400kV Lines under design	2024	600 km line, upgraded substations	1,000 km line, upgraded substations	138.5	NamPower/TEES project progress reports	NamPower
ACTION 3: Enact the NERA Bill and updated Electricity Act; operationalise the Modified Single Buyer model for private-to-private power trade. **INDICATOR:** *NERA Bill enacted; MSB model operational.*								
ECB, MIME	Unconditional	Electricity Bill (2017) and NERA Bill not yet enacted as of the NDC 2.0 stocktake	2026	NERA Bill enacted, with regulations	n/a	0.4	Government Gazette; ECB records	MIME; ECB
ACTION 4: Update the National Integrated Resource Plan (NIRP) to reflect the NDC 3.0 electricity generation portfolio, including renewable energy, BESS, waste-to-energy, biogas and other emerging low-carbon technologies. **INDICATOR:** *Updated NIRP approved; planned low-carbon generation capacity incorporated into NIRP.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
MIME	Unconditional	Current NIRP dated 2022, predates NDC 3.0 portfolio	2022	NIRP updated	NIRP updated	0.8	MIME/NIRP publication	MIME; ECB; NamPower
Measure 1C: Strategic Long-Term Generation								
ACTION 1: Scale up green hydrogen projects supporting renewable electricity generation and grid stability (e.g. Renewstable Swakopmund-HDF Energy). **INDICATOR:** *H_2-linked RE capacity operational (MW); projects reaching financial close.*								
NGH_2P	Conditional	Advanced development and pre-construction phase, ECC issued	2026	H_2 projects operational, 142 GWh/yr	H_2 projects operational, 142 GWh/yr	210	ECB, NamPower & NGH_2P reports	MIME; NGH_2P; private H_2 developers
Measure 1D: Explore Nuclear Energy as a Future Option *Total GHG emissions reduction:* N/A | *Total cost:* USD 56 Million (Conditional)								
ACTION 1: Establish a national position, conduct feasibility studies, develop initial legal/regulatory frameworks and designate a lead institution (by 2027). **INDICATOR:** *Feasibility assessment reports approved.*								
MIME, NRPA, NEPIO	Conditional	Nuclear Industry Strategy approved by Cabinet; NEPIO Phase 1 established late 2025	2025	Feasibility completed, national position established	Regulatory framework developed, local capacity strengthened	5	NEPIO reports; Cabinet records	MIME; NRPA, NEPIO
ACTION 2: Finalise and enact the 2026 amendments to the Atomic Energy and Radiation Protection Act to strengthen regulator independence (2027-2028). **INDICATOR:** *Amended Act enacted .*								
NRPA, MHSS	Unconditional	2005 Act currently in force; 2026 amendments in progress	2026	Amendments to the Act enacted	n/a	1	Government Gazette	National Radiation Protection Authority (NRPA), Atomic Energy Board (AEB) MHSS,
ACTION 3: Develop detailed regulations, strengthen the nuclear regulator, select a site, assess technology requirements and prepare tenders (2027-2030). **INDICATOR:** *Regulator operational; site selected; tender documents issued.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
NRPA, MHSS, IAEA	Conditional	NEPIO Phase 2 (2027-2030) projected; not yet started	2026	Regulations developed, regulator operational and strengthened	site established	25	NEPIO reports	MHSS; NRPA, AEB, MIME,
ACTION 4: Strengthen national capacity and international partnerships in nuclear science and technology. **INDICATOR:** *Number of technical staff trained; partnerships established.*								
NRPA, MHSS, IAEA	Conditional	IAEA-supported capacity-building programmes in place, but limited national technical capacity	2026	International partnerships strengthened	Local capacities strengthened	10	NEPIO/IAEA training records	NRPA; NEPIO; IAEA,
ACTION 5: Finalise contracting, construction permits, regulatory oversight, workforce training and infrastructure implementation (2030-2035). **INDICATOR:** *Construction permits issued; workforce trained.*								
NRPA, MHSS, IAEA	Conditional	NEPIO Phase 3 (2030-2035+) projected; not yet started	2026	N/A	Contracting finalised, construction commenced	15	NEPIO, NRPA, MIME reports	NRPA, MIME; NEPIO
Measure 2: National Energy Efficiency Programme *Total cost:* USD 175 million								
ACTION 1: Enforce gazetted Minimum Energy Performance Standards (NAMS 109:2022 lighting; NAMS 110:2025 AC; NAMS 111:2025 refrigeration) through customs and market surveillance. **INDICATOR:** *MEPS compliance rate (%); compliant appliances deployed by category.*								
MIME	Unconditional	NAMS 109:2022 gazetted 2022; NAMS 110:2025 and NAMS 111:2025 issued 2025	2025	MEPS enforced	MEPS enforced	20	Customs/NSI compliance and market surveillance records	MIME; NAMRA; Namibian Standards Institution (NSI)
ACTION 2: Implement a utility-led Demand-Side Management programme (LED replacement in public buildings) with annual savings reporting. **INDICATOR:** *Energy efficient bulbs installed, MWh saved/yr; MW peak demand reduced.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
MME, MWT	Conditional	~181,955 LED bulbs installed under earlier DSM campaign; ~8 MW peak demand reduction	2022	50,000 energy efficient bulbs installed	100,000 energy efficient bulbs installed	140	NamPower DSM annual savings reports	MWT, MIME
ACTION 3: Introduce mandatory energy audits for public sector buildings and large commercial consumers, with published performance benchmarks. **INDICATOR:** *Number of buildings with completed energy audits.*								
MIME, MWT	Unconditional	No mandatory energy audit programme currently exists.	2026	Regulation in place and enforced	Regulation enforced	10	MIME/ECB audit registry	MIME, MWT, NEI
ACTION 4: Establish a National Energy Efficiency MRV Protocol with annual public reporting on savings, peak reduction and compliance. **INDICATOR:** *Protocol established (Y/N); annual report published.*								
MIME	Unconditional	No EE MRV protocol currently exists	2026	National EE MRV protocol in place	National EE MRV protocol in place	2	Annual public MRV report	MIME, MWT
ACTION 5: Develop fiscal and regulatory incentives (e.g. VAT/duty exemption) on efficient appliances and promote green public procurement. **INDICATOR:** *Incentive framework adopted (Y/N).*								
MoF	Unconditional	No incentives framework exists	2026	Framework in place	n/a	3	Ministry of Finance policy record	Ministry of Finance; NAMRA, MIME
Measure 3A: Battery Electric Vehicle (BEV) Deployment *Total GHG emissions reduction:* ~0.60 $MtCO_2e$/yr | *Total cost:* USD 809 million								
ACTION 1: Ensure 45% of the standard annual passenger vehicle import are electric (20% EV/Hybrid passenger vehicles, 15% light commercial vehicles, 10% Medium/heavy vehicles) **INDICATOR:** *% of new vehicle imports are electric/hybrid.*								
RFA (22 May 2026)	Conditional	224 electric vehicles 113 hybrid, 0 heavy vehicles electric or hybrid	2026	45% annual vehicle imports electric/ hybrid	50% annual vehicle imports electric/ hybrid	300	NATIS vehicle registration data; Roads Authority reports	MIME; Ministry of Finance; Roads Authority (NATIS), consumers

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 2: Establish a national EV charging network and expand EV charging infrastructure along national corridors and in urban centres, **INDICATOR:** *Number of public charging points installed; utilisation (kWh/yr).*								
ECB (2024)	Conditional	ECB EV Charging Infrastructure Regulatory Framework adopted; 4-7 public charging network established nationally	2024	Charging infrastructure developed in major urban centres	Charging infrastructure developed in major urban centres	500	ECB charging infrastructure registry	ECB; NamPower; private charging operators
ACTION 3: Develop and implement national standards for EV charging interoperability, roaming, and time-of-use (ToU) tariffs to support efficient charging infrastructure and grid management. **INDICATOR:** *National EV charging interoperability standards adopted; ToU tariff framework implemented for EV charging.*								
ECB	Unconditional	National roaming and standardised EV-specific ToU tariffs not yet operational.	2024	Standards developed and enforced	Standards developed and enforced	2	ECB regulatory gazette	ECB
ACTION 4: Introduce fiscal incentives for BEV adoption (VAT/import duty exemption, 0% environmental levy on EVs and charging components). **INDICATOR:** *Incentive framework adopted (Y/N).*								
ECB, MoF	Unconditional	No incentives exist	2026	Incentive framework developed	Incentive framework developed	3	Ministry of Finance policy record	Ministry of Finance; MIME
ACTION 5: Develop a phased regulatory framework for a carbon penalty to be applied to high-emissions internal combustion engine (ICE) imports **INDICATOR:** *framework adopted and implemented.*								
	Unconditional	None currently in place nationally	2026	Regulations adopted	Regulations adopted	1	NATIS registration data	MoF, RFA
ACTION 6: Review and reform the Road Fund Administration (RFA) fuel levy revenue model to ensure fiscal sustainability during the transition to electric and hydrogen vehicles **INDICATOR:** *Revenue replacement mechanism adopted.*								
RFA	Unconditional	RFA relies predominantly on fuel levies. Feasibility study underway	2026	Alternative revenue mechanism implemented	Alternative revenue mechanism implemented	3	RFA/Ministry of Finance policy record	RFA; MoF

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 7: Integrate safe, dedicated None-Motorised infrastructure (cycling and pedestrian pathways) into municipal urban designs to encourage active, zero-emissions/ low carbon commuting **INDICATOR:** kilometres of dedicated cycling lanes and pedestrian pathways constructed in urban centres, Number of local authorities with None-motorised infrastructure								
Local authorities	Conditional	Dedicated cycling and pedestrian infrastructure limited and fragmented across urban centres.	2026	100km of dedicated none-motorised infrastructure (nationally)	200km of none-motorised infrastructure (nationally)	35	Roads Authority, local authority reports	RA, local authorities
Measure 3B: Low-Carbon Fuel Technologies for Heavy-Duty Transport and Freight *Total GHG emissions reduction:* 0.414 $MtCO_2e$| *Total cost:* USD 382.5 million, pilot phase (fleet-scale cost conditional on pilot outcomes)								
ACTION 1: Scaled conversion of commercial trucking fleets and long-distance buses to hydrogen fuel cells and battery-electric systems linked to domestic H_2 supply, refuelling infrastructure and safety standards (Cleanergy Solutions, Hyphen, H_2-DRI). **INDICATOR:** *Number of H_2 heavy vehicle pilots deployed; refuelling stations installed; tonnes diesel displaced; tCO_2e reduced.*								
	Conditional	One (1) H_2-powered bus piloted in Walvis Bay	2026	Pending; pilot 2027-2028	Pilot completed	250	Cleanergy Solutions/NGH_2P project reports	Roads Authority; H_2 industry
ACTION 2: Develop low-carbon charging and hydrogen refuelling hubs, integrated with BESS, along major national transport corridors and industrial hubs. **INDICATOR:** *Number of low-carbon charging hubs established.*								
Roads Authority	Conditional	Pilot *H_2* refuelling infrastructure exists at Walvis Bay	2025	At least 5 *H_2 refuelling hubs*	At least 10 *H_2 refuelling hubs*	120	Roads Authority project records	Roads Authority, MWT
ACTION 3: Assess and pilot Liquefied Biogas (BioLNG), HVO/renewable diesel as low-carbon fuels for heavy-duty transport, including evaluation of domestic feedstock potential, infrastructure requirements, and lifecycle emissions. **INDICATOR:** *Number of BioLNG pilot projects implemented.*								
MIME	Conditional	none	2025	Feasibility study completed, pilot implemented	Project fully implemented	12	MIME/MAFWLR feasibility study reports	MIME; MAFWLR;
ACTION 4: Establish enabling policies, standards, and investment incentives to accelerate the adoption of low-carbon fuels in heavy-duty transport.								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
INDICATOR: *National roadmap for low-carbon heavy transport approved.*								
MoF , MWT	Unconditional	none		Roadmap developed and implemented	Roadmap developed and implemented	0.5	MIME/MWT roadmap publication, MoF	MIME, MWT, MoF
Measure 3C: Sustainable Aviation Fuel (SAF) and Synthetic E-fuels *Total GHG emissions reduction:* ~0.07 $MtCO_2e$/yr | *Total cost:* USD 98 million (Conditional)								
ACTION 1: Complete technical, economic and regulatory assessments to enable the future production and use of Sustainable Aviation Fuel (SAF) for domestic aviation **INDICATOR:** *SAF feasibility study completed. SAF roadmap and regulatory framework developed*								
NCAA	Conditional	Feasibility study commissioned	2026	Feasibility completed, regulations developed	Pilot initiated	5	NCAA fuel uplift records; NID, BTR,	NCAA; MWT
ACTION 2: Promote integration of SAF, Biogenic CO_2 and other synthetic e-fuel production into Namibia's green hydrogen and renewable energy value chains. **INDICATOR:** *SAF/e-fuel and Biogenic CO_2 projects incorporated into green hydrogen and other renewable energy developments. Resources mobilised (USD); number of pilots initiated. Resources mobilised (USD); number of pilots initiated.*								
NCAA	Conditional	none	2026	1 pilot initiated	Pilot fully implemented	60	NID, BTR, NCAA reports	NCAA; MWT, MIME; private sector
ACTION 3: Implement flight path optimisation and air traffic management measures to improve fuel efficiency in domestic aviation **INDICATOR:** Number of airports/routes implementing optimised flight procedures								
NCAA	Conditional	none	2026	1 pilot initiated	Pilot fully implemented	3	NID, BTR, NCAA reports	NCAA; MWT, MIME;
ACTION 4: Electrify airport ground operations by promoting electric ground support equipment (GSE), fixed ground power, and other low-emission airport support infrastructure. **INDICATOR:** *Number of airports with electrified ground operations; electric GSE units deployed; tCO_2e reduced.*								
Namibia Airports Company (NAC)	Conditional	Not confirmed	2026	2 national airports	All national airports	30	NAC/NCAA operations reports	NAC, NCAA, MWT,

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 5: Integrate domestic civil aviation emissions into the national GHG inventory and BTR, applying the domestic/international bunker fuel split. **INDICATOR:** *Aviation emissions included in national inventory and BTR.*								
NCAA	Unconditional	Domestic aviation emissions not yet separately integrated into the national inventory	2026	Aviation emissions included in national inventory and BTR	Aviation emissions included in national inventory and BTR	0.5	National GHG Inventory/BTR	MEFT
Measure 4: Phased Rail Decarbonisation Programme *Total GHG emissions reduction:* 0.15 $MtCO_2e$/yr | *Total cost:* USD 321 million (conditional)								
ACTION 1: Complete the HyRail Namibia hydrogen-diesel dual-fuel locomotive pilot on the Walvis Bay-Windhoek corridor and publish verified operational, emissions, and performance results. **INDICATOR:** *Pilot completed; litres diesel displaced/yr; tCO_2e avoided.*								
TransNamib, HyRail	Conditional	No locomotive conversion completed; Board-approved pilot at planning/preparation stage	2026	Locomotive conversion pilot completed	10 locomotives converted	8.68	TransNamib pilot performance reports	TransNamib; NGH_2P, private sector
ACTION 2: Based on pilot outcomes, develop a National Rail Decarbonisation Roadmap evaluating hydrogen, rail electrification, and energy-efficiency options, supported by cost-benefit analysis and an implementation and financing plan. **INDICATOR:** *Roadmap completed and adopted.*								
TransNamib	Conditional	none	2026	Roadmap completed	Roadmap under implementation	1.0	TransNamib/MWT roadmap publication	TransNamib; MWT, NGH_2P,
ACTION 3: Convert existing fleet to cleaner or dual fuel operations (building on HyRail pilot) **INDICATOR:** *Number of locomotives converted.*								
TransNamib	Conditional	TransNamib fleet dominated by ageing diesel-electric locomotives; no conversions to date	2026	30% of existing locomotives converted	100% locomotives converted	300	TransNamib fleet registry	MWT, TransNamib; NGH_2P
ACTION 4: Assess and prioritise rail rehabilitation, axle-load upgrades, and strategic investments to increase freight movement from road to rail and reduce transport emissions.								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
INDICATOR: *Freight transported by rail (tonnes/year or %); estimated diesel displaced (litres/year); estimated GHG emissions reduced (tCO_2e/year).*								
TransNamib	Conditional	Cargo predominantly transported by road	2026	15% freight transported by rail	30% freight transported by rail	8	TransNamib/MWT freight statistics	MWT, TransNamib, MEFT
ACTION 5: Develop hydrogen-specific rail safety standards, refuelling infrastructure specifications and a regulatory framework. **INDICATOR:** *Standards/regulatory framework adopted; refuelling infrastructure commissioned at depots.*								
TransNamib NGH_2P	Conditional	none	2026	Framework and standards developed and adopted	Standards fully enforced	4	TransNamib/NGH_2P regulatory records	TransNamib; NGH_2P; MWT
Measure 5A: Solar Street Lighting Programme *Total GHG emissions reduction:* ~0.0053 $MtCO_2e$/yr | *Total cost:* USD 30 million								
ACTION 1: Upgrade municipal street lighting to high-efficiency solar-powered LED systems (target: 25,000 units over 5 years); upgrade remaining non-solar streetlights to LED. **INDICATOR:** *Number of solar streetlights installed (cumulative); functionality rate (%).*								
MIME, local authorities	Conditional	Fragmented, project-based rollout; no consolidated national count; deployment materially below the 25,000-unit target	2025	25,000 units upgraded	50,000 units upgraded	29	MIME, local authority records	Local authorities; MIME; Ministry of Urban and Rural Development (MURD)
ACTION 2: Develop a national registry for solar streetlight to enable verifiable progress tracking. **INDICATOR:** *Registry established and operational.*								
MIME	Unconditional	No registry currently exists	2026	National registry developed	To be updated	1	MIME registry (once established)	MIME
Measure 5B: Commercial and Institutional Solar Water Heating *Total GHG emissions reduction:* ~0.00826 $MtCO_2e$/yr | *Total cost:* USD 30 million								
ACTION 1: Develop and implement a national policy requiring SWH installation in new residential, commercial and public buildings (phased, starting in urban areas). **INDICATOR:** *Policy adopted.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
	Conditional	Cabinet Directive already mandates SWH in new public/parastatal buildings	2026	Policy adopted	To be updated	1	Government Gazette/Cabinet directive records	MIME; MURD
ACTION 2: Implement a national institutional SWH programme targeting public facilities (schools, clinics, hospitals; target: 5,000 units over 5 years). **INDICATOR:** *Number of institutional SWH units installed (cumulative); kWh displaced/yr.*								
SOLTRAIN monitoring data, MWT	Conditional	5,000-unit NDC 2.0 target not verified as achieved; fragmented deployment	2022-2023	5,000 units installed	10,000 units installed	28	To be updated	MIME; MWT, MHSS; MEIYSAC
ACTION 3: Develop a national registry for SWH installations to enable verifiable progress tracking. **INDICATOR:** *Registry established and operational.*								
	Unconditional	No registry currently exists	2026	Registry established and in use	Registry established and in use	1	MIME registry (once established)	MIME
Measure 6A: Household Electricity Access and Lighting Transition *Total GHG emissions reduction: 0.020* $MtCO_2e$/yr (targeting 200,000 households) | *Total cost:* USD 120 million; part of national rural electrification programme budget								
ACTION 1: Expand national Solar Home Systems (SHS) to accelerate rural electrification access. **INDICATOR:** *Number of households transitioning to electricity annually (rural/urban) (gender aggregated).*								
MIME	Conditional	Rural electrification rate is 19% (to 2022); an estimated 80% of rural households still lack reliable electricity; national connectivity currently below 60%	2022	70% national connectivity	90% national connectivity	78	NSA household surveys/census; MIME	MIME; local authorities
ACTION 2: Develop solar/BESS mini-grids in rural communities, tourism routes, health facilities and schools (non-grid areas). **INDICATOR:** *Number of mini-grids installed and operational.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
	Conditional	See Action 1	2022	70% of households	90% of households	42	MIME/NamPower mini-grid registry	MIME; NamPower
Measure 6B: Clean Cooking and Fuelwood Substitution *Total GHG emissions reduction: 0.*262 $MtCO_2e$/yr (using 200,000 households target) | *Total cost:* USD 50 million								
ACTION 1: Deploy decentralised affordable modern clean cooking appliances targeting rural and peri-urban households currently dependent on fuelwood and gas. **INDICATOR:** *Number/type of clean cooking solutions adopted; % households using fuelwood as primary fuel.*								
NSA, MIME	Conditional	Fuelwood as primary cooking fuel: 50% (2023) nationally; ~84% of rural households use fuelwood; 49% access to clean cooking (2023)	2023	Increase clean cooking access rate to 61%	Increase clean cooking access rate to 80%	50	NSA household census/surveys; MIME/MEFT programme monitoring	MIME; MEFT
Measure 6C: Household Solar Water Heating *Total GHG emissions reduction:* ~0.0222 $MtCO_2e$/yr | *Total cost:* USD 50 million								
ACTION 1: Scale up household SWH deployment through the MIME Solar Revolving Fund and EIF Green Credit Line Facility (target: 25,000 units); establish a programme registry. **INDICATOR:** *Number of household SWH units installed (cumulative); kWh displaced/yr.*								
MIME	Conditional	pre-NDC 2.0 baseline ~20,000 SWH units (NamPower DSM); SRF supports ~150-200 residential SWH installations/yr since 2021;	2021	15,000 units installed	25,000 units installed	50	MIME/EIF Solar Revolving Fund and Green Credit Line disbursement records; proposed national SWH registry	MIME; EIF
Measure 7: Renewable Energy in Agriculture, Agri-Food Value Chains and Rural Water Supply *Total GHG emissions reduction:* ~0.0322 $MtCO_2e$/yr (illustrative) | *Total cost:* USD 40 million (conditional & unconditional)								
ACTION 1: Replace diesel-powered water pumps with solar-powered systems in agricultural and rural water supply schemes (target: 20,000 pumps). **INDICATOR:** *Number of solar pumps installed (cumulative); litres diesel displaced/yr.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
MAFWLR	Conditional	Over 630 solar-powered water pumping systems deployed (2018-2024) through MAFWLR and projects	2026	20,000 pumps replaced/ installed	35,000 pumps replaced/ installed	35	MAFWLR/ NamWater borehole & pump registry; project completion reports	MAFWLR; NamWater; Agribank
ACTION 2: Increase incentives and financing mechanisms for Renewable Energy (RE) adoption by farmers and agribusinesses, building on pilots (e.g. Daures Net Zero Green Hydrogen Project). **INDICATOR:** *Incentive framework developed, Number of farmers/agribusinesses accessing financing.*								
MoF , MAFWLR	Unconditional	No incentives	2026	Incentives framework developed	Incentives framework fully implemented	2	Agribank/EIF/DBN financing records	MAFWLR, Agribank; EIF; DBN
ACTION 3: Establish a monitoring system to track RE uptake, diesel displacement and emissions reductions in the agricultural sector. **INDICATOR:** *Monitoring system operational.*								
MAFWLR	Unconditional	No monitoring system currently exists	2026	Monitoring system operational	Monitoring system reviewed and operational	1	MAFWLR monitoring system (once established)	MAFWLR
ACTION 4: Establish an Operations and Maintenance (O&M) support network for solar pump maintenance; train local technicians and waterpoint committees/associations. **INDICATOR:** *Number of technicians trained (gender-aggregated); O&M support network established.*								
MAFWLR	Unconditional	Waterpoint committee associations trained by MAFWLR on basic maintenance	2026	O&M support network functional	O&M support network functional	1.5	MAFWLR/NamWater training records	MAFWLR; NamWater
ACTION 5: Install water metering on solar-powered water infrastructure to track abstraction, safeguard sustainability, and support cost-recovery and asset management. **INDICATOR:** *Number of water points metered.*								
Namwater	Unconditional	No national baseline	2026	At least at 60% of national waterpoints	At least at 90% of national waterpoints	0.5	NamWater metering database	NamWater; MAFWLR

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
Measure 8: Phased Marine Decarbonisation Programme (Waterborne Navigation) *Total GHG emissions reduction: 0.0025 $MtCO_2e$	Total cost:* USD 15 million (Conditional)							
ACTION 1: Finalise institutional frameworks to support green bunkering at the port. **INDICATOR:** *Bunkering framework adopted.*								
MWT, NamPort	Unconditional	No framework currently exists	2026	Framework adopted	Framework fully implemented	1	NamPort/NGH_2P framework publication	NamPort; NGH_2P
ACTION 2: Promote demonstration of commercially viable low-carbon marine fuels (including BioLNG, hydrogen, ammonia and sustainable biofuels), bunkering infrastructure, and fuel-efficiency technologies for port operations and maritime transport. **INDICATOR:** *Number of low-carbon marine fuel/efficiency projects implemented; number of ports with low-carbon bunkering infrastructure; GHG reduced from maritime transport (tCO_2e/yr).*								
NamPort, project developers	Conditional	Cleanergy Solutions Walvis Bay (5 MW electrolyser, ~2 t H_2/day)	2026	Pilot project implemented	Low-carbon bunkering infrastructure installed	8	NamPort/MWT project reports	MWT, NamPort, private sector
ACTION 3: Reduce NamPort operational emissions by 10%: Adopt on-site solar PV installations, Pilot renewable energy-powered shore-side power (cold ironing) infrastructure at Walvis Bay and Lüderitz ports to reduce emissions from berthed vessels, E-mobility for passenger vehicles and low-density vehicles **INDICATOR:** *Ports with operational shore-side power facilities; number of vessels using shore power annually; estimated GHG emissions reduced (tCO_2e/yr).*								
NamPort	Conditional	No shore power infrastructure currently exists	2026	Pilot implemented	Scale renewable power, efficient equipment and shore-power use based on pilot results and demand	5	NamPort operations reports	NamPort, MWT, private sector
ACTION 4: Based on pilot results, deploy selected vessel demonstrations and develop the economic case for fishing vessel conversion. **INDICATOR:** *Number of fishing vessels using alternative low-carbon fuels.*								
NamPort	Conditional	None currently exist	2026	Pilot implemented	Conversion of Namport-operated tugboats, pilot vessels and dredgers to H_2, NH_3	1	NamPort/MAFWLR fleet records	NamPort, MAFWLR, private sector

2. IPPU Sector

4 measures; 15 actions; total cost ~USD 796.6 million (USD 3.1 million unconditional / USD 793.5 million conditional); total estimated emissions reduction ~0.019 $MtCO_2e/yr$ (conditional) and 0.01 $MtCO_2e/yr$ (unconditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
Measure 1A: Plant-Level Cement Sector Decarbonisation Programme *Total GHG emissions reduction: ~0.019 $MtCO_2e/yr$ (conditional) and 0.01 $MtCO_2e/yr$ (unconditional)*| *Total cost:* USD 91 million								
ACTION 1: Promote clinker reduction and establish mandatory reporting of clinker production, clinker-to-cement ratios and blended cement sales to track progress towards a 15% clinker reduction target. **INDICATOR:** *Clinker replaced (%) reported annually per plant.*								
Cement factories, National GHG Inventory (BTR1/NC5)	Conditional	75% clinker in cement is assumed (BTR/NC5)	2022	5% reduction	15% reduction	60	Annual plant-level clinker/cement production disclosure (proposed); National GHG Inventory	MIME; Cement producers
ACTION 2: Increase clinker substitution through the use of validated supplementary cementitious materials (SCMs), expand production of blended/low-clinker cement, and introduce minimum low-clinker cement requirements in public procurement. **INDICATOR:** *Tonnes of blended cement produced/yr; number/proportion of public projects specifying low-clinker cement.*								
MIME, Cement producers	Conditional	Ohorongo already produces CEM II A-LL/B-LL blended cement commercially	2023	5% reduction	15% reduction	30	Cement producer production records; public procurement specifications	MIME; Cement producers; Ministry of Works and Transport
ACTION 3: Develop a National Industrial Emissions Monitoring and Reporting Framework for the cement sector, with mandatory annual disclosure aligned to TACCC principles. **INDICATOR:** *Framework adopted; cement plants with mandatory annual disclosure.*								
MIME	Unconditional	No dedicated IPPU monitoring framework currently exists;	2026	Framework adopted	Annual disclosure of data by cement manufacturers	1	MEFT/MIME framework publication	MEFT; MIME
Measure 1B: Decarbonisation in Other Manufacturing and Mineral Industries *Total cost:* USD 695 million								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 1: Accelerate industrial decarbonisation through the adoption of low-carbon production technologies, energy-efficiency measures, renewable energy systems and low-carbon fuels across energy-intensive industries. **INDICATORS:** *Number of industries implementing low-carbon technologies; energy savings (MWh/yr); installed on-site RE/hybrid capacity (MW); tCO_2e avoided.*								
MIME	Conditional	Industrial production largely fossil fuel-dependent	2026	3 energy-intensive industries	5 energy-intensive industries	180	MIME records,	MIME
ACTION 2: Promote the deployment of green hydrogen and its derivatives as industrial feedstocks and fuels through early-stage industrial off-take projects and demonstration initiatives across energy-intensive industries. *(E.g. green iron (H_2-DRI), industrial thermal substitution, mineral processing and mining applications.)* **INDICATORS:** *Number of* H_2 offtake projects implemented, H_2 utilised in industrial processes (tonnes/year), Annual GHG emissions reduced *tCO_2e*								
MIME	Conditional	H_2 strategy adopted. Industrial use of H_2 at pilot/ demonstration phase. Green iron manufacturing	2025	2x H_2 offtake projects implemented	5x H_2 offtake projects implemented	300	MIME, NH_2P records	MIME
ACTION 3: Promote industrial fuel switching and process optimisation through increased use of alternative fuels, renewable fuels and low-carbon feedstocks in energy-intensive industries **INDICATOR:** *Alternative fuel share (%), Annual fossil fuels replaced (TJ/yr), Annual GHG emissions reduced (*tCO_2e/yr)								
MIME	Conditional	Fossil fuels is the dominant thermal energy source in industrial processes	2026	2% alternative fuel share	5% alternative fuel share	200	NID, MIME	MIME, MEFT
ACTION 4: Develop investment-ready projects and enabling frameworks to support the decarbonisation of hard-to-abate industries, including cement, mineral processing and other energy-intensive manufacturing sectors. **INDICATOR:** *Bankable projects developed, Number of projects submitted for climate financing*								
MIME, DBN	Conditional	Limited projects	2026	At least 3 projects approved	At least 5 projects approved	15	MIME, DBN, GCF records	MIME, MEFT
Measure 2: Reduction of Lubricant Use (IPPU) *Total GHG emissions reduction:* N/A: feasibility stage| *Total cost:* USD 0.6 million								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 1: Commission a study to identify viable reduction pathways for lubricant-related emissions (transport efficiency standards, extended drain intervals, recycled lubricant programmes). **INDICATOR:** *Feasibility study completed.*								
MIME	Unconditional	none	2026	Feasibility completed	Pilots implemented	0.4	MIME records, NID	MIME; MEFT
ACTION 2: Establish minimum data collection requirements for lubricant imports and consumption within the national inventory system. **INDICATOR:** *Volume of lubricants imported/consumed (tonnes/yr) tracked in NID/BTR.*								
MIME, MoF	Unconditional	none	2025	Data collection requirements established	Volume of lubricants fully tracked	0.2	National GHG Inventory/BTR; NAMRA import records	MEFT; NAMRA
Measure 3: National RACHP Sector Decarbonisation and Lifecycle Refrigerant Management Programme *Total cost:* USD 10 million								
ACTION 1: Design and implement a national Lifecycle Refrigerant Management Programme: import controls, transition to low-GWP alternatives (R290), technician certification, and recovery/destruction arrangements. **INDICATOR:** *Refrigerant recovered annually (tonnes, by gas type); % new equipment using low-GWP refrigerants.*								
National GHG Inventory (BTR1/NC5)	Conditional	HFC emissions from refrigeration/AC by +27% (2022); no functional recovery/reclamation infrastructure at scale	2022	5% new equipment using low GWP refrigerants	15% new equipment using low GWP refrigerants	4	National Ozone Unit (NOU) recovery/certification records (to be established)	MIME (NOU); MEFT
ACTION 2: Develop domestic F-gas regulations beyond ODS; extend the National Ozone Unit (NOU) mandate to HFC regulation; establish mandatory recovery at servicing/end-of-life. **INDICATOR:** *F-gas regulations adopted.*								
MIME	Unconditional	Namibia ratified the Kigali Amendment in 2019 (HFC consumption freeze from 2024); no domestic F-gas regulation yet in force	2019	Regulations adopted	Regulations fully implemented	0.5	Government Gazette	MIME (NOU)

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 3: Develop a National Cooling Action Plan (NCAP) integrating HFC phase-down, energy efficiency for RACHP equipment and equitable cooling access (including a feasibility and cost-benefit analysis). **INDICATOR:** *NCAP adopted.*								
	Conditional	No NCAP currently exists	2026	NCAP developed	NCAP fully implemented	1	NCAP publication	MEFT; MIME, MWT, MHSS
ACTION 4: Submit a comprehensive HCFC phase-out Management Plan (HPMP) II proposal to the Multilateral Fund (recovery infrastructure, technician training, awareness, regulatory support). **INDICATOR:** *HPMP II submission made.*								
MIME (NOU)	Conditional	none	2026	HPMP submitted for funding	HPMP II under implementation	2	Multilateral Fund submission records	MIME (NOU)
ACTION 5: Commission a national RACHP equipment bank survey to establish an emissions baseline for the national GHG inventory. **INDICATOR:** *Equipment bank survey completed; national refrigerant bank size established.*								
MIME (NOU)	Conditional	No national refrigerant bank survey has been conducted to date	2026	National refrigerant bank established	National refrigerant bank under utilization	1.5	Survey report (once completed)	MEFT, MIME
ACTION 6: Review and update vocational and technical training curricula to incorporate the safe handling, installation, servicing, and maintenance of low-GWP refrigerants and emerging RACHP technologies. **INDICATOR:** *Number of curricula revised and accredited; institutions offering updated RACHP programmes; technicians trained and certified (gender-aggregated).*								
NTA	Unconditional	Current VTC curriculum outdated; does not reflect low-GWP/RACHP technology requirements	2026	Curricula updated	Technicians trained and certified	1	NTA curriculum accreditation records	Namibia Training Authority (NTA)

3. AFOLU Sector

4 measures; 20 actions; total cost ~USD 943.5 million (USD 8.5 million unconditional / USD 935.0 million conditional); total estimated emissions reduction ~10.354 $MtCO_2e$/yr (conditional) and 6.203 $MtCO_2e$/yr (unconditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
Measure 1: Livestock Methane Mitigation through Improved Finishing Systems *Total GHG emissions reduction:* ~0.043 $MtCO_2e$/yr (conditional) | *Total cost:* USD 15 million								
ACTION 1: Develop Namibia-specific evidence and implementation pathways for lower-emission livestock production systems, including assessment of net GHG balances across feedlot, free-range and regenerative systems, and sustainable domestic fodder options. **INDICATOR:** *Country-specific livestock GHG assessment and mitigation options developed; sustainable fodder production options assessed; domestic fodder production (tonnes/yr).*								
LLPBN, MAFWLR	Conditional	None	2026	Namibia-specific livestock GHG assessment completed; priority mitigation and sustainable fodder options identified.	Priority lower-emission livestock and sustainable fodder measures implemented, with GHG impacts quantified and monitored.	3	LLPBN annual production and slaughter statistics	MAFWLR; LLPBN,
ACTION 2: Scale up sustainable cattle finishing systems and associated rangeland management practices to enhance livestock productivity and reduce associated GHG emissions **INDICATOR:** *% slaughtered via feedlot finishing, % finished under regenerative rangeland management systems, National emissions intensity for livestock production from different finishing systems;*								
LLPBN, MAFWLR	Conditional	~17,500 cattle currently in feedlots); number of cattle in regenerative rangeland management practices is unknown	2024-2026	73,500 finished under improved feeding and/or regenerative rangeland management	147,000 finished under improved feeding and/or regenerative rangeland management	8	LLPBN annual production and slaughter statistics	MAFWLR; LLPBN,
ACTION 3: Explore linkages between the Integrated Climate Resilient Community Based Rangeland and Livestock Value Chain Project (ICCBRLP) (GCF), and all abattoirs to create a coherent implementation pathway. **INDICATOR:** *Linkage framework established.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
NPC, MAFWLR	Unconditional	ICCBRLP (2021-2040) is an ongoing GCF-funded programme	2026	Linkage framework established	Framework implemented	1	ICCBRLP/ UNDP programme reports	MAFWLR; LLPBN, UNDP (ICCBRLP)
ACTION 4: Revise livestock sector regulations to include GHG emissions reporting and performance standards for feedlot operations. **INDICATOR:** *Revised regulations adopted; enterprises reporting under GHG framework.*								
MAFWLR	Conditional	none	2026	Regulations adopted	Regulations fully implemented	2	MAFWLR regulatory gazette; NID	MAFWLR, MEFT
ACTION 5: Ensure gender-responsive delivery of communal livestock programmes by directing at least 40% of feedlot-linked support to women and youth. **INDICATOR:** *% of feedlot-linked support directed to women and youth.*								
MAFWLR	Unconditional	Not known	2022	40% of beneficiaries from vulnerable groups supported	50% of beneficiaries from vulnerable groups supported	1	MAFWLR programme monitoring, gender-disaggregated reporting	MAFWLR, farmers
Measure 2: Sustainable Land and Forest Management - Reduce Unsustainable Wood Removal *Total GHG emissions reduction:* ~ 10.148 $MtCO_2e$/yr (conditional) and ~6.089 $MtCO_2e$/yr (unconditional) | *Total cost:* USD 286 million								
ACTION 1: Accelerate rural/peri-urban electrification and clean cooking access to reduce forest-fuelwood demand (coordinate with Energy Measure 6A & 6B); track via NSA and household energy surveys. **INDICATOR:** *% of households using fuelwood as primary cooking fuel.*								
NSA 2023 Census	Conditional	Fuelwood as primary cooking fuel: 50% (2023) nationally; ~84% of rural households use fuelwood; 19% rural electrification (2022), ~80% of rural households still lacking reliable electricity	2023	Coordinated with Energy Measure 6A & 6B	Coordinated with Energy Measure 6A & 6B	Shared with Energy Measure 6A & 6B	NSA Population & Housing Census; household energy surveys	MEFT; MIME; NSA
ACTION 2: Develop and implement a national fuelwood reduction programme in deforested areas with extension support, to promote clean cooking, and targeted subsidies for low-income households. **INDICATOR:** *Number of households adopting alternative cooking fuels/technologies.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
MIME, NSA	Conditional	See Action 1 baseline	2023	61% clean cooking access rate	80% clean cooking access rate	5	MEFT/ MAFWLR programme monitoring; NSA surveys	MEFT; MAFWLR
ACTION 3: Strengthen the national bush biomass information system and periodic resource assessments to support sustainable biomass management and evidence-based planning across biomass value chains. **INDICATOR:** *National Bush Biomass Information System updated and operational; sustainable bush biomass resource quantified (tonnes/year).*								
MEFT	Conditional	System not updated	2026	Information system operational	Information system operational	2	National Bush Biomass Information System (once updated)	MEFT, MAFWLR, Biomass Industry players
ACTION 4: Promote the use of sustainable, low-carbon and locally appropriate construction materials, including alternatives that reduce pressure on natural forests, through appropriate standards, responsible supply chains, public procurement and awareness, with safeguards for biodiversity and sustainable resource use **INDICATOR:** *% new dwellings using alternative sustainable materials, % households using wood as material for main outer-wall construction, Number of relevant public projects applying sustainable-material procurement criteria*								
NSA	Unconditional	Households using wood for outer walls = 6.5%;	2023	5% households using wood	2% households using wood	3	NSA Population & Housing Census	MEFT; MWT
ACTION 5: Promote the sustainable utilisation of encroacher bush from bush-thinning programmes as an alternative source of fuelwood, construction timber, and other wood products to reduce pressure on natural forests. **INDICATOR:** *Tonnes of encroacher bush utilised for fuelwood and construction materials.*								
MEFT, Biomass Industry	Conditional	~45 million hectares are bush encroached nationally	2024	5% reduction in encroached areas	15% reduction in encroached areas	250	MEFT, Biomass Industry bush-harvesting records	MEFT
ACTION 6: Integrate forest-sourced fuelwood and construction-wood reduction targets into the National Forestry Programme and the NBSAP revision process. **INDICATOR:** *Targets integrated into National Forestry Programmes and policies.*								
MEFT	Unconditional	none	2025	Wood reduction targets included in national biodiversity policies	Wood reduction targets being achieved	0.5	National Forestry Programme/	MEFT

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
							NBSAP publication	
ACTION 7: Implement integrated forest fire management in priority fire-prone landscapes to reduce the extent and intensity of uncontrolled wildfires through prevention, early and prescribed burning, strategic fuel management, community-based fire management and strengthened fire response. **INDICATOR:** Annual forest/woodland area affected by uncontrolled fire (ha/year), CO_2 emissions avoided through reduced forest-fire disturbance ($MtCO_2e$/year).								
MEFT	Conditional	~2.71 million ha/year affected by forest-fire disturbance	2024	~20% reduction in annual area burned	~50% reduction in annual area burned	10	MEFT fire-disturbance records and satellite/ remote-sensing monitoring	MEFT
ACTION 8: Establish remote-sensing-based monitoring for forestland cover change, deforestation rates, and woody biomass stock. **INDICATOR:** *Annual deforestation/degradation rate (ha/yr)*								
MEFT	Conditional	no remote-sensing monitoring system yet established	2020	System developed and implemented	System maintained	15	MEFT/NSA remote-sensing monitoring system (once established)	MEFT; NSA
Measure 3: Land and Forest Restoration- Reforestation and Afforestation *Total GHG emissions reduction:* 0.147 $MtCO_2e$/yr (conditional), 0.098 $MtCO_2e$/yr (unconditional)| *Total cost:* USD 260 million								
ACTION 1: Implement the National Strategy for Tree Planting through its national planting and restoration streams: agroforestry, reforestation on communal and state land, bush-control and replanting, assisted natural regeneration, and urban greening **INDICATOR:** Hectares of degraded land (communal, state, freehold) reforested/ restored, Estimated CO_2 removals from reforested/restored land (tCO_2e)								
MEFT-DoF	Conditional	5,000	2016	1 million trees planted cumulatively (200,000 trees/annum)	2 million trees planted cumulatively	260	MEFT-DoF Tree Planting	MEFT
Measure 4: Integrated Rangeland Management, Soil Organic Carbon (SOC) Sequestration *Total GHG emissions reduction:* ~0.02 $MtCO_2e$/yr (Conditional), 0.012 (unconditional) | *Total cost:* USD 386 million								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 1: Scale up regenerative agriculture, (including implementation of the Regenerative Agriculture Project), sustainable grassland and rangeland management, and ecosystem restoration to enhance soil organic carbon, agricultural productivity (through soil carbon sequestration and other eligible technologies), including improved grazing in open communal land and alignment with conservancies, community forests. **INDICATOR:** *area under regenerative agriculture practices (ha), SOC monitored (*$MtCO_2e/yr$) across croplands and grasslands								
MEFT	Conditional	Regenerative and conservation agriculture practices are being implemented, (but no consolidated national area estimate is available).	2026	Up to 6.9 million ha under improved rangeland management	Up to 10 million ha under improved rangeland management	220	MEFT/ MAFWLR land-use monitoring	MAFWLR, MEFT, NNFU
ACTION 2: Develop and implement a national biochar, biofertiliser and compost deployment roadmap to enhance SOC: commercial-scale production pathways, farmer adoption support and supply chain development (targeting degraded areas). **INDICATOR:** *Volume of biochar/compost applied to cropland and grassland (tonnes/yr), Area covered (ha) by type (e.g. cropland, grassland)*								
MEFT	Conditional	Biochar/compost application at early research/pilot stage nationally; no quantified area or volume data identified	2026	Roadmap developed	Biomass deployment implemented	20	MEFT/ MAFWLR land-use monitoring	MEFT; MAFWLR; Biochar producers
ACTION 3: Develop nationally appropriate MRV methodologies and country-specific emission factors for SOC sequestration, sustainable biomass utilisation and other priority AFOLU mitigation activities, consistent with IPCC guidance and applicable international standards. **INDICATOR:** *National methodology developed and approved;* country-specific sequestration/emission factors developed*; tCO_2e reductions quantified;*								
MEFT	Conditional	No national methodology exists	2024	National methodologies approved	Methodologies implemented	12	MEFT methodology publication	MEFT
ACTION 4: Promote the reinstatement of urban green spaces and riparian buffers reinstatement in urban planning guidelines and town development strategies, with dedicated local authority budgets. **INDICATOR:** *Area of urban green space created (ha); area of riparian buffer reinstated (ha).*								
Local Authorities, MAFWLR	Conditional	Urban greening limited, and on project-specific basis by local authorities. No national urban-greening programme exists	2026	3,550ha urban greening, 10,000ha riparian buffer reinstated	Urban greening and riparian buffer targets achieved and maintained	120	Local authority/MURD urban planning records (to be established)	Local authorities; MURD; MEFT

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 5: Establish a dedicated AFOLU MRV system linked to NSA, MAFWLR and MEFT data systems, including improved monitoring of grassland and rangeland SOC stocks, removals and management practices INDICATOR: AFOLU MRV system operational; SOC baselines and changes monitored, removals reported tCO_2e/yr).								
MEFT, NSA	Conditional	No unified AFOLU MRV system currently exists	2026	MRV System operational	MRV System operational	8	AFOLU MRV system (once established)	MEFT; MAFWLR; NSA
ACTION 6: Develop and implement the national carbon-market framework to facilitate the development, registration, authorisation, and monitoring of high-integrity carbon projects (consistent with Namibia's NDC accounting, Article 6 requirements to ensure transparent carbon accounting, corresponding adjustments, and avoidance of double counting). **INDICATOR:** *National carbon market framework operational; registry established, authorisation procedures adopted, authorised activities and corresponding adjustments tracked; regulations published.*								
MEFT	Unconditional	No national carbon market framework currently exists	2026	National carbon market framework adopted	National carbon market framework implemented	3	National carbon project registry (once established); BTR	MEFT, Ministry of Finance, MAFWLR

4. Waste Sector

3 measures; 14 actions; total cost ~USD 865.5 million (USD 14.5 million unconditional / USD 851.0 million conditional); total estimated emissions reduction ~0.058 million tCO_2e/yr (partial; several sub-targets pending).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
Measure 1: National Solid Waste Reduction, Recycling and Energy Recovery Programme *Total GHG emissions reduction:* ~0.039 $MtCO_2e$/yr: Kupferberg WtE, RDF and waste-biochar data | *Total cost:* USD 400 million								
ACTION 1: Update the National Solid Waste Management Strategy (2026-2030) with NDC targets, updated baselines and MRV provisions. **INDICATOR:** *Revised strategy adopted.*								
MEFT	Unconditional	NSWMS 2018	2026	Revised National SWMS adopted	Revised National SWMS fully implemented	2	Government Gazette; MEFT strategy publication	MEFT; MURD; local authorities
ACTION 2: Develop Waste Management Regulations, including Extended Producer Responsibility (EPR) and the Circular Economy Policy to reduce waste generation and finance the recycling economy (including plastic waste reduction). **INDICATOR:** *Waste management regulations, including EPR regulations adopted and Circular Economy Policy adopted*								
MEFT	Unconditional	Draft Waste Management Regulations, Draft EPR regulation and Draft Circular Economy Policy	2026	Waste Management Regulations, including EPR regulations developed and Circular Economy Policy developed	Waste Management Regulations fully enforced	5	Government Gazette	MEFT
ACTION 3: Advance the Kupferberg landfill methane capture and waste-to-energy project to investment-ready status by completing a bankable feasibility study, securing regulatory approvals, establishing a power purchase arrangement, and mobilising climate finance. **INDICATOR:** *Bankable feasibility study completed; generation licence issued; installed gas-to-energy capacity.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
City of Windhoek (CoW)	Conditional	2012 CDM/Carbon Check feasibility confirmed technical suitability; pre-feasibility study is underway;	2023	Feasibility completed, project commissioned	Project operational	150	CoW/ ECB licensing and feasibility study reports	CoW; ECB, MIME, NamPower
ACTION 4: Promote the separation and conversion of organic waste into biochar, renewable energy, biofertilisers, and other value-added low-emission products to reduce landfill disposal. **INDICATOR:** *Tonnes of organic waste diverted from landfill; tonnes of biochar produced; estimated methane emissions avoided (tCO_2e/year).*								
MEFT	Conditional	Composting/organic waste separation at an early stage; not systematically practised	2026	Pilot projects implemented	Pilot projects scaled	30	Landfill records	MEFT, local authorities
ACTION 5: Scale up the production and utilisation of refuse-derived fuel (RDF) from suitable waste streams to reduce landfill disposal and substitute fossil fuels in industry. **INDICATOR:** *Tonnes of RDF produced annually; tonnes of RDF utilised by industry;*								
Recycle Namibia Forum, RDF producers	Conditional	12,000 tonnes produced annually (Rent-a-Drum), with 1,020 tonnes utilised by Ohorongo Cement	2022	15,000 tonnes RDF produced and at least 12,000 tonnes utilised	25,000 tonnes RDF produced and at least 20,000 tonnes utilised	6	Waste management data	MEFT, Local authorities, Waste management companies,
ACTION 6: Establish a national waste data registry and am integrated MRV system for the waste sector. **INDICATOR:** *Registry/MRV system operational.*								
MEFT	Unconditional	No national waste data registry currently exists	2026	Registry developed and in use	Registry fully operational	5	National waste registry (once established)	MEFT; MURD, NSA
ACTION 7: Develop a national composting strategy with phased infrastructure rollout and roll out programme for composting household organic waste. **INDICATOR:** *Tonnes of waste composted/yr.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
MEFT, local authorities	Conditional	Composting at an early stage; 65,000 rural household composting bins required and have not been distributed	2022	112,000 tonnes composted	150,000 tonnes composted	150	Local authority/MEFT composting tonnage records (once established)	MEFT; local authorities
ACTION 8: Scale the Waste Buy-Back Centre (WBBC) model from Windhoek to other urban centres across the country. **INDICATOR:** *Number of WBBCs operational and coverage area.*								
Local authorities	Conditional	First WBBC (Katutura), Windhoek, opened in 2024, a second centre in Okuryangava began development in 2025	2025	3 additional WBBC commissioned	6 additional WBBC commissioned	40	City of Windhoek WBBC operational records	City of Windhoek; MURD; local authorities
Measure 2: Reduction of Open Burning *Total GHG emissions reduction:* Pending national open-burning reduction target | *Total cost:* USD 323.5 million								
ACTION 1: Regulate open burning and establish a formal national tracking system for open burning incidence through strengthened local authority reporting. **INDICATOR:** *Annual volume of waste open-burned (tonnes).*								
NSA 2023 Census	Unconditional	No dedicated system currently tracks national open-burning volumes; the 2023 Census captures general disposal trends but not burning specifically;	2023	Tracking system developed and implemented	Tracking system fully operational	2	Local authority reporting system (once established)	MEFT; MURD; local authorities
ACTION 2: Accelerate upgrading of municipal waste disposal sites to managed disposal standards **INDICATOR:** *% of disposal sites meeting managed disposal standards.*								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
MEFT, local authorities	Conditional	Many disposal sites currently operate below NSWMS minimum standards	2026	25% disposal sites upgraded	50% disposal sites upgraded	180	MEFT/MURD site inspection records	MEFT, Local authorities; MURD
ACTION 3: Strengthen and implement a nationally coordinated framework to reduce and prevent open waste burning through enhanced regulatory and enforcement measures and inclusive public awareness and behavioural change initiatives. **INDICATOR:** *National framework for reducing open waste burning adopted and implemented; regulatory and enforcement measures strengthened; population reached through gender- and socially inclusive awareness initiatives*								
MEFT	Unconditional	No action plan currently exists	2026	Action plan developed and implemented	Action plan fully implemented	0.5	MEFT action plan publication	MEFT
ACTION 4: Expand formal waste collection coverage in informal settlements and outside local authority areas. **INDICATOR:** *Number of municipalities with collection covering informal settlements.*								
Local authorities	Conditional	Inadequate waste collection in informal settlements and rural areas remains the main driver of open burning	2026	Formal waste collection services in informal settlements and rural areas	Formal waste collection services in informal settlements and rural areas	140	Local authority/Regional Council collection coverage records	Local authorities; Regional Councils
Measure 3: Expand Wastewater Management and Methane Recovery Programme *Total GHG emissions reduction:* Pending methane recovery target | *Total cost:* USD 155 million								
ACTION 1: Accelerate sewer network expansion in informal, peri-urban and newly urbanising areas. **INDICATOR:** *Number of households connected to sewer systems (cumulative).*								
NSA 2023 Census; National GHG Inventory (BTR1/NC5)	Conditional	38.4% of households connected to sewer systems (2023);	2023	45% households connected to sewer systems	60% households connected to sewer systems	140	NSA Census; NamWater/local authority sewer connection records	NamWater; local authorities

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 Target	2035 Target (extrapolated)	Cost (USD) (million)	Means of Verification	Lead Agency
ACTION 2: Explore methane recovery from wastewater treatment facilities as an additional mitigation measure (per LT-LEDS). **INDICATOR:** *Number of facilities with biogas/methane recovery systems installed.*								
	Conditional	Gammams WWTP captures methane for biogas	2026	Feasibility and pilot for 1 additional WWTP	Methane recovery system at 1 additional WWTP	15	Local Authorities, NamWater/MEFT feasibility study reports	Local Authorities, NamWater;

9.3. Annexure III Adaptation Measures

46 measures; 110 actions; total cost ~USD 7.56 billion (USD 904.15 million, 12% unconditional / USD 6.65 billion, 88% conditional) to 2030.

1. Agriculture and Food Security

11 measures; 24 actions; total cost ~USD 1.80 billion (USD 292.5 million, 16% unconditional / USD 1.50 billion, 84% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 1. Enhance livestock disease control, prevention and treatment facilities								
Action 1: Construct and operationalise 4 additional veterinary clinics in priority regions **INDICATOR:** *No of veterinary clinics constructed and operational; No. of animals treated and vaccinated annually*								
MAFWLR	Unconditional	24	2026	2 veterinary clinics	4 veterinary clinics	120	MAFWLR Reports	MAFWLR
Action 2: Establish and operationalise livestock inspection centres and quarantine facilities at high-risk border control points **INDICATOR:** *No of livestock inspection centres and quarantine facilities established and operational; No of animals inspected at border facilities annually*								
MAFWLR	Conditional	11 high-risk border points	2026	6 livestock inspection centres	11 livestock inspection centres	90	MAFWLR Reports	MAFWLR
Action 3: Strengthen integrated animal health surveillance and Early Warning Systems (EWS) at national level including disease reporting and response mechanisms **INDICATOR:** *Functional national animal health surveillance system in place; Frequency of disease surveillance reporting (e.g. quarterly)*								
MAFWLR	Unconditional	Moderate surveillance and EWS	2026	Adequate surveillance and EWS	Advanced surveillance and EWS	10	MAFWLR reports	MAFWLR
Action 4: Strengthen livestock market access in northern communal areas through improved animal health certification systems, traceability mechanisms, and supporting infrastructure **INDICATOR:** *Number of animals marketed through certified /formal systems; Number of farmers accessing formal livestock markets*								
MAFWLR	Unconditional	Weak and inconsistent formal markets. Farmers rely on informal markets with poor prices	2026	Adequate formal markets through LLPBN (e.g. Meatco, ZAMCO, Etunda feedlot) with improved prices	Strong formal markets through LLPBN (e.g. Meatco, ZAMCO, Etunda feedlot) with good prices	40	MAFWLR reports	MAFWLR
Measure 2. Breeding and conserving of climate resilient and commercially viable livestock and crop varieties'								
Action 1: Establish and implement a targeted adapted small stock distribution programme for vulnerable households with preferential allocation to women, youth, indigenous minorities and persons living with disabilities (PwDs)								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
INDICATOR: *Number of adapted small stock distributed to beneficiary households; Number of women, youth, indigenous minorities and PwDs reached; Number of beneficiary households accessing livestock advisory, veterinary or market-support services.*								
MAFWLR	Unconditional	40 communal farmers supported in Omusati and Erongo regions	2025	50 vulnerable households per region	100 vulnerable households per region	7.5	MAFWLR reports	MAFWLR
Action 2: Establish and operationalise a national breeding programme for climate-resilient livestock, focusing on indigenous and adapted breeds with improved heat tolerance, drought resistance **INDICATOR:** *Number of livestock breeding stations operationalised, Number of climate-adapted breeds*								
MAFWLR	Conditional	6 livestock research stations	2025	5 breeding stations	11 breeding stations	37.5	MAFWLR reports	MAFWLR
Action 3: Strengthen farmer capacity in livestock management through extension services and training, including livestock health, drought preparedness, fodder management, emergency water access and market-based destocking, with targeted participation of women, youth, indigenous minorities and persons with disabilities (PwDs). **INDICATOR:** Number of farmers trained in climate-resilient livestock practices (disaggregated by gender)								
MAFWLR	Unconditional	Unconfirmed	2026	100 farmers trained across all 14 regions per year	250 farmers trained services across all 14 regions per year	20	MAFWLR reports	MAFWLR
Measure 3. Improved fodder production and create fodder banks								
Action 1: Establish and operationalise fodder production sites (including irrigated fodder systems, drought-tolerant fodder crops, bush to feed and hydroponic units where appropriate) in drought-prone communal regions **INDICATOR:** Fodder production systems established and operational; Estimated annual fodder production capacity (tons/year)								
MAFWLR	Conditional	Limited fodder production sites	2026	Adequate fodder production sites	Enhanced fodder production sites	25	MAFWLR reports	MAFWLR
Action 2: Establish and operationalise community-based fodder banks in drought-prone communal areas **INDICATOR:** Number of fodder banks established and operational; Storage capacity (tons); Volume distributed annually (tons/year); Number of farmers accessing fodder banks annually								
MAFWLR	Conditional	None	2026	7 community-based fodder banks	14 community-based fodder banks	100	MAFWLR reports	MAFWLR
Measure 4. Value chain development of agricultural products								
Action 1: Strengthen diversification of crops, food products and associated agricultural enterprises and markets for priority agricultural commodities covering production, processing, marketing, and distribution **INDICATOR:** Number of value chain programmes established and operational by commodity; Number of farmers participating in value chain programmes annually; Volume of agricultural products marketed through formal value chains annually								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
NAB, LLPBN	Unconditional	Limited value chain programmes	2026	At least 1 value chain programmes per commodity per region	At least 2 value chain programmes per commodity per region	5	MAFWLR reports, NAB, LLPBN reports,	MAFWLR, NAB, LLPBN
Action 2: Establish and operationalise agro-processing units in priority production areas **INDICATOR:** Number of agro-processing units established and operational; Processing capacity of agro-processing units (tons/year)								
NAB, LLPBN	Conditional	Limited agro-processing units	2026	At least 1 agro-processing unit per product per region	At least 2 agro-processing unit per product per region	200	MAFWLR, NAB, LLPBN reports	MAFWLR, NAB, LLPBN
Action 3: Capacity building for value chain development, including farmers and processors, through training, extension services, and coordination platforms **INDICATOR:** Number of farmers/processors trained in value chain development (disaggregated by gender)								
NAB, LLPBN	Unconditional	Unconfirmed	2026	100 farmers/ processors trained across all 14 regions per year	250 farmers/ processors trained across all 14 regions per year	20	MAFWLR, NAB, LLPBN reports	MAFWLR, NAB, LLPBN
Measure 5. Upgrading infrastructure and monitoring systems for climate impacts								
Action 1: Develop and operationalise integrated ICT infrastructure and digital monitoring systems for agriculture and climate impacts within MAFWLR (including capacity building of stakeholders across the agricultural value chain) **INDICATOR:** Functional integrated ICT and monitoring system in place,								
MAFWLR	Unconditional	None	2026	ICT and monitoring system in place	ICT and monitoring system in place	5	MAFWLR reports	MAFWLR
Action 2: Develop and implement targeted interventions to improve market infrastructure, processing capacity, market linkages (e.g. abattoir access), and veterinary compliance **INDICATOR:** Interventions implemented to improve marketing infrastructure and access								
MAFWLR	Conditional	Moderate marketing infrastructure	2026	Adequate marketing infrastructure and access	Enhanced marketing infrastructure and access	100	MAFWLR reports	MAFWLR
Measure 6 Scale Conservation Agricultural Practices								
Action 1: Scale conservation agriculture through farmer adoption, extension support and access to appropriate equipment and inputs. **INDICATOR:** Area under verified CA; no. of farmers actively practising CA (gender-disaggregated)								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
MAFWLR	Conditional	7,710 ha using ripper furrows; 1,636 ha with soil cover; 665 documented CA adopters; CA extension/mechanisation support already exists.	2020	10,000 ha under verified CA; 20,000 active CA farmers.	20,000 ha under verified CA; 30,000 active CA farmers.	140	MAFWLR reports, CCAP reports	MAFWLR
Action 2: Strengthen national and regional coordination for conservation agriculture implementation. **INDICATOR:** National coordination framework operational; no. of regional implementation programmes operational.								
MAFWLR	Unconditional	CCAP II provides a national coordination/programme framework.	2024	Integrated national framework operational; 7 regional programmes operational.	Framework institutionalised/reviewed; 7 regional programmes sustained.	5	MAFWLR reports, CCAP reports	MAFWLR
Action 3: Review and reform the Fertilizer Act 1947 and associated fertiliser registration requirements to enable safe use and commercialisation of low-risk organic, biological and natural agricultural inputs **INDICATOR:** Regulatory reform completed; proportionate registration pathway operational; no. of eligible local products approved.								
MAFWLR	Unconditional	No comprehensive reform completed.	2025	Reform approved; registration pathway operational; ≥10 local products approved.	Framework maintained/reviewed; ≥20 local products approved.	10	Government Gazette	MAFWLR
Measure 7 Strengthen climate-resilient crop production and food systems								
Action 1: Rehabilitate priority Green Schemes and scale up water-efficient irrigation systems **INDICATOR:** No. of Green Schemes rehabilitated/upgraded; ha under water-efficient irrigation.								
MAFWLR	Conditional	9 Green Schemes operational; 5,600 ha irrigable, 3,403 ha cultivated; drip irrigation used in selected communal gardens.	2025	5 priority schemes upgraded; 5,000 ha under water-efficient irrigation.	Operational schemes progressively upgraded; 7,500 ha under water-efficient irrigation.	100	MAFWLR reports	MAFWLR
Action 2: Strengthen farmer access to climate-resilient crops, inputs, mechanisation and extension services to support resilient production, livelihoods and food security. **INDICATOR:** No. of farmers accessing climate-resilient seeds/inputs/services; no. adopting improved climate-resilient practices.								
MAFWLR	Unconditional	20,908 farmers accessed certified seed; ~15,000 producers accessed mechanisation; climate-smart exten-	2024	40,000 farmers/yr access climate-resilient seeds/inputs/services; ≥5,000 farmers adopt	60,000 farmers/yr access seeds/inputs/services; ≥10,000 farmers	50	MAFWLR reports	MAFWLR

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		sion/livelihood interventions already underway.		improved practices annually.	adopt improved practices annually.			
Action 3: Strengthen grain and seed storage infrastructure to reduce post-harvest losses and improve food-system resilience. **INDICATOR:** No. of operational storage facilities; tonnes stored/preserved annually.								
MAFWLR	Conditional	5 national grain storage facilities Tsandi, Omuthiya, Okongo, Rundu and Katima Mulilo; ~5,400t of grain procured into national reserve system.	2025	At least 8 facilities operational; At least 40,000 t/yr stored/preserved.	At least 10 facilities operational; At least 75,000 t/yr stored/preserved.	40	MAFWLR reports	MAFWLR
Measure 8 Scale climate-resilient horticulture								
Action 1: Scale up water-efficient net-house/protected vegetable production in priority areas, with integrated technical support and appropriate pollination management. **INDICATOR:** Ha under operational net-house/protected production; no. of farmers adopting supported systems.								
MAFWLR,	Conditional	Protected horticulture systems are in use, but no consolidated national area baseline; >3,000 farmers have received related technical training/support.	2023	400 ha operational; At least 3,000 farmers adopting new production systems.	600 ha operational; At least 6,000 farmers cumulatively adopting supported systems	110	MAFWLR reports	MAFWLR
Measure 9 Diversification of Crops								
Action 1: Scale diversification of climate-resilient crops and irrigation suited to local agro-ecological conditions. **INDICATOR:** Ha under diversified cropping systems; no. of farmers adopting climate-resilient irrigation.								
MAFWLR, UNDP	Conditional	At least 220 households documented using drip irrigation in communal gardens nationally.	2024	10,000 ha diversified; 3,000 farmers adopting climate-resilient irrigation.	20,000 ha diversified; 6,000 farmers adopting climate-resilient irrigation.	125	MAFWLR reports	MAFWLR - DAPEES
Measure 10. Scale climate-resilient urban and peri-urban agriculture								
Action 1: Scale climate-resilient urban and peri-urban agriculture using water-efficient production systems, local partnerships and market support. **INDICATOR:** No. of urban/peri-urban agriculture initiatives supported; participating households/producers; ha under supported production.								
MAFWLR, EIF	Conditional	57 urban/peri-urban projects supported; >1,300 households/microenterprises reached.	2021/22 - 2025	150 initiatives; 5,000 participants; 100 ha under supported production.	250 initiatives; 8,000 participants; 200 ha under supported production.	375	MAFWLR reports	MURD, local Authorities

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 11: Strengthen agricultural climate information and early-warning systems								
Action 1: Strengthen agricultural climate information and early-warning systems through modernised monitoring infrastructure, integrated data and institutional capacity. **INDICATOR:** No. of functional agrometeorological monitoring stations/systems; integrated national platform operational; no. of climate-informed advisories issued annually.								
MAFWLR, SASSCAL	Conditional	Monitoring networks exist through SASSCAL and the Namibia Meteorological Service, but datasets remain fragmented across institutions.	2025	Integrated platform operational with strengthened monitoring network; At least 12 integrated climate/early-warning products annually.	Platform institutionalised with near-real-time integration; ≥12 routine products plus event-triggered advisories	60	MAFWLR reports, NMS reports	MAFWLR, MWT

2. Water Resources

5 measures; 11 actions; total cost ~USD 3.84 billion (USD 405.0 million, 11% unconditional / USD 3.44 billion, 89% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 1 Review and Update the Integrated Water Resources Management Plan								
Action 1: Review and update the Integrated Water Resources Management (IWRM) Plan to strengthen climate responsive water governance, basin management and sustainable allocation of surface and underground water resources **INDICATOR:** National IWRM Plan reviewed and approved; Annual progress report on IWRM implementation produced; Surface and ground water resource allocation								
MAFWLR	Unconditional	1 IWRM Plan developed 2010 and in adopted 2012	2010	The existing IWRM plan is outdated and should be reviewed and updated handle severe water scarcity, recurrent droughts, and climate change through coordinated management of water, land, and related resources	IWRM plan is regularly updated,	2	MAFWLR reports	MAFWLR NamWater, Regional and local authorities
Action 2: Facilitate women's equal participation and leadership in national and local water governance and associations **INDICATOR:** Number of local and national water management committees (or associations) with at least 40% women in executive leadership positions and a formal gender-responsive mandate								
MAFWLR	Unconditional	Data specific to the Ministry of Agriculture, Water and Land Reform (or legacy water departments) historically shows women accounting for roughly 34% of general staff, reflecting a male-dominated technical environment.	2025	At least 40% women occupying leadership positions of national and local water governance and associations	At least 50% women occupying leadership positions of national and local water governance and associations	1	Basin Management Committee Reports, MAFWLR reports	MAFWLR
Action 3: Develop and implement a long-term water security strategy for the Central Areas of Namibia as part of the national IWRM planning process, including climate risk assessment, demand management, supply diversification, infrastructure resilience and groundwater protection.								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
INDICATOR: Long-term water security strategy for the Central Areas of Namibia developed and approved; Number of priority interventions for Central Areas water security identified and integrated into IWRM implementation planning; Annual progress on Central Areas water security interventions reported.								
MAFWLR	Unconditional	A 20 – year water security strategy approved for Windhoek backed by AfDB	2025	A long-term water strategy developed and finance secured for implementation to enhance water security, efficiency, and resilience of the Central Areas in the face of climate challenges such as droughts	1	2	MAFWLR reports	MAFWLR NamWater, Regional and local authorities
Measure 2 Develop a National Water Master Plan								
Action 1: Rehabilitate existing rural water points in priority drought-prone and underserved areas., including with protection measures to reduce the risk of HWC **INDICATOR:** Number of water points rehabilitated and operational with improved water security and integrated safety and protection features; Number of households and communities benefiting, including women, youth, and persons living with disabilities (PwDs)								
MAFWLR	Conditional	139 boreholes and rural water points targeted for rehabilitation (with a N$37.5 million allocation)	2025	At least 695 rural water points rehabilitated and operationalised	At least 1,390 rural water points rehabilitated and operationalised	10	MAFWLR reports, Regional Councils reports	MAFWLR NamWater, Regional and local authorities
Action 2: Install new solar-powered (low-carbon powered) boreholes in water-scarce rural areas. **INDICATOR:** Number of solar-powered boreholes operational; number of people/households benefiting (disaggregated by gender, age, disability status and other relevant vulnerable groups)								
MAFWLR	Conditional	The government of Namibia drilled 1,907 new boreholes across the country between April 2024 and mid-2026 alone	2026	Construct additional 2,500 boreholes equipped with solar installations to cut fuel dependency	5,000 boreholes constructed	20	MAFWLR reports, Regional Councils reports	MAFWLR NamWater, Regional and local authorities
Action 3: Extend and upgrade water supply infrastructure through construction of pipelines, reticulation networks, and distribution systems to improve water access and reliability **INDICATOR:** Kilometres of water supply infrastructure constructed or upgraded annually; Number of households and communities benefiting, including women, youth, and persons living with disabilities (PwDs)								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
MAFWLR	Unconditional	250 km of pipelines, reticulation networks, and distribution systems extended and upgraded. Namibia is expanding and upgrading its water supply infrastructure through major regional bulk water and pipeline replacement initiatives led by NamWater and the government	2026	1,250 km of pipelines, reticulation networks, and distribution systems extended and upgraded.	3,000 km of pipelines, reticulation networks, and distribution systems extended and upgraded.	200	MAFWLR reports, Regional Councils reports, NamWater reports	MAFWLR, NamWater, Regional Councils
Action 4: Construct or rehabilitate reservoirs and strategic dams to improve water security and drought buffering capacity. **INDICATOR:** Number of reservoirs / dams / storage systems constructed or rehabilitated; Additional water storage capacity created (m^3)								
MAFWLR	Conditional	8 reservoirs and 10 strategic dams were constructed and rehabilitated.	2026	At least 2 strategic dams identified and constructed	At least another 2 dams identified and constructed	1,000	MAFWLR reports, Regional Councils reports, NamWater reports, Local Authority reports	MAFWLR, NamWater, Regional Councils, Local Authorities
Measure 3 Water Reclamation								
ACTION 1: Develop and operationalise wastewater collection, reticulation, and secondary treatment systems in towns with populations of approximately 40,000 or more residents. **INDICATOR:** Number of towns with operational wastewater reticulation and treatment systems; Volume of wastewater treated annually (m^3), Volume and percentage of treated wastewater reused annually, Number of agricultural, industrial, landscaping or other users supplied with treated wastewater.								
Local Authorities	Conditional	Approximately 8 to 10 major cities and towns in Namibia operate advanced, fully managed mechanical or biological secondary wastewater collection, reticulation, and treatment systems	2026	At least 20 cities and towns in Namibia developed and operating wastewater collection, reticulation and secondary treatment systems	At least 30 cities and towns in Namibia developed and operating wastewater collection, reticulation and secondary treatment systems	400	Local Authorities, Regional Councils reports,	Local Authorities, Regional Councils

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
ACTION 2: Construct and operationalise tertiary wastewater treatment and reuse plants to supply treated water for agriculture, industry, and managed aquifer recharge. **INDICATOR:** Number of reuse plants operational; Volume of treated wastewater reused annually (m^3)								
Local authorities	Unconditional	1 tertiary wastewater reclamation and potable reuse are implemented by the City of Windhoek under a public-private partnership arrangement	2026	5 tertiary wastewater treatment and reuse plants constructed and operationalised	10 tertiary wastewater treatment and reuse plants constructed and operationalised	200	Local Authorities reports, Regional Councils reports	Local Authorities, Regional Councils
Measure 4 Establishment and restoration of riparian buffer								
ACTION 1: Establish, restore and protect riparian buffer zones in the Okavango, Kwando and Zambezi River systems through community-based stewardship to strengthen ecosystem and community climate resilience. **INDICATOR:** Area (ha) of riparian buffer zones established, restored or protected; length (km) of riverbank under active riparian protection or restoration; number of community-led riparian restoration and protection initiatives operational.								
MAFWLR and MEFT	Conditional	0 buffer zones established	2026	At least 3 buffer zones established to prevent ongoing challenges of unchecked agricultural expansion, deforestation, and uncontrolled fires from significantly degrading these fragile riverbanks.	2 more additional riparian buffer zones created along the Orange and Kunene River	8	Transboundary river basin reports, MAFWLR reports	MAFWLR, MEFT, KAZA TFCA
Measure 5: Desalination systems (sea water and saline ground water)								
ACTION 1: Develop and operationalise coastal and inland desalination systems for seawater and saline groundwater water in priority water-stressed areas to increase climate-resilient water supply. **INDICATOR:** Number of desalination systems commissioned and operational; additional water supply capacity created through desalination (m^3/day).								
Namwater, MAFWLR	Conditional	1 coastal desalination plant commissioned and operational, Approxi-	2026	At least 1 more desalination plant com-	At least 2 Coastal and inland desalina-	2,000	NamWater reports	MAFWLR NamWater

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		mately N$2.1 billion is being mobilised for another coastal desalination initiative (**Erongo SUNAM Desalination** project).		missioned and operational	tion plants commissioned			

3. Terrestrial Biodiversity and Ecosystems

5 measures; 13 actions; total cost ~USD 402.0 million (USD 120.0 million, 30% unconditional / USD 282.0 million, 70% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 1 Sustainable Land Management								
Action 1: Restore degraded and bush-encroached rangelands through sustainable bush control, grazing management and soil/water conservation. **INDICATOR:** Area (ha) of priority rangeland restored/under verified climate-resilient management								
MEFT/ NNFU	Conditional	~45 million ha affected by bush encroachment nationally	2024	30% of degraded lands restored (aligned with AFOLU measure 4)	40% of degraded lands restored	aligned with AFOLU measure 4	MEFT, NNFU, MAFWLR reports	MEFT MAFWLR
Action 2: Strengthen integrated natural resource management in climate-vulnerable production and conservation landscapes through coordinated restoration, grazing, biodiversity conservation, and community stewardship. **INDICATOR:** Area (hectares) under integrated natural resource management; Number of landscapes / communities implementing integrated resource management plans								
MEFT	Conditional	National climate-adaptation coverage not systematically tracked	2026	Climate-informed management plans operational	Expanded national coverage	24	MEFT Reports	MEFT
Action 3: Strengthen early detection and control of invasive alien species (IAS) in priority ecosystems **INDICATOR:** priority invasive species under surveillance/control; ha of priority sites treated								
MEFT	Conditional	National IAS management exist, but not systematically tracked	2026	IAS surveillance and control systems operational in priority ecosystems	Measurable reduction in spread in priority ecosystems	18	MEFT Reports	MEFT
Measure 2 Climate-resilient State-Protected Areas and Wildlife Systems								
Action 1: Integrate climate-risk information into protected-area management to strengthen the resilience of priority habitats and ecosystems, ecological connectivity, water availability and management of Human-Wildlife Conflict (HWC) **INDICATOR:** Protected area systems with climate-risks integrated in management,								
MEFT	Unconditional	13 approved park management plans	2024	19 approved park management plans with climate-risk incorporated	Periodic review of these risks incorporated into protected-area planning	12	MEFT Reports	MEFT

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Action 2: Strengthen HWC management and ecological connectivity in priority areas **INDICATOR:** Number of HWC incidences; value of mobilised compensation funding								
MEFT	Unconditional	NDP6 reports 3,380 HWC cases	2024	50% reduction in reported HWC cases	sustain reduction in reported HWC cases	24	MEFT Reports	MEFT
Action 3: Upgrade water supply, sanitation, sewage and waste management infrastructure in priority protected areas and associated tourism facilities to reduce environmental pressure, strengthen ecosystem resilience and support sustainable tourism. **INDICATOR:** Number of protected areas and tourism facilities with upgraded infrastructure, Reduction in water consumption at upgraded facilities, Improved wastewater and waste management performance								
MEFT	Conditional	Outdated water, sewage and waste management infrastructure	2026	At least 5 priority protected areas and associated tourism facilities upgraded.	Upgrades expanded to additional priority protected areas	60	MEFT reports, NWR reports	MEFT NWR
Action 4: Implement adaptive species management, including translocations and other relevant interventions to support biodiversity conservation, reduce human–wildlife conflict, strengthen communal conservancies, and improve ecosystem resilience. **INDICATOR:** Number of wildlife individuals translocated annually, by species								
	Unconditional	Wildlife translocations undertaken on a need basis, guided by species management and conservation priorities	2026	Translocations and other species management interventions undertaken as required, based on ecological and management assessments	Adaptive species management institutionalised & implemented across priority species and landscapes	18	MEFT reports.	MEFT
Measure 3 Integrated Fire Management								
Action 1: Establish and maintain strategic firebreaks and fuel management across priority fire-prone landscapes, protected areas, and communal areas. **INDICATOR:** Length (km) of firebreaks established, rehabilitated, and maintained;								
MEFT OPM	Unconditional	Approximately 1.2 million hectares burned and 1,300 fire incidents recorded	2025	2,000 km of strategic firebreaks established, and maintained	Strategic fire breaks maintained	24	MEFT reports, Integrated State of Environment Report	MEFT, OPM

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Action 2: Strengthen community and institutional capacity for wildfire prevention, preparedness, and response through provision of appropriate equipment, vehicles, training, awareness, and community outreach programmes. **INDICATOR:** Number of community members and personnel trained in fire management (gender aggregated), Number of fire awareness and preparedness campaigns conducted annually								
MEFT OPM	Unconditional	National capacity and equipment are limited	2026	All regions have trained, and equipped teams	Regular readiness and preparedness maintained	18	MEFT, OPM reports	MEFT, OPM
ACTION 3: Promote ecosystem-appropriate controlled burning for rangeland management and ecosystem restoration, supported by clear guidelines and community participation. **INDICATOR:** National guidelines for controlled and prescribed burning developed and approved; Number of controlled or prescribed burns implemented; Area of land managed or restored through controlled burning; Number of communities or land managers trained to apply controlled burning practices.								
MEFT	Unconditional	Controlled burning is practised in some areas (limited national monitoring)	2026	Controlled burning guidelines strengthened and implemented in priority fire-prone landscapes	Ecosystem-appropriate controlled burning integrated into fire, rangeland and ecosystem management in priority landscapes.	6	MEFT reports	MEFT
Measure 4 Management of sensitive natural habitats (Karoo, Wetlands and Desert)								
Action 1: Strengthen adaptive management of priority sensitive ecosystems, including Karoo, wetland and desert ecosystems. **INDICATOR:** Priority sensitive ecosystems under adaptive management								
MEFT	Unconditional	Limited adaptive management	2026	Adaptive management strengthened in sensitive ecosystems	Adaptive management institutionalised across priority fragile ecosystems	18	MEFT Reports	MEFT
Measure 5 Integrate Ecosystem-based Adaptation (EbA) into Community-Based Natural Resource Management (CBNRM) Programmes								
Action 1: Integrate Ecosystem-based Adaptation into the management plans of communal conservancies, community forests and other CBNRM areas. **Indicators:** No of CBNRM areas integrating and implementing EbA in their management and monitoring systems, Area (ha) of CBNRM landscapes under improved climate-resilient ecosystem management through EbA								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
MEFT, CBNRM organisations	Conditional	EbA is implemented in selected CBNRM areas (not across national CBNRM programme)	2026	At least 50% of CBNRM areas integrate EbA into management and monitoring	At least 80% of CBNRM areas integrate EbA into management and monitoring	120	MEFT reports, State of Community Conservation Reports	MEFT, CBNRM organisations
Action 2: Strengthen climate-resilient landscape-based tourism and other nature-based livelihood activities in CBNRM areas to support ecosystem conservation, community income and local employment. **INDICATORS:** Number of landscape-based tourism and nature-based livelihood initiatives supported, Number of CBNRM areas implementing landscape-based tourism activities, Number of beneficiaries (disaggregated by benefit type, gender and social group)								
MEFT, CBNRM organisations	Conditional	Nature-based livelihood initiatives exist across CBNRM areas (limited integration of climate-resilience)	2026	Climate-resilient nature-based livelihood opportunities expanded across priority CBNRM areas	Climate-resilient livelihood diversification sustained and scaled across priority CBNRM areas	60	State of Conservancy Report	MEFT, CBNRM organisations

4. Marine Biodiversity, Fisheries, and Aquaculture

5 measures; 6 actions; total cost ~USD 127.0 million (USD 9.0 million, 7% unconditional / USD 118.0 million, 93% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 1 Support sustainable use of fisheries resources by local communities								
Action 1: Strengthen climate resilience of small-scale fisheries (SSF) communities through vulnerability assessment, adaptation planning and implementation of priority adaptation measures. **INDICATOR:** Number of priority fisheries communities with vulnerability/adaptation plans; Number of adaptation actions implemented; fishers/households benefiting.								
MAFWLR, MEFT	Conditional	4 vulnerability assessments; at least 1 documented SSF adaptation intervention; 45 community members directly supported.	2022	At least 5 priority assessments/plans; 6 adaptation actions implemented; 1,500 fishers/households benefiting.	10 assessments/plans completed or updated; 12 cumulative actions; 3,000 fishers/households benefiting.	20	MAFWLR, MEFT reports	MAFWLR
Measure 2 Adaptive management and protection of aquatic ecosystems/biodiversity								
Action 1: Strengthen adaptive management and protection of priority aquatic ecosystems through community fisheries reserves, climate-responsive management plans and participatory monitoring. **INDICATOR:** No. of fisheries reserves/protected areas operational; no. of climate-responsive management plans under implementation.								
MAFWLR	Conditional	20 gazetted community fisheries reserves (~2,500 ha) with 20 associated management plans; community fish guards/monitors already active.	2025	26 reserves operational with 26 climate-responsive management plans under implementation.	Established reserves sustainably managed; plans periodically reviewed and updated.	12	MAFWLR reports	MAFWLR -
Measure 3 Develop mariculture within blue economy								
Action 1: Scale up environmentally sustainable mariculture in suitable coastal zones, including prawn and oyster production, supported by environmental monitoring and technical/enterprise support. **INDICATOR:** Number of operational/expanded mariculture farms; production capacity; % of operational sites under environmental monitoring.								
MAFWLR	Conditional	0 prawn farms; 6 operational oyster	2025	2 prawn pilots (4 ha); 5 oyster farms	4 prawn farms (8 ha); 10 oys-	33	MAFWLR reports	MAFWLR -

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		farms/companies; 11 mariculture farms covered by monitoring programmes.		developed/expanded (+~750,000 oysters/yr); 100% of operational sites monitored.	ter farms cumulatively developed/expanded (+~1.5m oysters/yr); 100% monitoring maintained.			
Measure 4 Strengthen aquatic food systems								
Action 1: Strengthen climate-resilient aquaculture production through upgraded farms, reliable fingerling/feed supply and integrated extension and farm-management support. **INDICATOR:** No. of aquaculture facilities upgraded/operational; annual production; producers accessing production inputs/services.								
MAFWLR, FAO	Conditional	8 government aquaculture centres; ~34.18 tonnes/yr freshwater production; 779,359 fingerlings distributed/yr.	2023	5 priority centres upgraded; ≥100 tonnes/yr production; 500 producers/yr supported; 2m fingerlings/yr.	8 centres upgraded; ≥200 tonnes/yr production; 750 producers/yr supported; 3m fingerlings/yr.	53	MAFWLR reports	MAFWLR -
Measure 5 Building Climate Change Resilience in Marine and Freshwater Fishing and Aquaculture Communities								
Action 1: Strengthen climate-informed fisheries and aquaculture planning and early-warning systems. **INDICATOR:** No. of sector plans/projects integrating adaptation; frequency/coverage of fisheries and aquaculture climate advisories/warnings.								
MAFWLR	Unconditional	1 sector plan integrates adaptation; no systematic sector-specific early-warning dissemination demonstrated.	2022	3 sector plans/projects; ≥2 scheduled advisories/yr and all relevant event-triggered warnings disseminated.	5 sector plans/projects; ≥4 scheduled advisories/yr and all relevant event-triggered warnings disseminated.	5	MAFWLR reports	MAFWLR
Action 2: Upgrade and climate-proof priority fisheries landing sites, storage facilities, hatcheries, ponds, and aquaculture infrastructure against floods, storms, heat, and water stress. **INDICATOR:** Number of fisheries or aquaculture facilities upgraded for climate resilience; Number of communities benefiting from resilient infrastructure								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
MAFWLR (MFMR National Plan of Action for Small-Scale Fisheries 2022-2026)	Unconditional	1 existing fisheries infrastructure site (Katima fish market); climate-resilience status not documented.	2022	5 priority facilities upgraded/climate-proofed	10 facilities cumulatively upgraded/climate-proofed,	4	MAFWLR reports	MAFWLR

5. Health and Sanitation

6 measures; 19 actions; cost ~USD 349.65 million (USD 9.15 million, 3% unconditional / USD 340.5 million, 97% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 1 Climate-Resilient Health Infrastructure and Service Readiness								
Action 1: Climate-proof priority health facilities and essential service systems, including maternity, neonatal and SRH services, against floods, heat, water stress and other climate-related disruptions. **INDICATOR:** Health facilities climate-proofed and equipped with resilient energy, water, sanitation, diagnostic and digital systems.								
MHSS	Conditional	Existing network: 4 intermediate hospitals, 34 district hospitals, 56 health centres and 322 clinics; resilience upgrades incomplete.	2026	2 intermediate hospitals, 15 district hospitals, 28 health centres and 161 clinics climate-proofed/upgraded.	4 intermediate hospitals, 34 district hospitals, 56 health centres and 322 clinics climate-proofed/upgraded.	315	MHSS reports	MHSS
Action 2: Strengthen climate-resilient outreach and digital health services to maintain access to essential SRH and GBV services during climate-related disruptions. **INDICATOR:** Climate-resilient mobile/outreach and telehealth services operational during climate events.)								
MHSS	Conditional	1,150 mobile clinics outreach points, 14 GBV Protection Units (GBVPUs);	2026	1,150 mobile clinics outreach points, 14 GBVPUs	1,150 mobile clinics outreach points, 14 GBVPUs	10.5	MHSS reports	MHSS
Measure 2 Integrated Disease Surveillance, Early Warning and Health Security								
Action 1: Develop and operationalise integrated climate-induced disease surveillance and early warning systems for epidemic-prone and climate-sensitive diseases integrating SRHR, HIV, and GBV information to promote early action and mitigate negative impacts on vulnerable communities **INDICATOR:** Functional climate induced disease surveillance / early warning system in place; Number of regions connected to the surveillance system and early warning system								
MHSS	Unconditional	Lack of an effective climate induced disease surveillance and early warning system	2026	All 14 regions connected to the climate induced disease surveillance and early warning system	All 14 regions connected to the climate induced disease surveillance and early warning system	1.5	MHSS reports	MHSS
Action 2: Strengthen epidemiology, laboratory, surveillance, and early warning capacity through specialised training of health personnel, incorporating indicators of climate stress linked to major health impacts, including those related to reproductive, maternal, neonatal, child and adolescent health (RMNCAH)								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
INDICATOR: Number of health personnel trained in surveillance, epidemiology, laboratory, and early warning functions; Number of regions with trained surveillance personnel meeting staffing standards								
MHSS	Unconditional	4 personnel (nurse, midwife, medical officer, and pharmacists) per 1,150 outreach posts responsible for early warning and outreach services for climate linked health impacts including RMNCAH (the number of health personnel should increase proportionate to the population growth and disease outbreaks)	2026	4,600 key health personnel from all 14 regions trained in climate linked health impacts including RMNCAH	9,200 key health personnel from all 14 regions continuously trained in climate linked health impacts including RMNCAH	1.5	MHSS reports	MHSS
Action 3: Improve disease surveillance and early warning systems to detect and respond including for maternal and newborn mortalities **INDICATOR:** Functional and effective integrated Early Warning Systems that utilise climate-health data to generate automated alerts for spikes in maternal/newborn risks and disease outbreaks in climate-vulnerable districts.								
MHSS	Unconditional	Lack of a functional and effective integrated early warning system, using climate-health data to generate automated alerts for spikes in maternal/newborn risks and disease outbreaks	2026	Functional and effective integrated early warning system, using climate-health data to generate automated alerts for spikes in maternal/newborn risks and disease outbreaks	Advanced and effective integrated early warning system, using climate-health data to generate automated alerts for spikes in maternal/newborn risks and disease outbreaks	0.6	MHSS reports	MHSS
Measure 3 Climate-Sensitive Malaria and Vector-Borne Disease Control								
Action 1: Implement targeted vector control interventions in endemic and high-risk areas **INDICATOR:** Number of constituencies covered by vector control interventions; Percentage coverage of households in endemic/ hotspot areas								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
MHSS	Unconditional	Number of constituencies covered out of the 121 Constituencies in Namibia unconfirmed	2026	90% of households in vector control endemic / hotspot areas across all 121 constituencies covered	100% of households in vector endemic / hotspot areas across all 121 constituencies covered	1.2	MHSS reports	MHSS
Action 2: Strengthen diagnosis, treatment, and case management of vector-borne diseases in endemic areas. **INDICATOR:** Number / of vector-borne disease cases tested and treated annually								
MHSS	Unconditional	Namibia recorded about 7 vector-borne diseases including: Malaria, Dengue fever, West Nile virus, Congo fever, tick-bite fever, Leishmaniasis, bilharzia, etc	2026	7 (100%) vector-borne disease cases tested and treated annually	7 (100%) vector-borne disease cases tested and treated annually	0.3	MHSS reports	MHSS
Action 3: Strengthen community sensitisation and behaviour change programmes for prevention, early detection, and timely treatment seeking for vector-borne diseases **INDICATOR:** Number of vector-borne disease sensitisation campaigns conducted annually; Percentage of people treated through vector-borne disease awareness programmes								
MHSS	Unconditional	about 1 campaign done per disease vector including: mosquito, tick, sandfly and flatworms	2026	1 campaign done per disease vector annually (mosquito, tick, sandfly and flatworms), and 100% of infected people treated	1 campaign done per disease vector annually (mosquito, tick, sandfly and flatworms), and 100% of infected people treated	0.3	MHSS reports	MHSS
Measure 4 Climate-Sensitive Public Health and Disease Prevention								
Action 1: Strengthen prevention, control, and treatment of climate-sensitive communicable diseases, including diarrhoeal diseases, respiratory infections, hepatitis A, tuberculosis, and other priority risks. **INDICATOR:** Percentage detection of priority climate-sensitive diseases; Number of affected persons treated annually								
MHSS	Unconditional	Annual incidence of priority climate-sensitive diseases, includ-	2026	100% detection of priority climate-sensitive diseases and	100% detection of priority climate-sensitive diseases and	0.3	MHSS reports	MHSS

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		ing diarrhoeal diseases, respiratory infections, hepatitis A, tuberculosis		100% of infected people treated	100% of infected people treated			
Action 2: Strengthen water, sanitation, hygiene (WASH), nutrition, and community health interventions to reduce climate-related health vulnerabilities **INDICATOR:** Number of constituencies reached through WASH and nutrition interventions; percentage of health facilities with improved WASH systems								
MHSS	Unconditional	Number of constituencies reached out of the 121 Constituencies in Namibia unconfirmed	2026	all 121 constituencies reached annually and 100% of health facilities with improved WASH systems	all 121 constituencies reached annually and 100% of health facilities with improved WASH systems	0.3	MHSS reports	MHSS
Action 3: Strengthen health workforce capacity and community sensitisation on environmental health promotion through outreach campaigns **INDICATOR:** Number of health campaigns per constituency; percentage of people reached per constituency, including women, youth, and persons living with disabilities (PwDs)								
MHSS	Unconditional	Number of health campaigns done in constituencies out of the 121 Constituencies in Namibia unconfirmed	2026	Health campaigns done in all 121 constituencies annually and 70% of people reached per constituency	Health campaigns done in all 121 constituencies annually and 90% of people reached per constituency	0.3	MHSS reports	MHSS
Action 4: Establish and operationalise a national heat-health preparedness and response system to reduce illness and mortality associated with extreme heat, linking climate and meteorological early warnings with health surveillance, public-health advisories, health-facility preparedness and targeted protection of vulnerable populations and high-risk workers. **INDICATOR:** Heat-related illness admissions, Number of regions with operational heat-health preparedness and response arrangements linked to national meteorological heat warning, Percentage of designated priority health facilities implementing heat-health readiness and response protocols during extreme-heat alerts								
MHSS	Unconditional	No dedicated heat-health action plan identified	2026	Heat-health action operational in all 14 regions	At least 90% of priority health facilities implement heat-response protocols during extreme-heat alerts	0.45	MHSS reports, Meteorological services reports	MHSS

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Action 5: Conduct gender-transformative review of national and subnational vulnerability assessment tools to better integrate SRH, GBV risks and youth priorities **INDICATOR:** Number of national and sub-national climate and heath vulnerability and adaptation assessments that incorporate age- and gender-disaggregated data and specific risk-mapping for GBV and SRHR service disruptions during climate shocks.								
MHSS	Unconditional	Number of national and sub-national regional health campaigns done across 14 regions on risk mapping for GBV and SRHR during climate shocks un-confirmed	2026	1 national and 14 sub-national regional health campaigns done on risk mapping for GBV and SRHR during climate shocks conducted annually	1 national and 14 sub-national regional health campaigns done on risk mapping for GBV and SRHR during climate shocks conducted annually	0.3	MHSS reports	MHSS
Measure 5: Strengthening Health System Capacity for Disaster Preparedness and Continuity of SRH and GBV Services								
Action 1: Institutionalise the Minimum Initial Service Package (MISP) for sexual and reproductive health within national and district health strategies to ensure the immediate, uninterrupted delivery of lifesaving reproductive health and GBV response services during and after climate-induced disasters **INDICATOR:** Number of health facilities with functional disaster preparedness and response plans that include measures to ensure the continuity of SRH and maternal health services; Number of district preparedness plans that integrate the MISP								
MHSS	Conditional	34 District hospitals, 56 health centres have foundational capacity to implement MISP and 322 permanent Clinics have no capacity to implement	2026	34 District hospitals, 56 health centres have intermediate capacity to implement MISP and 322 permanent Clinics have foundational capacity to implement MISP	34 District hospitals, 56 health centres have advanced capacity to implement MISP and 322 permanent Clinics have intermediate capacity to implement MISP	15	Ministerial reports	MHSS
Action 2: Enhance the capacity of the health workforce in climate-vulnerable areas by integrating climate change, SRHR, and GBV response into pre-service midwifery curricula and in-service training materials **INDICATOR:** Number of health providers in climate-vulnerable and fragile areas trained/capacitated on climate-resilient healthcare delivery including MISP								
MHSS	Unconditional	Namibia has about 15,000 healthcare workers (3 per 1,000 people (trained number un-confirmed)	2026	All 15,000 healthcare workers in Namibia trained in climate resilient	All 15,000 healthcare workers in Namibia trained in climate resilient	0.6	Ministerial reports	MHSS

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
				healthcare and MISP	healthcare and MISP			
Action 3: Develop/adopt training modules and courses for the health workforce in Namibia on sexual reproductive health and climate change **INDICATOR:** Number of health providers trained using nationally approved modules on SRHR and climate change								
MHSS	Unconditional	Namibia has about 15,000 healthcare workers (3 per 1,000 people (trained number un-confirmed)	2026	All 15,000 healthcare workers in Namibia trained the link between SRHR and climate change	All 15,000 healthcare workers in Namibia trained the link between SRHR and climate change	0.6	MHSS, NTA, Training Institution reports	MHSS
Action 4: Integrate climate change education in pre-service midwifery curricula and in service training materials for enhanced maternal health outcomes **INDICATOR:** Number of accredited midwifery training institutions that have integrated climate change and health (including maternal and neonatal risks) into pre service curricula and in-service training programmes								
MHSS	Unconditional	Only 2 of the accredited midwifery training institutions have integrated climate change and health into the curricula	2026	All accredited midwifery training institutions have integrated climate change and health into the curricula	All accredited midwifery training institutions have integrated climate change and health into the curricula	0.3	MHSS, NTA, NQA, Training Institution reports	MHSS
Measure 6: Health National Adaptation Plan								
Action 1: Develop and operationalise the Health National Adaptation Plan (HNAP), integrating sexual and reproductive health (SRH), maternal and neonatal health, gender-based violence (GBV), and mental health and psychosocial impacts associated with climate-related hazards, disasters, displacement and livelihood disruption, with linkages to disaster risk reduction and emergency preparedness frameworks. **INDICATOR:** HNAP developed and operationalised								
MHSS	Unconditional	No operational HNAP	2026	HNAP endorsed and operational	Review of the HNAP	0.3	MHSS report	MHSS
Action 2: Mainstream climate resilience intersections in health sector policies, strategies and plans. **INDICATOR:** Number of costed national and sub-national health sector policies, strategies and disaster-response plans addressing the health impact of climate-induced disasters including on essential services continuity								
MHSS	Unconditional	No costed national health sector policies, strategies and disaster-response plans	2026	50% of national health sector policies, strategies and disaster-response	100% of national health sector policies, strategies and disaster-response	0.3	MHSS report	MHSS

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		addressing climate-induced health impacts		plans addressing climate-induced health impacts costed	plans addressing climate-induced health impacts costed			

6. Cross-cutting Issues

6 measures; 23 actions; cost ~USD 200.0 million (USD 52.0 million, 26% unconditional / USD 148.0 million, 74% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 1 Disaster Risk Management								
Action 1: Develop and operationalise a National Emergency Operation Centre: covering droughts, floods, storms, heatwaves, wildfires, disease outbreaks, and other priority hazards, including community dissemination mechanisms **INDICATOR:** National emergency operation centre established, Number of priority hazards integrated into the national multi-hazard early warning system; Number of regions covered by functional early warning dissemination systems								
OPM - DDRM	Conditional	Directorate of Disaster Risk Management (DDRM), Emergency Operation Centre (EOC), under OPM National Public Health Emergency Operations Centre (PHEOC), and the Multi-Hazard Emergencies Preparedness and Response Plan (MHEPRP), under MHSS	2026	National emergency centre operational, Priority hazards integrated in the multi-hazard early warning system for all 14 regions	National emergency centre operational, Priority hazards integrated in the multi-hazard early warning system for all 14 regions	12	OPM - DDRM, MHSS reports	OPM - DDRM
Action 2: Identify, assess, and map high-risk areas and vulnerable populations through climate and disaster risk assessments at national, regional, constituency, and community levels utilising gender, Age, and Disability Disaggregated Data (SADDD) **INDICATOR:** Number of high-risk areas mapped and updated; Number of regions or constituencies with completed multi-hazard risk profiles that incorporate age- and gender-disaggregated data								
OPM - DDRM	Conditional	National Risk Profile, Hazard and Climate Risk Profiles, and Mapping Social & Population Vulnerability	2026	Number of regions with completed multi-hazard risk profiles, incorporate age- and gender-dis-	Number of regions with completed multi-hazard risk profiles, incorporate age- and gender-dis-	8	OPM- DDRM Reports	OPM- DDRM

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		Data-driven, risk-based planning to mitigate the impacts of severe droughts, floods, and health epidemics		aggregated data for all 14 regions	aggregated data for all 14 regions			
Action 3: Strengthen and operationalise national and subnational crisis and disaster management systems for preparedness, coordination, response, and recovery. **INDICATOR:** Functional national disaster management system operational; Number of regions with operational disaster preparedness and response structures								
OPM - DDRM	Unconditional	Fully Functional, legally established, national disaster risk management system (NDRMS), Operational disaster preparedness and response structures across all 14 regions	2026	Agile disaster preparedness and response structures across all 14 regions	Dynamic disaster preparedness and response structures across all 14 regions	4	Vulnerability Assessment and Analysis Report	OPM – DDRM
Action 4: Strengthen technical capacity of institutions and improve community access, understanding, and use of early warning information through training, awareness, and preparedness programmes **INDICATOR:** Regions trained in forecasting, risk analysis, and early warning operations; Constituencies reached through community early warning awareness and preparedness programmes, including gender aggregated data								
OPM - DDRM	Conditional	Institutions such as the Namibia Flood and Drought Monitor (FDM) supported by UNESCO, and SASSCAL has capacity to provide	2026	Key institutions trained in climate forecasting, risk analysis and early warning operations	Training in climate forecasting, risk analysis and early warning operations for Key institutions, strengthened	8	OPM – DDRM. MAFWLR, NMS	OPM – DDRM.

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		high-level data-driven training on drought, rainfall, fire indicators using tools like WeMust						
Action 5: Strengthen national capacity to assess, address and respond to climate-related Loss and Damage through coordinated institutional, information and financing mechanisms. **INDICATOR:** National Loss and Damage framework operational, Loss and Damage coordination and financing mechanism operational.								
OPM	Unconditional	None	2026	National Loss and Damage framework operational	Loss and Damage coordination and financing mechanism operational.	4	OPM, MAFWLR, NMS reports	OPM – DDRM.
Measure 2 Education, training and public awareness								
Action 1: Strengthen the climate change subjects in formal and informal education systems, including schools, vocational education, tertiary institutions, and community learning programmes. **INDICATOR:** Climate change subjects in education curricula strengthened, across all levels (from primary school to university)								
MEIYSAC	Unconditional	National plan integrating climate change in the education system developed and implemented	2026	Climate change subjects in education curricula strengthened	Climate change subjects in education curricula reviewed and continuously strengthened	4	MEIYSAC, MEFT, MAFWLR reports	MEIYSAC
Action 2: Strengthen capacity of lecturers, teachers, trainers, extension officers, and relevant institutions to deliver climate change education, adaptation, and resilience skills **INDICATOR:** Number of training programmes for lecturers, teachers, trainers, and officials in climate change education and awareness								
MEIYSAC	Unconditional	Education for Sustainable Development (ESD) Policy, developed	2026	Education for Sustainable Development (ESD) Policy implemented	Education capacity on climate change education enhanced, in accordance with the ESD Policy	4	MEIYSAC, MAFWLR, NIED reports	MEIYSAC

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Action 3: Implement targeted climate awareness and behaviour change programmes for vulnerable constituencies and communities **INDICATOR:** National awareness campaigns extended to regional level; Number of constituencies reached through climate awareness programmes, including gender disaggregated data								
MEIYSAC	Unconditional	National Climate Change awareness campaigns conducted (Think Namibia Campaign)	2026	National Climate Change awareness campaigns conducted, extended to regional level	National Climate Change awareness campaigns conducted, extended to regional and constituency level	4	Regional Council reports	Regional Councils, MEFT – DEA
Measure 3 Research and systematic observations								
Action 1: Strengthen and modernise national hydrometeorological monitoring, forecasting, and early warning systems covering weather, climate, surface water, groundwater, and water quality **INDICATOR:** Functional national hydrometeorological forecasting system in place; Number of monitoring stations / observation sites operational and reporting								
MWT, NMS, MAFWLR	Unconditional	National hydrometeorological forecasting system in place (operating as a single piece of software, combining field infrastructure, satellite-driven data models, and multi-agency communications)	2026	National hydrometeorological forecasting strengthened	National hydrometeorological forecasting reviewed and continuously strengthened	4	MWT, NMS, MAFWLR Reports	MWT, NMS
Action 2: Develop and operationalise integrated national climate data, statistics, and information management systems for real-time access, analysis, and cross-sectoral data sharing **INDICATOR:** Functional national climate statistics and data management system in place; Number of institutions connected to the integrated climate information platform								
MWT, NMS, MAFWLR	Unconditional	National climate statistics and data management system in place	2026	National climate statistics and data management strengthened	National climate statistics and data management reviewed	4	MWT, NMS, MAFWLR Reports	MWT, NMS

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		(upgrade from fragmented institutional databases into a centralised, system tracking climate statistics and environmental metrics)			and continuously strengthened			
Action 3: Strengthen national capacity for climate research, risk modelling, forecasting, and decision-support applications for adaptation planning **INDICATOR:** Functional Climate risk assessments, forecasts, or modelling products								
MWT, NMS, MAFWLR	Unconditional	Namibia has sophisticated catalogue of climate risk assessments, operational forecasting tools, and advanced climate modelling products	2026	Climate modelling products strengthened	National climate statistics and data management reviewed and continuously strengthened	4	MWT, NMS, MAFWLR Reports	MWT, NMS
Measure 4 Integrated Land Use planning								
Action 1: Develop Integrated Regional Land Use Plans (IRLUPs) for all regions, aligned with climate risk management, environmental sustainability, and resilient development objectives **INDICATOR:** Number of regional land use plans completed or updated; Number of regions implementing approved IRLUPs								
MAFWLR	Conditional	Integrated Regional Land Use Plans (IRLUPs) developed for 8 out of 14 regions	2026	Integrated Regional Land Use Plans (IRLUPs) developed for all 14 regions	Integrated Regional Land Use Plans (IRLUPs) reviewed and complied with	8	MAFWLR, MEFT, MURD, RC'S	MAFWLR
Action 2: Strengthen inclusion of women, youth, indigenous minorities and PwDs in land planning, land access, and development programmes **INDICATOR:** GESI and PwDs inclusion in land reform, land use planning, land access or development programmes implemented; Number of beneficiaries accessing land, including Gender aggregated data								
MAFWLR	Unconditional	GESI and PwDs consideration for land reform land use planning and land access	2026	GESI and PwDs measures for land reform land use planning and land access implemented	GESI and PwDs measures for land reform land use planning and land access strengthened	4	MAFWLR, MEFT, MURD, RC'S	MAFWLR

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Action 3: Promote sustainable and climate-resilient land management in communal areas through secure land rights, restoration, productive land use, and community-based natural resource management **INDICATOR:** Area (hectares) of communal land under sustainable land management								
MAFWLR	Unconditional	Proportion of the 28.7Mha communal land under sustainable land management unconfirmed	2026	20% of the 28.7Mha communal land under sustainable land management	40% of the 28.7Mha communal land under sustainable land management	4	MAFWLR, MEFT, MURD, RC'S	MAFWLR
Action 4: Identify, map and designate flood-prone, erosion-prone and fire-prone areas **INDICATOR:** Number of regions with updated flood-prone, erosion-prone, fire-prone maps, other high-risk or development-free zones as indicated in the IRLUPs								
MAFWLR	Conditional	Number of regions with updated flood-prone, erosion-prone, fire-prone maps un-confirmed	2026	Updated flood-prone, erosion-prone, and fire-prone maps developed for all 14 regions	Updated flood-prone, erosion-prone, and fire-prone maps for all 14 regions reviewed and continuously improved	8	MAFWLR, MURD reports	MAFWLR,
Action 5: Implement priority flood, riverbank and coastal protection measures identified through land-use plans, including dikes, groynes, bank protection and sea walls **INDICATOR**: Priority measures such as dikes, groynes, sea walls, river bank protection structures constructed or rehabilitated								
MURD	Conditional	Priority measures such as dikes, sea walls, river bank protection structures and sea walls constructed or rehabilitated unconfirmed	2026	50% Priority measures such as dikes, sea walls, river bank protection structures (as indicated in the IRLUPs and climate risk priority maps) constructed or rehabilitated	50% Priority measures such as dikes, sea walls, river bank protection structures (as indicated in the IRLUPs and climate risk priority maps) constructed or rehabilitated	40	MURD, MAFWLR, RC'S, LA's	MURD

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Action 6: Restore and protect beaches, dunes and other natural barriers in areas exposed to coastal erosion and flooding INDICATOR: Length of coastline (km) protected or restored, Number of projects or area of beaches replenished and restored (ha)								
MURD	Conditional	Length of coastline (km) protected or restored, Number of projects or area of beaches replenished and restored (ha), unconfirmed	2026	50% of the priority coastline length (km) protected or restored, Number of projects or area of beaches replenished and restored (ha)	50% of the priority coastline length (km) protected or restored, Number of projects or area of beaches replenished and restored (ha)	40	MURD, MAFWLR, RC'S, LA's	MURD
Measure 5 Gender Equality, Social Inclusion and Resilient Livelihoods								
Action 1: Promote diversified and climate-resilient livelihood opportunities for vulnerable groups (including women, youth and PwDs) through vocational skills development, climate-smart agriculture, sustainable enterprises and trade, to strengthen economic resilience and reduce dependence on climate-sensitive livelihoods. **INDICATOR:** Regional GESI disaggregated data (including: women, youth, PwDs and other vulnerable groups) trained or supported through climate resilience programmes annually								
MGECW	Conditional	GESI disaggregated data for people trained or supported through climate resilience programmes per region unconfirmed	2026	GESI disaggregated data for people trained or supported through climate resilience programmes per region	GESI disaggregated data for people trained or supported through climate resilience programmes per region, strengthened	8	MGECW, MEFT	MEFT, MGECW
Action 2: Promote climate-resilient livelihood opportunities, enterprise development, and access to productive resources, finance, land, and skills for women, men, youth, PwDs, and other vulnerable groups **INDICATOR:** Regional GESI disaggregated data (including: women, youth, PwDs and other vulnerable groups) benefiting from climate-resilient livelihood programmes								
MGECW	Conditional	GESI disaggregated data for people benefiting from climate-resilient livelihood pro-	2026	GESI disaggregated data for people benefiting from climate-resilient livelihood	GESI disaggregated data for people benefiting from climate-resilient liveli-	8	MGECW, MURD	MEFT, MGECW

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		grammes per region unconfirmed		programmes per region	hood programmes per region strengthened			
Action 3: Establish and strengthen shelters, gender responsive safe spaces, and provision of support and GBV prevention and response services for vulnerable populations affected by climate shocks, displacement, or gender-based risks. **INDICATOR:** Regional GESI disaggregated data (including women, youth, PwDs, and other vulnerable persons) accessing protection and support services (including number of new women and girls' friendly spaces established; Number of women and girls receiving GBV prevention and response services in these spaces)								
MHAISS	Unconditional	GESI disaggregated data for people accessing protection and support services per region unconfirmed	2026	GESI disaggregated data for people accessing protection and support services per region	GESI disaggregated data for people accessing protection and support services per region strengthened	4	MHAISS, MGECW	MHAISS
Action 4: Integrate alternative livelihood programmes such as vocational training, climate-smart agriculture, and sustainable trade prioritising women and girls to reduce financial dependency. **INDICATOR:** Regional GESI disaggregated data (youth, PwDs, and other vulnerable persons) who have transitioned to low economic dependency livelihoods, through vocational training, climate-smart agriculture, or sustainable trade, resulting in a measurable increase in non-seasonal, independent income								
MGECW, RCs, reports	Conditional	GESI disaggregated data for people transitioned to low economic dependency livelihoods per region unconfirmed	2026	GESI disaggregated data for people transitioned to low economic dependency livelihoods per region	GESI disaggregated data for people transitioned to low economic dependency livelihoods per region strengthened	8	MGECW, RCs, reports	MEFT MGECW
Action 5: Implement the National Climate Change Gender Action Plan (ccGAP) to mainstream gender-transformative climate action across all of Namibia's climate-priority sectors. **INDICATOR:** National Climate Change Gender Action Plan (ccGAP) is adopted and operational; (Yes/No) ccGAP is informed by gender-responsive evidence from a comprehensive gender needs assessment; Number of climate-priority sectors actively implementing ccGAP priorities								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
MGECW, MEFT, reports	Unconditional	National Climate Change Gender Action Plan (NCCGAP) in place	2026	National Climate Change Gender Action Plan (NCCGAP) implemented	National Climate Change Gender Action Plan (NCCGAP) implementation, strengthened	4	MGECW, MEFT, reports	MEFT MGECW
Measure 6: Social Protection and Shock-Responsive Support								
Action 1: Implement shock-responsive social protection strategies to protect vulnerable households exposed to severe shocks (such as devastating droughts, floods, or macroeconomic crises) and GBV survivors/victims **INDICATOR:** *Vulnerable households covered by shock-responsive social protection (No./%);*								
MGECW, MHAISS, reports	Unconditional	National Social Protection Policy (2021–2030), to provide shock-responsive social protection to support vulnerable populations exposed to severe shocks in place	2026	National Social Protection Policy (2021–2030), implemented	National Social Protection Policy (2021–2030), reviewed and strengthened	4	MGECW, MHAISS, reports	MEFT MGECW,

7. Infrastructure

2 measures; 5 actions; cost ~USD 312.0 million (USD 12.0 million, 4% unconditional / USD 300.0 million, 96% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Measure 1 National infrastructure resilience system								
Action 1: Develop and implement a national climate vulnerability assessment framework for critical infrastructure sectors, including transport, water, energy, health, education, and public facilities. **INDICATOR:** National infrastructure climate vulnerability assessment framework completed; Number of infrastructure sectors covered by the assessment								
MWT	Unconditional	0, Namibia does not have a single, standalone "National Climate Vulnerability Assessment Framework" document	2026	1 National standalone climate Vulnerability Assessment Framework document developed	Revise and update the framework to strengthen a short-coming	1.5	Annual Reports	Ministry of Works and Transport, OPM – DDRM
Action 2: Develop and implement national standards, guidelines and screening tools for climate-resilient and resource-efficient infrastructure and buildings, incorporating forward-looking climate projections and circular construction principles into planning, siting, design, construction, operation and major rehabilitation **INDICATOR:** National climate-resilient and resource-efficient infrastructure framework/standards adopted; Number of public infrastructure projects applying climate-risk screening/resilience standards; Number of public infrastructure projects applying resource-efficiency/circular construction criteria.								
MWT, Local Authorities	Unconditional	No mandatory national standards, unified building codes, or official screening tools specifically legislated for climate-resilient and resource-efficient infrastructure and buildings	2026	1 National standard developed and adopted	National standards under implementation	6	Annual Reports	MWT in collaboration with Local Authorities, Banking institutions

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
Action 3: Strengthen cross-sectoral coordination and financing mechanisms for climate-resilient infrastructure development and retrofitting. **INDICATOR:** Functional cross-sectoral infrastructure resilience coordination mechanism in place; Amount of finance mobilised for resilient infrastructure annually								
MWT	Unconditional	Namibia does not yet have a fully unified, comprehensive cross-sectoral coordination and financing mechanism for climate-resilient infrastructure	2026	A unified, comprehensive cross-sectoral coordination and financing mechanism for climate-resilient infrastructure developed	Sectoral coordination implemented	1.5	Annual Reports	MWT, MEFT, NPC, MoF
Measure 2 Climate-Resilient Road Infrastructure								
Action 1: Upgrade and climate-proof road infrastructures in flood-prone and climate-vulnerable areas. **INDICATOR:** Length (km) of roads upgraded or climate-proofed; Number of regions benefiting from upgraded road infrastructure								
MWT	Conditional	700 km. Projects like the "Access Roads for Climate Adaptation in Northern Namibia" have built over 700 km of climate-resilient roads in Oshana, Ohangwena and Omusati	2026	100 km of upgraded and climate proofed roads	200 km of upgraded and climate proofed roads	300	Ministerial reports	MWT, RA, RFA
Action 2: Integrate climate risk assessments and resilient engineering standards into road planning, maintenance, and rehabilitation programmes. **INDICATOR:** Number of road projects applying climate-resilient design standards; Percentage of priority road maintenance programmes incorporating climate risk measures								
MWT, RA	Unconditional	Considered but formal, system-	2026	10 road designs and con-	10 road designs and con-	3	MWT and RA reports	MWT, RA

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD (million)	Means of verification	Lead Agency
		atic integration across all routine road planning, maintenance, and rehabilitation remains an ongoing transition constrained by funding, data gaps, and traditional practices		struction projects fully integrating climate risk assessments and resilient engineering standards	struction projects fully integrating climate risk assessments and resilient engineering standards			

8. Coastal Zone

1 measure; 1 action; total cost ~USD 32.0 million (100% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD ('000)	Means of verification	Lead Agency
Measure 1 Integrated coastal zone management plan								
Action 1: Operationalise a national Integrated Coastal Zone Management (ICZM) framework integrating climate-risk monitoring, cross-sectoral coordination and priority coastal adaptation measures. **INDICATOR:** National ICZM plan operational; integrated coastal monitoring system operational; no. of priority coastal adaptation/management actions implemented.								
MAFWLR	Conditional	National Coastal Management Policy (2012) in place; regional BCC monitoring/data systems exist; cross-sectoral coordination previously established through NACOMA.	2020	National ICZM plan and integrated monitoring system operational; ≥5 institutions implementing the framework; ≥5 priority coastal actions implemented.	ICZM framework reviewed and fully institutionalised; monitoring system upgraded; ≥10 priority coastal actions implemented cumulatively.	32	MAFWLR reports	MAFWLR, MEFT, Local Authorities

9. Urban Settlements

5 measures; 8 actions; total cost ~USD 497.5 million (USD 4.5 million, 1% unconditional / USD 493.0 million, 99% conditional).

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD ('000)	Means of verification	Lead Agency
Measure 1: Strengthen climate risk assessment, vulnerability analysis and flood mapping for urban areas								
Action 1: Strengthen climate-risk-informed urban planning through integrated risk assessments, hazard mapping and incorporation of climate risk into spatial, land-use and investment planning. **INDICATOR:** % of local authorities with updated climate-risk/flood assessments; % applying climate-risk information in spatial/development planning.								
MURD, Local Authorities and Regional Councils	Unconditional	Climate-risk assessments exist for selected cities; participatory urban profiling/planning processes are already used in some settlements.	2026	At least 70% of local authorities with climate-risk/flood assessments; ≥80% integrate climate risk into spatial/land-use plans and development screening.	At least 90% maintain updated assessments/maps and integrate findings into planning and investment decisions.	4.5	Local authorities report	MURD, Local Authorities and Regional Councils
Measure 2: Implement climate-resilient upgrading of informal settlements								
Action 1: Upgrade priority informal settlements through climate-resilient drainage, flood-management infrastructure, improved roads and pathways, resilient housing materials, and reliable water and sanitation services **INDICATOR:** No. of informal settlements upgraded; households benefiting from climate-resilient infrastructure.								
MURD, Local Authorities, Regional Councils	Conditional	419 informal settlements; 160 have undergone some formalisation, but upgrading is not classified by climate-resilience criteria.	2026	At least settlements upgraded; At least 20,000 households benefiting.	At least 150 settlements cumulatively upgraded; At least 40,000 households benefiting.	200	Local authorities report	Local Authorities
Action 2: Support climate-resilient, energy- and water-efficient housing for low-income and climate-vulnerable households. **INDICATOR:** No. of households supported; % benefiting from energy, water or thermal-resilience measures.								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD ('000)	Means of verification	Lead Agency
MURD, National Housing Policy 2023	Conditional	Policy framework promotes affordable, climate-resilient and resource-efficient housing.	2026	At least 15,000 households supported; ≥70% benefit from resilience/resource-efficiency measures.	At least 30,000 households cumulatively supported; ≥90% benefit.	10	Local authorities report	Local Authorities
Measure 3: Promote water-sensitive urban design and green and blue infrastructure								
Action 1: Integrate water-sensitive urban design and green-blue infrastructure into municipal planning and development. **INDICATOR:** % of local authorities applying water-sensitive urban design; % of urban projects using sustainable drainage, rainwater harvesting, infiltration or reuse.								
MURD, MAFWLR, Local Authorities, Regional Councils	Conditional	Water-efficiency and reuse measures are already applied in Windhoek; national application remains limited.	2026	At least 40% of local authorities apply principles; At least 30% of urban development projects use water-sensitive systems.	At least 70% of local authorities; ≥60% of urban development projects.	250	Local authorities report	Local Authorities
Action 2: Establish and implement national water-efficiency requirements for new and substantially renovated buildings. **INDICATOR:** National building water-efficiency standard adopted; % of local authorities applying requirements.								
MURD, MWT, Local authorities	Unconditional	National Housing Policy (2023) promotes water efficiency; local authorities can regulate water fittings/conservation	2026	National standard adopted; ≥40% of local authorities apply requirements.	Standard reviewed/fully implemented; ≥70% of local authorities apply and verify compliance.	5	MURD/NSI records; local-authority development-control and building-plan approval records;	MURD, NSI
Measure 4: Strengthen coastal adaptation and resilience in Walvis Bay and Lüderitz								

Primary Source	Conditionality	Baseline Value	Baseline Year	2030 target	2035 target (extrapolated)	Cost (2030) USD ('000)	Means of verification	Lead Agency
Action 1: Update coastal climate-risk assessments and hazard maps for Walvis Bay and Lüderitz. **INDICATOR:** Risk assessments updated; coastal hazard maps maintained.								
MURD, Local Authorities, MAFWLR	Conditional	Climate-risk/vulnerability assessments exist for both towns.	2026	Assessments updated for both towns; At least 6 coastal hazard maps developed/updated.	Assessments and at least 6 hazard maps regularly updated and maintained.	5	Local authorities report	Local Authorities
Action 2: Implement priority structural and ecosystem-based coastal adaptation measures in Walvis Bay and Lüderitz. **INDICATOR:** Coastal adaptation plans under implementation; km of priority coastline protected/restored/managed.								
MURD, Local Authorities, MAFWLR	Conditional	Adaptation planning foundations exist for both towns.	2026	2 adaptation plans under implementation; at least 20 km priority coastline protected/restored/managed.	Plans reviewed and implementation sustained; At least 20 km maintained under active adaptation management.	10	Local authorities report	Local Authorities
Measure 5: Green spaces and urban corridors								
Action 1: Establish, rehabilitate and sustain green spaces and urban ecological corridors in priority urban areas through community stewardship. **INDICATOR:** Area (ha) of green spaces/corridors established or rehabilitated; no. of urban areas maintaining greening initiatives.								
MURD, Local authorities	Conditional	Urban greening/community initiatives exist, but no consolidated national area baseline is available.	2026	5,000ha established or rehabilitated, with stewardship arrangements in participating urban areas.	Green spaces/corridors maintained and expanded in priority urban areas.	13	Local Authorities reports,	MURD, Local Authorities









